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ABSTRACTS

George W. Corner
President
Carnegie Institution of Washington
Baltimore

Marion Hines
Second Vice-President
Johns Hopkins Medical School
Baltimore

Edward A. Boyden
First Vice-President
University of Minnesota
Minneapolis

Normand L. Hoerr
Secretary-Treasurer
Western Reserve University
Cleveland

PAPERS PRESENTED

(The numbers indicate the order in which the papers appear on the program of the meeting.)

40. *Origin and activity of primary trophoblast giant cells during implantation (days 4–6) in the albino rat.* Roland H. ALDEN, Division of Anatomy, University of Tennessee, College of Medicine.

These cells are believed to be identifiable in the preimplantation stage as enlarged polar (antimesometrial) trophoblast elements, with characteristic hyperchromatic nuclei, at 4½ to 5 days of age. As the embryo becomes situated in the egg-chamber, toward the end of the fifth day, the polar trophoblast cells increase in size and contact the uterine epithelium at the antimesometrial base and sides of the egg-chamber. There follows a very rapid hypertrophy and hyperplasia of the trophoblast elements, not only from the antimesometrial portion of the trophoblast, but from the base of the ectoplacental cone as well. Just where, and only where, primary trophoblast cells contact the maternal epithelium there is evidence of a disturbance and a subsequent destruction and dissolution of this layer. Careful study of the primary giant cells in normal implantation, investigation of abnormal implantation sites, and employment of artificial ova provide evidence that trophoblast giant cells: (a) are truly of embryonic origin; (b) migrate along the mesometrial-antimesometrial axis of the decidual crypt; (c) phagocytize extravasated blood; and (d) phagocytize uterine epithelial cells.

The frequently mentioned difficulty of distinguishing giant cells from decidual cells, after the disappearance of the uterine epithelium, has been wholly resolved by the employment of an eosin-methylene blue stain, after which the marked and persistent basophilia of the embryonic cells contrasts with the delicate eosinophilia of the decidual tissue.

60. *Age changes in the deep cervical lymph nodes of 100 Wistar Institute rats.*[1] Warren ANDREW, Department of Anatomy, Southwestern Medical College.

A series of Wistar Institute rats, ranging in age from 21 days to 1170 days, has been studied. The nodes have been found to present fairly characteristic pictures for the periods which may roughly be called youth, middle-age, and senility. The germinal centers (reaction centers) are just beginning to appear in the 21-day animals. They are large and conspicuous in animals from 50 days up through 300 days and become very definitely reduced in old age, although they are present to some extent in about half of the senile animals (over 700 days of age) and are found even in an animal of 1100 days. In old age the medullary part of the node is much increased at the expense of the compact cortical tissue. Wide sinusoids and even large rounded or ovoid cavities, lined with endothelium, are found in a high proportion of senile rats. Much of the tissue formerly composed of small lymphocytes comes to consist of plasma cells. The capsule of the senile rats frequently is infiltrated with mast-cells and these may enter the node itself.

[1] Part of a program on aging sponsored by Dr. Edmond J. Farris, Executive Director of The Wistar Institute, and supported by the Samuel S. Fels fund. This work was aided by grants from the John and Mary R. Markle Foundation and the Winthrop Chemical Company.

118. *Crypt-like analogues in the pharyngeal tonsil.* Leslie B. AREY, Department of Anatomy, Northwestern University Medical School.

Descriptions of the pharyngeal tonsil are either erroneous on the subject of crypts by explicit statement, or they fail to indicate the true conditions that exist. The epithelial furrows arise by frank folding and are far more complicated than can be judged from the surface; none is comparable to the crypts of the palatine or lingual tonsils.

Glandular ducts traverse the lymphoid tissue and open either into the furrows or onto the free surface. Many are markedly dilated so that they are tubular, spindle-shaped or conical. Their diameter compares well with the crypts of the adult lingual tonsil and with all but the largest in the palatine tonsil. These expanded ducts are definitely crypt-like, since they extend as diverticula into the substance of the tonsil and increase its epithelial area greatly. Although these dilatations doubtless function as crypts in the palatine tonsillar sense, they are not homologous structures. Their development is fundamentally that of ducts and they continue normally to serve as outlets from the subjacent glands.

Some blind diverticula occur as well. These are perhaps nothing more than local manifestations of invagination in a region notable for its tendency toward major and minor folding. Much less numerous than the expanded ducts, they conform better to the conventional conception of a tonsillar crypt. It seems probable, however, that their occurrence is incidental.

39. *Studies in gynecologic cytology.* J. E. AYRE and P. CHEVALIER, Gynecytology Laboratory, Royal Victoria Hospital, McGill University. (Introduced by C. P. Martin.)

Using the cervical and vaginal smear techniques, studies have been carried out on the individual cells desquamated from the genital tract. These revealed that normally cyclic changes occurred and that hormone therapy could also induce such changes.

A correlation between the cornified cell count, and the concentration of estrogen in different parts of the genital tract was attempted.

57. *Thymic reaction following injection of colloidal pigments.* Ralph N. BAILLIF, Department of Anatomy, Louisiana State University School of Medicine.

It has been found that after repeated intraperitoneal injections of colloidal pigments (Biebrich's scarlet, mercury sulphide or chlorazol black e) the thymus undergoes involution. In young rats the degree of response is dependent on the amount of pigment injected in a given time; severe reactions appear only when dosage is at the toxic level.

Involution begins with marked destruction of cortical thymocytes and hypertrophy of the reticulum. In this phase there is an accumulation of macrophages and mast cells in the perithymic connective tissue; a few phagocytes are also found in the perivascular medullary reticulum.

The macrophage cells appear to be of 2 functional types. One type, first seen at the cortical periphery, effects pigment granule coalescence with some difficulty, i.e., the ingested globules remain deeply colored, small and discrete. The second type of macrophage, most frequently found in the thymic medulla, is characterized by the presence of large, partially digested pigment enclosures.

Late in the involution process, the lobular cortices become completely freed of their thymocytes. This destruction is carried out by cytolysis following karyorrhexis, or after drawing out the nuclei into tenuous strands. The cortical reticulum is gradually replaced by collagenous fibers interspersed with dye-filled phagocytes and tissue mast cells. During the terminal phase, reticular stroma is confined almost entirely to the perivascular medullary zones. Most of the Hassall's corpuscles undergo complete regression.

81. *The effects of small doses of diethylstilbestrol on the anterior hypophysis of thyroidectomized rats.* Burton L. BAKER, Department of Anatomy, University of Michigan.

It was demonstrated previously that small doses of diethylstilbestrol administered to immature rats for short periods stimulated the mitotic activity of acidophiles increasing their number and inducing hypertrophy of the Golgi apparatus in these cells. Since these changes are similar to those described in hyperthyroidism, the experiment being reported was designed to test the possibility that estrogen might act on the acidophiles through the thyroid. Forty-four adult female rats were divided into the following groups: ovariectomized, non-treated; ovariectomized, stilbestrol-treated; ovariectomized, thyroidectomized, non-treated; ovariectomized, thyroidectomized, stilbestrol-treated. Six weeks after the operations, 2 μg of stilbestrol were injected daily for 10 days. The hypophyses were fixed in Bouin's fluid and stained with the Masson technique. Differential cell counts revealed: (1) a marked increase in the number of acidophiles in ovariectomized adult rats following estrogen treatment, (2) accelerated mitotic activity did not play as significant a role in this increase as in the immature rat, (3) combined ovariectomy and thyroidectomy almost completely removed the acidophiles, (4) stilbestrol treatment of these animals increased the number of acidophiles to about the level found in the ovariectomized, non-treated controls showing the action of estrogen on the hypophysis to be at least partially independent of the thyroid, and (5) small remnants of thyroid tissue did not affect the response.

65. *Some effects of amputation of the chick wing bud on the early differentiation of the associated motor neuroblasts.* Donald H. BARRON, The Laboratory of Physiology, Yale University.

Amputation of the wing bud of the chick at circa 72 hours produces no observable effect, in silver impregnated embryos, on the subsequent development and differentiation of the associated segments of the spinal cord until toward the end of the fifth day of incubation when the lateral motor column normally makes its appearance. Thereafter, in the absence of the wing bud, there is a reduction in the number of primary unipolar neuroblasts that become bipolar and lay down the lateral motor column. And apparently as a result of this reduction, fewer indifferent mantle cells differentiate into neuroblasts, for it appears that only those indifferent cells which are in close association with bipolar cells or others more advanced in their differentiation, develop into neuroblasts. As a consequence of the failure of the unipolar neuroblasts and the indifferent cells to complete their differentiation in normal numbers there is a resultant neuron hypoplasia in the lateral motor column as development advances. These results suggest that the axon of a unipolar motor neuroblast by establishing contact with peripheral elements — in this case myoblasts — enables that neuroblast to proceed in its development, and that the bipolar neuroblasts in turn provide a stimulus to indifferent cells which appears to be essential for their differentiation into neuroblasts.

69. *The effect of epinephrin on the pulsation frequency of embryonic heart muscle.* Alexander BARRY, Department of Anatomy, University of Michigan Medical School.

This report covers one phase of an investigation into the development of the physiological properties of muscle. It is concerned with the effect of epinephrin on the rate of pulsation of myocardial tissue of chick embryos between the ages of 40 hours and 7 days. The hearts were excised and placed in aereated, sugar-free Ringer-Locke's Solution at 37.5°C. The entire heart, or specific portions of the heart were mounted in such a way that they could be observed and their pulsations recorded by means of an optical level and slit camera.

When used on slowly beating tissues, it was found that epinephrin in concentrations as low as 1 part in 10 million caused an abrupt acceleration, reaching a maximum rate within a few beats. Usually within 30 seconds this rapid rate began to slow until in 10 to 15 minutes it levelled off. This deceleration was not due to a drop in the concentration of the drug. The degree of this maximal acceleration depended upon the initial rate of pulsation before the addition of epinephrin. The slower the initial rate, the greater was the acceleration; but if the initial rate was faster than 180 beats per minute, the addition of epinephrin produced no significant acceleration. This relationship held true for entire hearts and portions of hearts regardless of the age of the embryo from which the tissue was taken.

95. *Effect of intermittent exposure to 30,000 feet simulated altitude on memory in animals.* R. Frederick BECKER, Department of Anatomy, University of Washington.

Twenty-four young, male guinea pigs were subjected to a simulated altitude of 30,000 feet in a decompression chamber 6 hours daily, 6 days a week for a total of 250 hours. Prior to ascent all animals learned an alternation maze to the point of perfection. Four controls were picked at random from the superior half of initial learners and 5 from the inferior half. All other animals were exposed to decompression. All were retested after 100 hours exposure. No differences in retention of the learned task between controls and experimentals were noted. Groups of 4 experimentals and 2 controls were retested after 150, 200 and 250 hours decompression and sacrificed for study of brain pathology. In all retests controls exhibited perfect retention, while all altitude animals required retraining. The most significant impairment of memory function occurred in the 150- and 250-hour samplings. This impairment occurred without specific correlation with the brain pathology observed.

35. *Ipsilateral conduction in the medial lemniscus of the cat.* C. M. BERRY, R. C. KARL and J. C. HINSEY, Department of Anatomy, Cornell University Medical College.

Cathode-ray records were taken from unipolar electrodes placed in the thalamus and midbrain with the aid of a Horsley-Clarke instrument. Stimulation of both sciatic and both saphenous nerves produced large, positive deviations conducted at a maximum velocity of 50 m.p.s. when the electrodes were located in the medial lemniscus, in the *pars externa* of the ventral nucleus of the thalamus in the internal capsule. Ipsilateral stimulation produced spikes of about one-half the amplitude of spikes produced by contralateral sciatic and saphenous stimulation.

The exact location of the recording needle in each experiment was determined from serial sections of the brain stem. In the mid-brain, positive responses were localized in an extremely small area on both the ipsilateral and contralateral sides. At the thalamic level, responses were found diffusely in the *pars externa* of the ventral nucleus and in the internal capsule.

22. Phosphatase glycogen relationships in integumentary derivatives. Gerrit BEVELANDER and P. L. JOHNSON, Department of Anatomy, New York University, College of Dentistry.

In several previous studies we have shown that alkaline phosphatase and glycogen are present in the developing tooth and hair; we also pointed out a number of correlations in regard to growth, differentiation and spacial relationships of glycogen and phosphatase during development. We have now extended our observations to include a study of the interrelationships between phosphatase and glycogen in the developing feather. In this latter structure we observed that phosphatase is intimately associated with the mesodermal components of the developing feather, particularly during the period of active growth and differentiation. Glycogen makes its appearance on the thirteenth day of development. It is confined to the barb cells and by the seventeenth day it disappears completely. The similarities and differences of the phosphatase glycogen relationships of these integumentary derivatives together with certain aspects of functional significance which these substances may play in development will be discussed.

2. Experimental implantation in the rat and guinea pig. Richard J. BLANDAU, Department of Anatomy, University of Rochester, School of Medicine and Dentistry.

The current concepts of the mechanism of implantation include 2 opposite views regarding the specific role of the ovum in the implantation process. One group of investigators considers the blastocyst an active agent facing a passive endometrium, whereas another group holds that the ovum acts as an inert object and that the reactions of the endometrium are largely responsible for implantation. In order to obtain further information on this problem, glass or paraffin beads of the approximate size of the blastocyst, were inserted in the uterine lumina of rats and guinea pigs between the second and sixth day after the onset of behavioral estrus. The animals were subsequently killed between the eighth and twelfth days after the onset of heat and their uteri were examined to determine the fate of the beads. A definite species difference was observed in the endometrial response. In the rat the formation of the decidual crypts and the implantation of the beads into them occurred in the same manner and rate as did implantations of ova in the controls. In the guinea pigs, on the other hand, the beads did not penetrate the uterine epithelium and the decidual reactions induced by their presence in the uterine lumina were minimal.

66. *Growth of the lumbosacral nervous system of chick embryos after the substitution of tumors for the limb periphery.* Elmer D. BUEKER and Morris BELKIN, Departments of Anatomy and Pharmacology, Medical College of the State of South Carolina.

Hind limb primordia were removed from the right side of the 48–60-hour embryonic chick. Small pieces of Rous fowl sarcoma, mouse sarcoma 37 and 180, and mouse adenocarcinoma were then fitted into the limb region. After a total of 8–9 days incubation animals were fixed in Bouin's fluid, imbedded, sectioned at 10 μ and stained with hematoxylin. Studies were then made of the right and left sides of the lumbosacral nervous system.

Results of 50 experiments were as follows: (1) mouse adenocarcinoma transplants are mostly absorbed and the growth pattern of the lumbosacral nervous system at 8–9 days total incubation was similar to cases of radical limb extirpation; (2) mouse sarcoma 37 and 180 and Rous fowl sarcoma develop tumor masses at the limb site; (3) in these cases the lumbosacral nerves are well developed and spread out irregularly in the tumors, cells of the lateral motor column are reduced 30–60% while spinal ganglion cells approach the normal number or become hyperplastic. General results indicate that a sarcoma may be an adequate environment for the differentiation of spinal ganglia.

7. *Studies on the sebum on the forehead.* E. O. BUTCHER and J. P. PARNELL, Department of Anatomy, New York University.

Sebum was collected at the hairline weekly and daily from several individuals. Associated with an oily scalp was the most fat and the lowest percentage of cholesterol. Associated with scalps having dandruff was less fat and a greater percentage of cholesterol.

The amount of sebum which could be collected weekly or daily at the hairline from the individuals varied considerably. Much more sebum could be collected from an area which was covered by a shallow cup for several hours than from the same area after it had been uncovered for a similar interval.

Investigations showed that the decreased amount from the uncovered area was not due to votalization of the sebum, that the increased accumulation on the covered area was not due to the stimulation of the moisture under the cup, but it was due to the increased temperature resulting from the covering.

54. *The cell lineage of the inflammatory cells of acute encephalitis with a discussion of the significance of the plasma cell.* Berry CAMPBELL, Department of Anatomy, University of Minnesota Medical School.

A study was made of the development of the inflammatory cells of the early stages of herpetic encephalitis in rabbits. The formation of the perivascular cuff with the differentiation of infiltrating lymphocytes to form polyblasts and finally the macrophages has best been understood through the combined study of convential H. and E. stains and air dried Romanowsky-stained imprints. In early encephalitis there is differentiation of some of the polyblasts to form plasma cells. This cytomorphic sequence is of importance in that it links the earliest stages of this disease to allergic processes. Reference to other experimental work supporting this interpretation will be made.

Progressive changes in both the oligodendroglia and the microglia occur in the early stages of encephalitis. These consist of an "activation" of the cells resulting in the formation of a larger stainable cytoplasm, a rounding up of the cell, an increase in the chromatin content of the nucleus, and thickening of the nuclear membrane. A close similarity of the oligodendroglia to the inactive reticulo-endothelial cells of the connective tissue (the resting wandering cells) is suggested by these findings.

111. *Effects of use and disuse on neurosomes in skeletal muscle of the chameleon.*[1] Eben J. CAREY, Leo C. MASSOPUST, Walter ZEIT, and Eugene HAUSHALTER, Department of Anatomy, Marquette University School of Medicine.

The effects of unilateral tenotomy and immobilization by adhesive tape casts were observed on the neuromuscular apparatus of the control and experimental gastrocnemius muscles of the chameleon. The normally used and innervated gastrocnemius muscle has dark, granular, and light, agranular muscle fibers. The dark, granular muscle fibers progressively disappeared during the atrophy of disuse. The altered nerve ending during atrophy discharged small and giant fusiform neurosomes. The giant neurosomes were the product of retardation in rate of discharge, diffusion, and disappearance by hydrolysis during the atrophy of disuse. The experimental evidence suggested that one factor in the atrophy of muscle by disuse was a substantial loss of the normal discharge of neurosomes from the nerve endings in skeletal muscle.

[1] Aided by a grant from the National Foundation for Infantile Paralysis, Inc., and the Baruch Committee on Physical Medicine.

62. *Some preliminary observations on the descent of the testis in man.* Simon B. CHANDLER, Department of Anatomy, West Virginia University School of Medicine.

The differential growth between the trunk and the gonad is said to bring about a caudal shift of 10 segments in the position of the testis. Does the gubernaculum cause further descent by active as well as relative shortening? Moscowicz concluded that the gubernaculum is composed of mucoid connective tissue which first swells up to dilate the inguinal canal and then loses its sustaining power so that the testis sinks downward by a normal herniation process carrying the processus vaginalis with it.

This report is based on dissection of the inguino-hypogastric region in 21 fetuses whose crown-rump length varied between 9.5 cm and 23.5 cm. In fetuses from 9 to 10 cm long the processus vaginalis was about 1 mm in length. In those 17 to 18 cm long the testes were at or near the internal orifice of the processus, and inside the canal in those near 23 cm long.

Grossly the gubernaculum resembles umbilical cord. Histologically it is apparently mucoid connective tissue. In no specimen was it as large as the testis so it probably does not dilate the canal. Since it is small the testis probably could not sink down into its substance nor herniate the processus vaginalis.

13. The architecture of the Achilles tendon. K. S. CHOUKÉ, Department of Anatomy, Graduate School of Medicine, University of Pennsylvania.

The architecture of tendons has not received as much consideration as that of bones and muscles. This may be due to the difficulty of working out the structural patterns in tendons and in part to the lack of appreciation of their value. The tendon fibers from the lateral head of the gastrocnemius insert in front of the tendon fibers from the medial head. This slight torsion has been diagramed as an exaggerated helix. The anterior-posterior arrangement in the human has been studied by White who made use of it in devising a technic for tenotomy. The textbooks and handbooks of anatomy do not describe this torsion.

In 62 cadavera the arrangement of the tendon into 2 distinct, separable parts was found bilaterally in every instance; generally the tendon from the lateral head was anterior to the medial head tendon. In one cadaver the tendon relations were reversed bilaterally. This reversed condition was found in 3 other cases; 1 on the left side and 2 on the right. The operative technic recommended by White for tenotomy would work equally well in these variants. This reversal would rule out a correlation of this twist in the tendon with the torsion of the limb in development. A ready explanation for the occurrence of this pattern or its reversal is not evident.

112. The "interstitial cell net" of Cajal in the skin of mouse. C. H. U. CHU, Department of Anatomy, Washington University and The Barnard Free Skin and Cancer Hospital, St. Louis. (Introduced by E. V. Cowdry.)

Two kinds of nerve elements are observed in mouse skin fixed in formic acid and stained with hematoxylin. The one conforms with the interstitial cell net first described by Cajal (1889); the other appears as subcutaneous and subepithelial plexuses.

The interstitial cells (Cajal) vary from fusiform to multipolar according to the number of processes, which, often varicose and tortuous, anastomose with those of neighboring cells at different levels forming a network throughout the entire thickness of the dermis. Where there are few processes, the cell resembles a Remak's knot. At bifurcations, a fibroblast is often lodged. The interstitial cells differ from Schwann cells in that the width of the nucleus of the former often exceeds that of its processes.

The question whether the interstitial cell net is intrinsic has a significant bearing on functional interpretation. The spinal nerve forms a myelinated subcutaneous plexus which continues peripherally as an amyelinated subepithelial plexus. The subepithelial plexus intertwines repeatedly with the interstitial cell net, yet no continuity has been observed. The pseudopodial extension of fibroblasts at bifurcations might account for a different interpretation. Further observation reveals that the spinal nerves also possess an interstitial cell net which is seen to be continuous with that in the dermis.

This study leads to the belief that the interstitial cell net is of extrinsic origin and that the cells are lemmocytes.

6. *Arterial anastomoses — some factors concerned in their formation.* Eliot R. CLARK and Eleanor Linton CLARK, Department of Anatomy, University of Pennsylvania.

The great variation in arterial anastomoses in different parts of the body — a matter of great clinical importance — stimulates the question: why?

The governing factor appears to be Thoma's histomechanical principle, that the size of the lumen of an artery is regulated by the amount of blood flow. In the absence of flow, the lumen is reduced to zero and the artery obliterated. In order that anastomoses may survive, conditions must be such as to provide a flow of blood through the final connecting part.

Observation of living vessels both preformed and newly-formed in the rabbit's ear, as seen microscopically in artificially installed chambers, reveals, as an immediate factor for maintenance of arterial anastomoses, frequent reversal of blood flow in the terminal connecting portions. This is caused by unsynchronized contractions of arteries or arterioles.

This internal vasoformative factor must be added to the factor of varying external pressures that undoubtedly operates in causing reversals of blood flow in the exposed parts of the body.

If correct, one could predict that organs with very small or no arterial anastomoses, that are protected from varying outside pressures, will show little if any unsynchronized arterial contraction. This has been borne out in observations of the cerebral circulation (Wentzler); and of certain ear chambers in which arteries failed to receive a nerve supply, and were therefore without nerve-controlled contractions.

29. *Visual disturbances following frontal ablations in the monkey.* George CLARK and K. S. LASHLEY, Yerkes Laboratories of Primate Biology, Orange Park, Florida.

Confirming the report of Kennard and Ectors ('38) and of Kennard ('39) it has been possible to produce an homonymous hemianopsia by ablations of portions of the frontal lobes. The destruction must be more extensive than indicated in the original reports. In addition to the area within the arcuate sulcus much of the tissue mediad and rostrad must be removed. It was also possible to produce an homonymous hemianopsia by a transverse lesion of the medullary substance of the cerebrum severing the superior longitudinal fasciculus. Deviation of the head and eyes and circling do not necessarily accompany the visual defect. The duration of the visual defect is correlated with the amount of destruction. Eye movements in all directions in response to visual stimulation in the intact field (ipsilateral to lesion) are normal as are movements in response to auditory stimulation. It is suggested that the visual defect represents a traumatic disorganization of reentry circuits producing interaction between the frontal and occipital regions.

108. *Gross structure of the supra-pubic and pre-pubic subcutaneous layer in the male.* E. D. CONGDON, Department of Anatomy, Chicago Medical School.

Photographic records of serial sections in 27 male bodies show no superficial fascia passing down in front of the pubis. Instead of this, a rather complex and specialized system of fibrous structures was found. They include a tract in the upper pre-pubic region lying in the mid-plane of the body and dividing to form a loop around the free part of the penis. It corresponds roughly to the fundiform ligament of the penis as described by authors from dissections. This ligament is clearly distinguished from the suspensory ligament of the penis.

Traced cranially, the median tract is seen to break up into laminae in the supra-pubic region. These laminae, taken with the masses of fatty areolar tissue they enclose, form, when well developed, a clearly delimited body of pattern different from that of the panniculus adiposus lying superficial to it. This body tapers cranially and soon terminates.

The structures described show connections not only with the penis but with the scrotum. Fibrous condensations partly invest the spermatic cords in the pre-pubic region. They connect, toward the mid-line, with the median tract and they go over on either side distally into the superficial fascia of the corresponding half of the scrotum. Also in the mid-line, the fascia in the scrotal septum may be attached to the loop.

83. *Effects of hypophysectomy upon gastric acidity of adult female rats.* Roger C. CRAFTS and Burnham S. WALKER, Departments of Anatomy and Biochemistry, Boston University.

Hypophysectomy produces a microcytic hypochromic anemia in the rat. Injections of thyroxine, iron, and copper alleviate this anemia. In a study of the possible effects of hypophysectomy on iron metabolism, gastric acidity was the first aspect investigated.

Adult female rats of the Long-Evans strain and of 3 to 4 months of age were divided into 3 groups, (1) 11 normal control rats, (2) 12 hypophysectomized rats, and (3) 10 rats bled from the heart until anemic. After anemia had developed in the last 2 groups of rats, stomach contents were studied immediately after anesthetization with ether (gastric sample no. 1), ½ hour after anesthesia (gastric sample no. 2), ½ hour after an injection of 0.04 mg of mecholyl per kilo body weight (gastric sample no. 3), and 1 hour after the above injection (gastric sample no. 4).

Determinations were made on pH, total acid, free acid, hemoglobin, and total erythrocyte count. Erythrocyte counts and hemoglobin values for the normal control rats were 8.40 million cells per mm^3 and 15.1 gm per 100 cm^3; for the hypophysectomized rats, 5.60 million and 9.3 gm; for the bled rats, 5.42 million and 9.1 gm. In all samples the hypophysectomized rats had lower gastric acidities than the control rats. The bled rats had normal or elevated gastric acidities. Free acid was found in 8 samples from control rats, in 2 samples from hypophysectomized rats, and in 14 samples from the bled rats.

Hypophysectomy decreased gastric acidity in adult female rats. This decrease was not due to the anemia which was induced by hypophysectomy.

104. *Genetic differences in the ossification pattern of the rabbit.* D. D. CRARY and P. B. SAWIN, Department of Biology, Brown University.

The possible effect of genetic and environmental influences upon the pattern of first appearance of ossification is at present controversial, largely due to inadequate genetic control of the material previously used. A series of roentgenograms of 101 rabbits of 2 inbred races (III and X) has been made during the first 20 days post-partum, during which time most of the epiphyseal centers of the limbs are established. Forty of the race X individuals are heterozygous for the dwarf gene (dw). The order of first appearance of the centers in the 3 different populations is essentially the same except in carpals and tarsals, but time of appearance varies between the 2 races, the larger initiating ossification the earlier. In spite of a significant difference in body weight between the homozygous and heterozygous populations of race X, there seems to be little significant difference in the time of first appearance of the centers. Thus whereas the specific gene (dw) appears to have no influence upon the time of appearance of centers, the genes determining the racial differences in body size do seem to have a specific effect. Since it has been previously shown that ossification of the ribs occurs earlier and more anteriorly in race X than in race III, it becomes apparent that the specific genes controlling rib ossification are different from those controlling limb ossification.

101. *Anomalous right subclavian artery originating from the descending aorta.* Arthur DALTON, Resident in Surgery, St. Louis City Hospital, and William F. ALEXANDER, Department of Anatomy, St. Louis University School of Medicine.

The right subclavian artery arises from the right side of the descending aorta at the level of the lower border of the fourth thoracic vertebra. It crosses the vertebral column transversely, partially separated from the posterior surface of the esophagus by the thoracic duct.

At the right side of the fourth thoracic vertebra it executes a right angle and ascends along the lateral aspects of the ver brae paralleling the esophagus, but on a more posterior plane. As it turns lateralwards to leave the thorax it lies directly anterior to the first thoracic sympathetic ganglion.

Other relations were essentially normal except the origin of the right common carotid which arose from the right part of the aortic arch.

This variation of the origin of the right subclavian artery is the first encountered in observations on 193 cadavers, a percentage of 52. Its topography is such that it falls within Piersols' Group II of these variations.

115. *Histogenesis of the argentophile cells of the proventriculus and gizzard of the chicken.* Alden B. DAWSON and Sue L. MOYER, The Biological Laboratories, Harvard University and Radcliffe College.

These cells are demonstrable by the Bodian protargol method (even without reduction or gold toning) and the Gros-Schultze method, but are not chromaffin positive and are not impregnated by the Masson-Hamperl technique. Accordingly they are referred to as argentophile rather than argentaffin. In contrast, the majority of morphologically similar cells in the intestine are argentaffin. Argentophile cells first appear in the superficial epithelium of the proventriculus on the eighth day of incubation when the compound glands are seen as simple sacs, and reach their maximum concentration by the thirteenth day. Their distribution is modified by the development of superficial plicae, apparent by the sixteenth day, and they are, thereafter, progressively reduced in number. Argentophile cells of the compound glands develop in situ and are present in the primary glandular sacculations by the twelfth day. They increase in number rapidly as each saccule becomes multilobed and acquires secondary evaginations which eventually become tubular glands. Argentophile cells become most numerous on the periphery of the compound glands and at first are rounded. Shortly after hatching they become greatly elongated, lie scattered over the surfaces of the compound glands or are radially arranged between the distal ends of the constituent tubules. Argentophile cells appear in the epithelium of the gizzard 3 or 4 days before hatching. They never become numerous and always remain rounded in form.

129. *Changes in the adrenal cortex and thymus induced by different B-vitamin deficiencies.* Helen Wendler DEANE, Department of Anatomy, Harvard Medical School, and James H. SHAW, Harvard School of Dental Medicine.

Dietary deficiencies of thiamine, riboflavin, and pyridoxine were produced in young male rats. With riboflavin and pyridoxine deficiencies growth continued, although slowly, throughout a 10 weeks' period. With thiamine deficiency growth ceased after 2 weeks, and loss of weight ensued until death at 4 to 5 weeks. All 3 groups showed inanition due to loss of appetite. Of 2 control groups on a normal diet, one was fed ad libitum, the other pair fed with the thiamine deficient animals.

With thiamine deficiency the adrenals hypertrophied, while the thymus atrophied. Cytochemical preparations showed that the ketosteroids (sudanophilic, Schiff-positive, birefringent, and fluorescent lipids) of the zona fasciculata of the adrenal cortex increased in quantity by 2 weeks and then disappeared by 4 weeks. The mitochondria became swollen and difficult to stain. This stimulation, followed by exhaustion, of the fasciculata is attributed to a high level of secretion of adrenotropin under stress. Inanition controls experienced less atrophy of the thymus and slower stimulation of the adrenal cortex; consequently the deficiency changes were not the result of inanition alone.

With the absence of riboflavin or pyridoxine the adrenals hypertrophied only temporarily (probably a result of mild inanition), and no marked cytochemical alteration was observed. With pyridoxine deficiency, however, the thymus atrophied rapidly, presumably independently of adrenal stimulation.

1. *Electromyograms of wrist muscles correlated with wrist statics.* Wilfrid T. DEMPSTER, Department of Anatomy, University of Michigan, and John C. FINERTY, Department of Anatomy, Washington University.

In holding and supporting weights with the hand and forearm horizontal, various combinations of adjacent forearm muscles must resist the pull of gravity. Neither anatomical nor mechanical analyses tell how the various muscles share in the task. An action potential study was made of various (10) regions of the forearm muscle mass using surface electrodes over specific muscles. Motor points served as loci for the proximal electrode. Curves showing potentials correlated with the torque due to the weight of the empty hand and torques to 32x or 64x show increasing potentials with increased strength of isometric contractions.

Rotation of the hand and forearm through 12 stations 30° apart permitted potentials to be recorded for different directions of gravitational pull. Agonist, antagonist and lateral stabilizing actions of 10 muscles (?) could be compared. Although potentials of adjacent muscles are sometimes superimposed in one direction or another, a basic pattern of activity appears. Typically, the activity of a muscle is maximal when that muscle is in the direct agonist position and it decreases as the direct supporting function changes toward lateral stabilization. Action at a right angle to the pull of gravity may be as low as half the maximal agonist strength. As a muscle again approaches the vertical, lateral stabilization gives way to antagonist activity and this may be as low as a third of maximal agonist values for the weight used.

52. *The development of plasma cells in the organs of animals receiving various antigens.* Thomas F. DOUGHERTY and Harald GORMSEN, Department of Anatomy, Yale University and Department of Pathology, University Institute of Forensic Medicine, Copenhagen.

Single injections of antigen (staphylococcus toxin) resulted in an increase in the numbers of highly basophilic lymphoid cells in the spleen and lymph nodes of treated mice. These cells have been identified by various authors as immature plasma cells, reticular lymphocytes or acute splenic tumor cells. These immature cells are capable of differentiating to typical plasma cells and lymphocytes.

Prolonged treatment of rabbits with polyvalent pneumococcus or salmonella vaccines and of mice with either sheep cells or staphylococcus toxin also resulted in intensive proliferation of the immature basophilic cells in the lymphoid tissues. The number of typical mature plasma cells in the lymphoid tissues was far greater in the animals that received successive dosages of antigens.

Formation of immature basophilic lymphoid cells or of mature plasma cells was not observed in the thymi of animals receiving single or successive doses of antigen.

Plasmacytic reticular cells in the bone marrow increased markedly following injection of various antigens and appeared to differentiate to typical bone marrow plasma cells.

Extracts of non-lymphoid tissues of immunized animals containing numerous plasma cells yielded large amounts of antibody. In this respect plasma cells are similar to lymphocytes. The progressive increase in numbers of immature, basophilic lymphoid cells and mature plasma cells following successive doses of antigens is correlated with the development of the hyper-immune state.

43. *Fibrous connective tissue of the rabbit ovary.* Kenneth L. DUKE, Department of Anatomy, Duke University School of Medicine.

Differential stains for reticular, collagenous and elastic fibers were used on sections of rabbit ovaries covering the period from sex differentiation to maturity.

Reticular fibers appear first and are clearly seen in 22-day embryos. The fibers, in the mesovarium, extend into the epithelial core, insinuating between groups of cells. The ingrowing sex-cords are partially encased with a reticular capsule and the anlagen of the tunica albuginea is visible. The medullary fibers increase at the expense of the epithelial constituents, and radiating trabeculae increase in number and size as development proceeds. The theca interna is invaded by reticular fibers and interstitial cells are enmeshed in a reticular network.

Collagenous fibers show clearly in 27-day embryos. The distribution is similar to the reticular fibers but not so extensive. In older stages collagenous fibers are most pronounced in the tunica albuginea and the medulla.

Elastic fibers do not appear in the ovary until 4 days after birth when the proximal part of the ovarian artery develops an internal elastic membrane. An elastic interna in veins and externa in arteries does not develop until later: about 20 days after birth. Not until shortly before maturity do the distal portions of the vessels exhibit elastic fibers. Ovaries containing masses of interstitial cells and/or regressing corpora lutea show arteries with a laminated external elastic membrane.

103. The femoral venous system. Edward A. EDWARDS, Department of Anatomy, Harvard Medical School.

The femoral venous system was examined in 60 limbs, with reference to its operative interruption.

The *deep femoral vein* was constituted by the confluence of the perforating veins. It entered the femoral vein at an average of 8 cm below the inguinal ligament. It communicated with the femoral just above the adductor opening in several instances.

The *medial and lateral femoral circumflex veins* ordinarily terminated independently in the femoral above the deep femoral termination. The medial was the higher, entering at the same level as the saphenous, or above.

Collaterals of the femoral veins were often seen below the level of the deep femoral termination. *Anastomoses* between the femoral and pelvic veins consisted of 2 pathways: a medial, between the medial femoral circumflex, and the obturator, and inferior gluteal veins: and a lateral, consisting of an anastomotic circle from the lateral femoral circumflex vein, or the femoral above this level, to the deep circumflex iliac and external iliac veins. The valves of these anastomotic routes have been noted in this study.

Obstruction to venous return is least when the femoral is interrupted below the deep femoral termination. The anastomotic routes are small and devious when the femoral is interrupted above the deep femoral but below the circumflex femoral veins. The pathways described, constitute large and direct anastomoses, when the interruption is done above the circumflex femoral veins. The limitations imposed by their valves disappear when the gradual venous dilatation brings about a valvular insufficiency.

20. The effect of testosterone propionate on the alkaline phosphatase of the adrenal cortex of the mouse. Herbert ELFTMAN, Department of Anatomy, College of Physicians and Surgeons, Columbia University.

Alkaline phosphatase is present in marked concentration in both nuclei and cytoplasm of the adrenal cortex of the adult male mouse. In the adult female this enzyme is present only in traces in the adrenal cortex. The distribution characteristic of the adult male appears during post-natal development concurrently with the regression of the X-zone. The dependence of this distribution on androgen is indicated by its disappearance in the adult male after castration and its appearance in castrated males and females and in juvenile males and females after the administration of testosterone propionate.

116. Visualization of histological structure: The liver. Hans ELIAS, U. S. Public Health Service.

The teaching of histology is perhaps the most difficult part of medical education because the approach to this subject differs from that to all subjects with which students have previously come in contact.

Histological structure is a 3 dimensional affair. But it is customary to present it in the form of 2 dimensional slices. Much experience and a special stereographical talent are required to visualize the complicated relationships of the elements of an organ.

Modern histology teaching ought to be physiologically and embryologically oriented. The histology teacher ought not to spend all his time on the explanation of spatial relationships, but he ought to use as much of his effort as possible to explain the reasons for the existing structure.

In order to free teachers from lengthy discussions of structure, a series of slide films is being created to provide a stereographic synthesis of the architecture of organs. The first unit chosen is the liver because, according to the writer's experience, this is one of the most difficult chapters of a histology course, for students as well as for teachers.

44. Regenerative processes induced by gonadotrophic hormones and pituitary implants in x-ray damaged ovaries of mice. J. M. ESSENBERG, Department of Anatomy, The Chicago Medical School.

The ovaries of mice seem to be more easily damaged by x-ray than those of other mammals. This experiment was designed to test the regenerative power of damaged ovaries of young mice by follicle-stimulating hormones.

Four litters of young mice of ages 3 to 4 weeks were given 244 to 300 Roentgen units of x-rays. About one-third of these were set aside as treated controls. The remaining mice were divided into 2 groups; one to receive injection of gonadophysin (G. D. Searle and Company) and the other group for pituitary implant. From 10–15 R.U. of gonadophysin was given to each mouse of the first group twice weekly for 3 months; the mice of the second group received pituitary implant weekly for 3 months. During the 3 months samples were taken from all groups for study. At the end of the 3 months the remaining females were mated to normal males.

With the dosages used the ovaries are seriously damaged and remain so in most cases for life. The gonadophysin treated ovaries show active germinal epithelium and the formation of young follicles. The ova of these follicles soon atrophy. The same is true of the ovaries of pituitary implanted mice but less consistently.

It is concluded that in the albino mouse definitive germ cells arise from the germinal epithelium.

12. *The behavior of the femur as an elastic body.* F. Gaynor EVANS and H. R. LISSNER, Departments of Anatomy and Engineering Mechanics, Wayne University.

The use of the "stresscoat" technique provides a new approach applicable to both static and dynamic studies of the problems of bone and skeleton architecture and mechanics. "Stresscoat" is the trade name for a brittle lacquer which responds to tension deformation occurring in the object upon which the lacquer is sprayed. The lacquer is applied under known temperature-humidity conditions. The lacquered femur is placed in a materials-testing machine, and during the test cracks appear in the lacquer where fracture would eventually occur.

In the present tests cracks appear almost simultaneously on the superior surface of the neck and the lateral surface of the shaft about 6 inches distal to the gluteus minimus insertion. When a lacquer with a sensitivity of 0.0018 inches per inch was used cracks first appeared when the load reached 652.5 pounds (average). The nearer the load and the anatomical axes coincide the greater the load borne before cracks appear in the lacquer. Femurs become noticeably bent during the tests but quickly return to normal after removal of the load.

The method of determining the various reference points on the femur and the orientation of the bone in the testing machine are illustrated.

85. *Gigantism produced in normal female rats by chronic treatment with pure pituitary growth hormone. II. Skeletal changes.* H. M. EVANS, H. BECKS, C. W. ASLING, and C. H. LI, Institute of Experimental Biology and Division of Dental Medicine, University of California.

Gigantism was induced in 8 normal female rats by daily injection of pituitary growth hormone for 437 days, after their growth curves had plateaued. Their skeletons were examined roentgenographically and histologically, and compared with those of 5 normal controls.

Linear increase in the bones measured was proportional to that of the entire body. The average length of tibias was 44.6 mm for injected rats, and 39.7 mm for controls. Caudal vertebrae increased in length, the ninth segment measuring 10.6 mm in injected rats and 9.4 mm in controls.

Metacarpal and metatarsal maturation had occurred before onset of injections, and overgrowth of paws was not seen.

Endochondral ossification was studied at the proximal end of the tibia, at the costochondral junction and in caudal vertebrae. Injected rats had an average tibial epiphyseal cartilage width of 154 μ, compared with 126 μ in controls. In all epiphyses examined there was persistence of some zonal arrangement in the cartilage, erosion of cartilage at point of contact with capillary tufts, and formation of delicate trabecular bone, indicating continuing chondrogenesis and osteogenesis. Controls showed atrophic cartilage subject to little or no erosion; capillary tufts were rare; bone trabeculae were coarse and few in number.

Intramembranous ossification was stimulated, as shown by lines of accretion at the sagittal suture as well as on inner and outer tables of parietal bones.

19. *Factors influencing accumulation of histochemically detectable cholesterol in the corpus luteum of the rat.* John W. EVERETT, Department of Anatomy, Duke University School of Medicine.

In the rat corpus luteum widespread accumulation of cholesterol, demonstrable by the Schultz reaction, apparently can occur only after very recent luteotrophic stimulation. In the presence of the hypophysis, cholesterol deposition in pregnancy corpora lutea has regularly followed injection of estradiol benzoate in oil (2 μg or more) and has occurred approximately coincidentally with induced ovulation (ca. 36 hours after injection on the fourth day of pregnancy in our standard procedure). The identical chronological pattern of events was caused by minute estradiol crystals implanted at a similar time, provided that they remained in the animals longer than 24 hours. LH injected on the fifth day of pregnancy regularly caused both ovulation and cholesterol deposition within 18 hours, even when the hypophysis was removed at the time of injection. However, if the gland had been excised earlier (24 hours or more) neither event transpired. In hypophysectomized rats in which corpora lutea were maintained by LH-free lactogen, injection of LH induced cholesterol deposition, while administration of estradiol benzoate failed.

Estrogen probably brings about this accumulation of cholesterol by liberating large amounts of hypophyseal LH. In some manner LH affects actively secreting luteal cells to deposit cholesterol therein. If this latter substance is a natural precursor of progesterone as certain evidence implies, its accumulation suggests interference with, or possibly undue acceleration of, early steps in elaboration of corpus luteum hormone.

77. *Autoplastic and homoplastic transplantation of rat adrenal components to the kidney.* Newton B. EVERETT, Department of Anatomy, University of Washington.

In an attempt to determine the capacity of adrenal components to become functional in a heterotopic position transplants of the adrenal capsule, and of combined glomerular and fascicular zones were made to the kidney of the same animal, to litter mates and to less closely related animals. Pieces of the medulla were autotransplanted. One adrenal was removed from each of the hosts receiving an autoplastic graft. All transplants remained on the host for a minimum of 3 weeks.

The cortical transplants to non-related hosts were negative, 25% of those to littermate hosts were positive, and 75% of the autotransplants were positive. Both capsular, and combined glomerular and fascicular tissue were incorporated as successful grafts. These increased in size, showed mitoses, and the cells appeared physiologically active.

Thirty-three per cent of the autoplastic medullary transplants were positive in that the tissue persisted and was recognized as typical medullary tissue, but there were no indications of an increase in the component parts.

17. *Anatomical studies on jet penetration of human skin for subcutaneous medication without the use of needles.* Frank H. J. FIGGE and Robert P. SCHERER, Department of Anatomy, University of Maryland School of Medicine.

When it became known that the human skin could be penetrated by high velocity fine bore oil jets from Diesel engines, one of us (R.P.S.) designed and perfected an instrument to utilize this principle for subcutaneous administration of drugs without the use of needles. The present paper will describe the investigations on the anatomical potentialities of this instrument. A number of factors were found to influence the practical and physical aspects of skin penetrability. The velocity of the jet could be regulated by the instrument spring tension. The diameter of the jet could be regulated by the size of the opening in the ampule pressure chamber. By using jets of very small diameters it was found that materials could be injected subcutaneously with far less pain than is elicited by the introduction of a needle.

A variety of substances were injected into fresh unembalmed cadaver extremities and other parts of the body to determine depth of penetration, deposition pattern, and the practical limitation of the method. The instrument injected aqueous solutions, oils, colloidal suspensions, emulsions, and even mercury. The material injected with a jet was deposited over a wider area than that injected with a needle. Skin tension was found to be a more important factor in regulating penetrability than skin thickness. Kodachrome pictures will be projected to facilitate presentation of this data.

48. *A method for the bio-assay of minute amounts of progestin.*[1] Thomas R. FORBES and Charles W. HOOKER, Department of Anatomy, Yale University.

A method has been devised for application to unconcentrated biological materials which involves intra-uterine injection of mice. By micrometer advancement of the plunger of a syringe, 0.0006 ml of material is injected into a 5 mm uterine segment, defined by ligatures, in a young adult mouse ovariectomized 16 days earlier. A shorter period after ovariectomy decreases the uniformity of the response. The test animals are killed 48 hours after injection; shorter intervals between injection and autopsy decrease the sensitivity of response. The injected segment is fixed in Lavdowsky's fluid; 6 micron paraffin sections are stained with hematoxylin. A positive response to progestin consists of transformation of dense, fusiform stromal nuclei into enlarged, vesicular nuclei with fine chromatin particles and distinct nucleoli.

Under these conditions the minimum effective dose of progesterone dissolved in sesame oil has consistently been 0.0002 μg. The response has not been evoked by intrauterine injections of estradiol, testosterone, desoxycorticosterone acetate, or serum from ovariectomized mice, and it was not modified by estradiol given simultaneously with progesterone. A crude extract of sows' corpora lutea, liquor folliculi of sows, and serum from ovariectomized mice given 2 mg of progesterone 6 to 7 hours earlier each provoked a positive response.

[1] Aided by grants from the Committee for Research in Problems of Sex, National Research Council, and from the James Hudson Brown Memorial Fund of the Yale University School of Medicine.

88. Spike potentials in sensory fibers from the knee joint of the cat. Ernest GARDNER, Department of Anatomy, Wayne University.

It has been shown that some myelinated fibers in an articular twig of the posterior tibial nerve are derived from Ruffini-type endings in the posterior part of the capsule, while others accompany blood vessels to other regions of the knee joint. In the initial study of functions of these fibers, 3 different methods were used. (1) The sciatic was cut and all distal branches except the posterior tibial and its articular branch severed. When the sciatic was stimulated, no muscle contractions resulted. The spike recorded from the articular nerve following such stimulation was the compound potential of all A fibers in the nerve. (2) Spikes were recorded after stimulation of dorsal roots. The inflow thus determined was over lumbar 6 and 7, and sometimes sacral 1. Results indicated, therefore, that most if not all myelinated fibers in the articular nerve were afferent. The temporal dispersion allowed the recording of numerous elevations corresponding to fibers conducting at rates of 8 to 85 meters a second. This fits with the grouping of fiber sizes determined previously. (3) The articular branch was severed from the sciatic. Spontaneous potentials were usually present. Trains of spikes were recorded following pressure on the capsule. The receptors stimulated appeared to be slowly adapting and were probably the Ruffini-type endings described previously.

45. Studies on germinal epithelium in ovarian transplants in mice. W. U. GARDNER and M. H. LI, Department of Anatomy, Yale University.

Previous studies from this laboratory revealed that ovarian transplants into the spleens of castrated male and female mice gave rise to granulosa cell tumors, luteomas, or mixed cell tumors (Li and Gardner, Science, 105: 13–15, 1947). Ovarian transplants into the subcutaneous tissues or testes rarely or never became tumorous in either intact or castrated hosts.

Non-tumorous proliferation occurred in ovarian grafts at all sites (testis, subcutaneous tissues, spleen and kidney), and differed depending upon the sites and endocrine status of the hosts. All well developed grafts were partially surrounded by a bursa-like space or spaces of variable sizes lined by epithelium. Transplants in all sites, except subcutaneously in intact hosts, showed tubular ingrowths of germinal epithelium into the graft. Cysts formed by germinal epithelium sometimes contained blood and the lining cells contained pigment (hemosiderin) indicating phagocytosis. Stages indicating transformation of germinal epithelial cells into macrophages were noted. Ingrowths of germinal epithelial cells appeared to transform into interstitial and granulosa cells. Subcutaneous grafts of ovary in intact males showed no germinal epithelial ingrowths and had "hypophysectomy-like" interstitial cells. Luteinized follicles were found only in grafts in females. Qualitative and quantitative differences in the gonadotrophic hormones are considered responsible for part of the activity of the germinal epithelium and cellular constitution of the ovarian grafts.

130. *Histological observations on the vitamin E-deficient rabbit heart.* Arthur J. GATZ, Department of Anatomy, Loyola University Medical School, and O. Boyd HOUCHIN, Department of Biochemistry, University of Louisville School of Medicine.

Previous studies by the authors have shown that degeneration areas occur in the cardiac muscle of rabbits reared on vitamin E-deficient diets. The present series of E-deficiency experiments in which vitamin D was supplied as cod liver oil in some series, and as Viosterol in other series, confirms our previous results. The electrocardiograms of the experimental animals consistently show a reversal of the QRS interval in lead I.

The portions of the heart which show necrosis of cardiac muscle are listed according to the frequency and severity of the lesion: papillary muscles, columnae carneae, apex of heart, ventricular walls and septum and infrequently atria. The Purkinje fibers appear to be unaffected.

The affected cardiac muscle fibers may show wide contraction bands, hyaline degeneration, basophilia with the appearance of numerous droplets in the muscle fiber (which fail to respond to acid fast dyes and are unstained by osmic acid) and finally complete necrosis caused by the coalescence of the aforementioned droplets.

90. *Sensory and parasympathetic nerve terminations in the nasal mucosa of the cat.* Edwin F. GILCHRIST and Kermit CHRISTENSEN, Department of Anatomy, St. Louis University School of Medicine.

The mucous membrane was taken from various parts of the nose of normal animals and of 2 groups of operated animals: one, in which each animal had a unilateral cervical sympathectomy; the other, in which each had the nerve of the pterygoid canal and the sphenopalatine nerves removed on one side. Sections were prepared by the activated protargol technic.

In the epithelium of normal and sympathectomized animals single nerve fibers, or sometimes plexuses of nerve fibers, with bulblike enlargements were observed coursing from the base of the epithelium toward the surface. Similar nerve terminations were also seen in the tunica propria. These nerve terminations are considered to be sensory and the nerve fibers reach the nasal mucosa as components of the nasal and ethmoidal nerves. A part of these nerve terminations disappear following removal of the nerve of the pterygoid canal and the sphenopalatine nerves.

In both normal and operated animals smaller nerve fibers with beadlike endings terminated in both glands and arterioles. Plexuses of these small fibers were observed paralleling and frequently entering the walls of the blood vessels. The terminations of the small nerve fibers are regarded as parasympathetic, especially because they are present in the sympathectomized animals.

53. *Differentiation on the basis of plasma cell formation of allergic from non-allergic inflammation in the cerebral cortex of the rabbit.* Robert A. GOOD, Department of Anatomy, University of Minnesota Medical School. (Introduced by Berry Campbell).

In an attempt to elucidate the nature of inflammatory cells and the cytomorphic mechanisms in the brains of hypersentitive and normal rabbits the following experiments were carried out. Fifteen rabbits sensitized to egg white and 10 normal rabbits were used. The Romanowsky-stained air dried imprint method previously used was applied to this study. In both sensitive and nonsensitive rabbits sections of micro-lesions and imprints of the inflamed cerebral cortex were studied at interval's of 6, 12, 18, 24, 36, 48, 72, and 96 hours after the introduction of egg white.

In these studies previous author's findings that the acute inflammatory cycle in the central nervous system of non-sensitive rabbits is similar to that seen in loose connective tissue and that hemic lymphocytes are a source of phagocytic cells in central nervous tissues were substantiated. In rabbits sensitized to the irritant the time scale of the inflammatory cycle was altered, the inflammation was more extensive, and production of marked plasmacytosis in the lesions was noted. Alterations of the glial cells were seen in both the sensitive and non-sensitive rabbits.

98. *Vessels and nerves in the hypophysial stalk and median eminence in man.* J. D. GREEN, Department of Anatomy, Wayne University. (Introduced by Gordon H. Scott.)

Three groups of vessels may be distinguished in the median eminence and hypophysial stalk: (1) arterioles and small arteries in the pars tuberalis possessing smooth muscle and elastic tissue in their walls; (2) tufted glomerular-like loops of capillaries, chiefly within the median eminence which project into connective tissue sheaths composed of reticulum fibres, collagen, and undifferentiated mesenchymal cells; (3) portal vessels similarly ensheathed passing distally to the adenohypophysis and containing some smooth muscle in their walls. Nerve fibres are associated with all 3 types of vessel. Fibres may be seen to accompany the vessels in the tuberalis; these lie close to the wall and appear to be mainly vasomotor. The tufted capillary loops show many irregular fibres in their sheaths which are believed to derive mainly from the hypothalamico-hypophysial tract. They terminate in delicate branches and in small knob and ring endings. Similar endings and fibres may be seen about the portal vessels. Some of these endings are probably vasomotor since they are seen to end on smooth muscle cells. Others terminate blindly within the sheaths and may be secretomotor or even secretory. Nerve endings of similar nature have been found in a wide variety of forms, associated with corresponding blood vessels. So far the sheaths have only been observed in man and the macaque.

107. *An x-ray study of the growth and development of the sacrum of girls during puberty and early adolescence.* William Walter GREULICH and Herbert THOMS, Department of Anatomy and the Brush Foundation, Stanford University and Department of Obstetrics and Gynecology, Yale University.

Lateral roentgenograms of the pelvis were repeated at intervals of approximately 12 months, over a period of several years, on 115 young White girls. In the majority of cases, the first films were made either shortly before, or soon after, the menarche and the observations were continued for from 2 to 5 years.

A comparison of successive films of individual girls shows that the increase in the length and the change in the shape of the sacrum proceed more slowly than the changes in the dimensions and configuration of the superior aperture of the same pelvis. The sacrum seems to attain its adult shape and size within about 2 years after the menarche.

The paper is illustrated by lantern slides in which the sacral changes are compared with those which occur concomitantly in the size and shape of the superior pelvic aperture.

113. The localization of thyroxine containing radioactive iodine in the tissues of the rat. J. GROSS, Department of Anatomy, McGill University. (Introduced by C. P. Leblond.)

Radioactive iodine (I^{131}) was introduced into the thyroxine molecule by both chemical and biological methods. The chemical method yielded relatively large amounts of thyroxine (1–7 mgm), while with the biological method, microgram amounts of the labelled compound were obtained, these being within the physiological range.

Female albino rats were injected intravenously with the large dose of radiothyroxine and sacrificed at 2 and 24 hours. The radioactive hormone was traced by Geiger counter and autographic methods.

A predominant role of the gastro-intestinal system in thyroxine metabolism was shown by these experiments. At 2 hours after the injection, nearly half of the administered dose had found its way into the lumen of the gastro-intestinal tract and at 24 hours the major portion of the injected dose was found in the feces. Butyl alcohol extractions showed that a considerable proportion of the radioactivity was present as thyroxine with the exception of the stomach contents in which the radioactivity was predominantly as the non-thyroxine form.

The relative concentrations of radio-thyroxine in over 30 organs and tissues following large and small doses will be reported.

64. Deviation of axial organs as a cause of sirenoid malformations. Peter GRUENWALD, Department of Pathology, Long Island College of Medicine.

A series of sporadic malformations in chick embryos of the first week of incubation shows ventral deviation of the notochord, or of all axial organs into the yolk sac or its derivatives. The deviated parts may disappear later on by degeneration of their tissues. This results in a defect of median organs which may be classified as a sirenoid malformation, usually of a low degree. In some cases there is fusion of normally bilateral parts in the midline. One of these cases resembles a human siren, 19 mm long, which was previously described.

This mode of origin of sirenoid malformations, as well as the degeneration of previously normal appearing tissues in the development of hereditary malformations of similar types, demonstrates that the embryogenesis of sirenoid malformations cannot be determined by examination of the fully developed condition.

123. Effect of aging on 17-ketosteroid output of normal males. Howard B. HAMILTON, Department of Anatomy, Long Island College of Medicine. (Introduced by George H. Paff.)

Urinary 17-ketosteroid values in 48–144 hour collections from normal males aging 20–73 years were studied. The procedure followed is Dobriner's method of acid hydrolysis and continuous ether extraction. The extract is partitioned into acidic, phenolic (estragenic), and neutral (androgenic) fractions. After further purification, the 17-ketosteroids in the neutral fraction are measured colorimetrically.

The steroid values for 24 hours show a gradual but definite decline with aging. The average is 10.2 mgm (range 16.5–5.6; mode 9.1) for 11 men in the third decade; 10.7 mgm (range 22.8–5.2; mode 9.6) for 9 men in the fourth decade; 8 mgm (range 15.2–5.1; mode 7) for 11 men in the fifth decade; 5.9 mgm (range 10.4–3.5; mode 4.7) for 10 men in the sixth decade; 4.9 mgm (range 6.7–3.2; mode 4.1) for 7 men in the seventh decade; and 3.4 mgm (range 4.2–2.8; mode 3.3) for 3 men in the eighth decade.

The slow decline in average hormone output indicates gradual lessening of secretion of 17-ketosteroid precursors but at no age is there a sudden drop in titer. These data are inconsistent with postulates of an abrupt "male climacteric," but definite conclusions are not to be drawn until a complete study of these men with regard to testicular biopsies, urinary gonadotrophins, androgens and estrogens is made.

124. *Growth changes induced by androgens in the connective tissues, sebaceous glands, hairs, muscle, and melanoblasts of the skin.* James B. HAMILTON, Department of Anatomy, Long Island College of Medicine.

Kupperman (1944) reported that androgenic stimulation results in pigmentation of 2 cutaneous spots in hamsters. Histological study with Dr. William Montagna, using Mallory stains of the tissues of which this spot is composed, shows the following rapid changes and acceleration of numerous growth processes after androgenic treatment of ovariectomized hamsters: (1) Pronounced increase in vascularization. (2) Proliferation of hairs and increase in their size and pigmentation. (3) Greatly-augmented melanogenesis particularly in hair roots and regions surrounding hair follicles and sebaceous glands. (4) Many-fold increase in volume of sebaceous glands and marked fatty changes in their cells. (5) Enlargement of muscle fibers and fibrils of the panniculus carnosus accompanied by denser staining and accentuation of cross-striation. (6) Great increase in coarseness of fibers and fibrils in dermal collagenous tissues. The collagenous fibers appear light blue or gray, in contrast to their deep blue color in control animals, and after 1 week of treatment acquire orange-colored streaks that do not correspond necessarily with complete collagenous fibers. Cells in the dermis (particularly mast cells?) appear larger and in some areas are much more numerous. In some areas intercellular spaces are enlarged, suggesting that the amount of ground substance may be increased.

These changes induced by testosterone propionate are in the direction of the status of normal adult males, but with continued treatment the growth of some constituents surpasses that in normal males.

70. *Embryology of the molossid bat, Eumops.* George W. D. HAMLETT, Department of Anatomy, Louisiana State University School of Medicine.

Study of a series of embryos of the molossid bat Eumops has shown several points in which this relatively unknown neotropical family differs embryologically from other bats. The uterus is bicornuate, but only the right horn and ovary are functional. The single blastocyst attaches in the extreme upper end of the right cornu and expands into contact with the endometrium on all sides. The first chorionic villi are in the region of the yolk sac. This early yolk sac placenta is soon replaced by an allantoic placenta, and the yolk sac degenerates; the definitive endotheliochorial placenta covers the entire surface of the chorion until a late stage in development and never becomes definitely discoidal. The inner cell mass reaches the surface of the blastocyst to form an exposed embryonic shield which is then enclosed by the overgrowth of amniotic folds.

In its exposed embryonic shield Eumops differs both from European bats, in which the trophoblast is described as roofing over shield and primitive amniotic cavity at all times, and from the phyllostomids and Megachiroptera in which the amniotic cavity forms by hollowing out of the entypic inner cell mass. The diffuse, endotheliochorial placenta apparently differs from all other bats, although one worker has claimed a discoidal, endotheliochorial placenta for 2 European vespertilionids. Even in the phyllostomids and Megachiroptera, which attach over the entire chorionic surface, the villi soon disappear except in the definitive placental disk. The small allantoic sac resembles that of the vespertilionids and differs from both the Megachiroptera, where it is large, and the phyllostomids, where there is only an urachus.

55. *Hematopoiesis in the bone marrow of rats recovering from nutritional anemia.*
Christopher J. HAMRE, Department of Anatomy, University of Minnesota. (Introduced by Hal Downey.)

Sixty rats were made anemic by being fed a diet of cow's milk only. Seven of these were killed as untreated anemic control animals while 53 animals were fed 0.5 mgm iron and 0.003 mgm copper per gram body weight to stimulate recovery hematopoietic processes. The treated animals were killed at hourly intervals during the first 48 hours after feeding of copper and iron. Studies of blood values and of sections and imprints of the marrow of the femur and humerus of all animals were carried out.

The bone marrow of all untreated animals was hyperplastic and hematopoiesis of the homoplastic type was arrested through a failure of maturation of blood cells. The marrow responded to recovery treatment by the development of extravascular heteroplastic erythropoietic tissue, including numerous pronormoblasts. Development and proliferation of hemocytoblasts from reticular stroma cells was observed. The greatest development of the new erythropoietic tissue, the greatest number of instances of migration of normoblasts and granulocytes into the venous sinuses, and the highest values for total number of leucocytes, normoblasts, heterophiles and eosinophiles of the circulating blood were found for animals killed 18 to 26 hours after the first feeding of copper and iron. We regard this as the critical period of regenerative hematopoiesis. By 36 to 48 hours following beginning of therapy, heteroplastic erythropoiesis had disappeared and an active homoplastic erythropoiesis had been established.

99. *Quantitative analysis of the brain-isocortex relationship in Mammalia.* Pinckney Jones HARMAN, New York University College of Medicine and Georgetown University Brain Research Institute.

The volumes were estimated on stained and sectioned material by planimetric procedures. The values in cubic millimeters for total brain and isocortex respectively are as follows: Primates: Pan 81,400; 38,100. Papio 48,800; 22,400. Ateles 28,600; 13,800. Cercocebus 28,100; 15,300. Macaca 21,100; 10,600. Cercopithecus 19,500; 11,400. Cebus 17,200; 9,130. Perodicticus 2,930; 1,140. Lemur 1,980; 797. Galago 1,260; 577. Carnivora: Lynx 17,600; 7,310. Vulpes 11,500; 4,380. Cercoleptes 6,750; 2,650. Felis 6,690; 2,690. Canis (young) 5,630; 2,500. Mustela 4,470; 1,850. Putorius 776; 302. Ungulata: Tagassu 29,700; 8,390. Rodentia: Erethizon 4,320; 1,260. Cynomys 1,190; 341. Cavia 867; 243. Marsupialia: Didelphis 1,310; 162.

Curves have been constructed and regression formulae, t values and coefficients of correlation (r) have been computed for the log brain-log cortex relationship in Primates (including man), Carnivora, and Mammalia (using 1 representative from each Order). The r values are very high for Primates (0.9986) and Carnivora (0.9996), but only moderately so for Mammalia (0.7691). The indications are that the relationship within Orders is linear, although inspection of the primate plot reveals a tendency toward cortical maxima in the middle range, suggesting a curve of the second degree. Some other function (apparently parabolic) probably best obtains for the Class.

The above observations are consistent with the findings of previous investigators on the brain weight-body weight relationship in Mammalia.

131. *Histological changes in kidneys of choline-deficient rats.* W. Stanley HARTROFT, Banting and Best Department of Medical Research, University of Toronto. (Introduced by Charles C. Macklin.)

Young rats on a diet containing all known essential food factors other than choline, develop the lesion known as "haemorrhagic kidney." The present paper, confirming and extending reports of others[1,2] concerning this renal acholinopathy, is based on histological observations of paraffin and frozen microsections of kidneys of 75 animals, for 7 days on such a diet, and initially weighing 35 to 45 gm. Animals were killed at daily intervals during this period. Survivors, then given a normal diet, were sacrificed on reaching maturity.

Previously reported lesions in the acute stage, including subcapsular haemorrhage, postglomerular capillary plexus congestion, proximal convoluted tubular necrosis and tubular cast formation at the cortico-medullary junction, were confirmed. Preceding the development of these lesions, intracytoplasmic, sudanophilic droplets in the proximal convoluted tubular epithelial cells were noted. The possible pathogenic rôle of this lipoid substance will be discussed. Alterations of the renal vascular flow, demonstrated by intra-arterial india-ink injections, are considered an important factor in the production of tubular necrosis.

Extensive cortical parenchymal destruction during the acute stage produces characteristic sequelae in the survivors attaining maturity. Lesions of nephrons resemble those of subacute and chronic glomerular nephritis in man.

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[1] Christensen, K., Arch. Path. 1942, *34:* 633.

[2] György, P., and Goldblatt, H., J. Exp. Med., 1940, *72:* 1.

16. *The participation of various portions of the urinary bladder in normal and induced activity.* Erling S. HEGRE and E. H. INGERSOLL, Department of Anatomy, Medical College of Virginia, Richmond, Virginia.

A study has been made of the activity of the uriary bladder of the female cat under light nembutal anesthesia. Various points on the exposed surface of the bladder were marked by white spots. From a photographic record on 16-mm film, the point to point activity of the various regions was plotted. During each experiment, a continuous kymographic record of the total bladder response was also obtained.

A comparison was made of the activity observed in each quadrant of the lateral surface of the bladder during rhythmic contraction, after the intravenous injection of adrenalin, and on stimulation of the hypogastric nerves. Contraction and relaxation per unit length in the observed areas were compared as to amplitude, rate, duration and sequence.

All parts of the bladder may participate in a given spontaneous contraction but all regions may not begin to contract at the same time. On stimulation of the hypogastric nerves, all parts of the bladder under observation show an initial, unsustained contraction which is followed by a marked relaxation. After an injection of adrenalin, all parts of the bladder may not exhibit the same response. A sustained contraction has been observed in the proximal portion while all other parts of the bladder have relaxed.

56. *Effect of synthetic Lactobacillus casei factor on the blood changes induced by gastrectomy in the rat.* George M. HIGGINS and Olive R. JONESON, The Mayo Foundation; Dwight J. INGLE, Upjohn Laboratories.

Synthetic L. casei factor has proved effective in controlling certain clinical as well as experimental anemias. Granulocytopenia, induced by giving sulfonamides, was corrected by L. casei factor. Granulocytopenia, occurring spontaneously, was corrected by the factor. Neutropenias induced by bone marrow depressants, however, did not respond. Anemias, induced by changing the bacterial flora, improved by giving L. casei factor. Total removal of the stomach from rats incites microcytoses, lowered hemoglobin levels and lowered granulocyte counts. This report covers the results of giving L. casei factor to gastrectomized rats in which a microcytic hypochromic anemia had persisted for 5 months.

Twenty-four healthy adult male rats, weighing about 300 gm were selected. Sixteen were gastrectomized; 8 were unoperated. All were fed a purified diet, of sucrose, vitamin-free casein, corn oil and salt mixture. Five months after operation L. casei factor was provided in the diet, at a level of 100 μg per gm, for 14 days. L. casei factor was then given intraperitoneally (400 μg per day) for 14 days. Administration of L. casei factor, either in diet or parenterally did not change red cell count, red cell size or hemoglobin level of gastrectomized rats. Total number of granulocytes increased from 1,900 per mm^3 to 7,200 when the vitamin was given orally; and to 11,100 when given parenterally.

74. *The mouse ovary and its adrenocortical functions.* R. T. HILL, Department of Anatomy, Indiana University.

Large numbers of inbred mice (CHI strain) have served in these experiments. Ovaries were grafted in either the ears of donor animals (homografts) or in the ears or testes of male animals (heterografts). Adrenals were removed in 2 stages with approximately a 7-day interval between operations. Grafted ovaries were removed in some of the animals at various intervals following adrenalectomy. Death follows graft removal at intervals which are more variable and more extended than the survival interval in non-grafted adrenalectomized controls. When animals bearing grafts are adrenalectomized, part of the animals die within the period of survival of the controls. Others live more or less indefinitely, and at this writing, some graft bearing adrenalectomized animals are alive and in apparent good health at more than 250 days following adrenalectomy. The longest survival of adrenalectomized control animals (males and females) to date has been 29 days. Sodium chloride added to drinking water in concentrations of 1% to 2.5% does not prolong survival time of the controls.

24. *The movement of human fetal brain cells in tissue cultures.* M. J. HOGUE, Department of Anatomy, University of Pennsylvania.

Tissue cultures were made from the cerebellum of human fetuses whose crown-rump measurements varied from 27 mm to 185 mm. They were grown in roller tube and hanging drop cultures. The first cells to move out from the explanted tissue were small oligodendroglia cells which did not live long when alone. Later when granule cells and other nerve cells appeared the oligodendroglia cells lived and flourished in indentations of these cells and on their processes. This suggests a symbiotic relationship. Macrophages appeared at this time. These were soon followed by large nerve cells. In 10- to 12-day-old cultures Purkinje cells appeared and worked themselves free from the mass of migrating cells. They moved by means of their 2 large branched anterior dendrites. The long slender axon trailed behind. Other large cells, probably from the dentate nucleus, moved out by means of their processes which were numerous and much branched.

80. *The inhibition of mitotic activity in the anterior hypophysis during pregnancy and after injections of gonadotropins.* Thomas E. HUNT, Department of Anatomy, Medical College of Alabama.

Mitotic activity in the anterior hypophysis of 3-month-old ovariectomized rats can be increased from an average of 1.2 to 22.1 mitoses per mm^2 of section with 2 injections of 25 μg of estradiol benzoate (Progynon-B) made 48 and 72 hours before death. In pregnant animals of the same age such injections result in approximately the same mitotic activity (17.33, S.E., $\pm$ 3.27) if made on the fourth to sixth day of pregnancy, but if made on the tenth to thirteenth day, after the placenta is well established, mitotic activity is reduced to 4.35 $\pm$ 1.41. To test whether this inhibition of mitotic activity is due to gonadotropins, extract of pregnancy urine (Pranturon) and extract of pregnant mare serum (Anteron) were injected into ovariectomized animals which also received estrogen. Injections of 25 and 50 I.U. of the gonadotropins did not significantly inhibit the mitotic activity caused by the estrogen, but with 4 injections of 150 I.U. of Pranturon the activity was reduced to 4.9 $\pm$ 2.2 and with 3 injections of 500 I.U. of Anteron, it was reduced to 3.8 $\pm$ .87. In comparison with the animals receiving estrogen only (average 22.1 $\pm$ 5.5), the probability of the difference being significant is less than .01.

92. *Hypothalamic obesity in the cat.* W. R. INGRAM, Department of Anatomy, State University of Iowa.

Although hypothalamic obesity has been experimentally produced and extensively studied in the rat, the occurrence of this type of obesity in cats has not yet been reported. After production of bilateral hypothalamic lesions of rather specific location cats display altered behavior which ranges from defensive attitude to actual savageness, and includes considerably increased avidity of appetite. Most of these cats will gain weight quite rapidly. This occurs to some extent even while they are fed the same measured diet on which their preoperative weights were kept constant. This period of gain may level off after some weeks during which the weight may have increased 400 to 1000 gm, or in extreme cases 2000 gm. If the ration is than increased 50% a long period of increasing obesity ensues. In the most striking case the weight increased from 3100 gm initial to 7800 gm. This obesity may be reduced by drastic restrictions of diet. The fat deposits, while general, are greatest in the omenutum and retroperitoneal regions. The hypothalamic lesions are quite restricted and confined to the region of the ventromedial nuclei. In most cases the latter are completely destroyed; the ventral portion may be preserved, however, in fat and savage animals. These cats appear to have a tendency to fatty alteration of the liver.

26. *Tissue cultures of cells from ascitic fluid.* James B. IVERS and C. M. POMERAT, Department of Anatomy, University of Texas, Medical Branch.

Cells from ascitic fluid were collected from the bottom of 250 ml bottles after centrifugation for 15 minutes at 1500 r.p.m. Hanging drop and roller tube cultures were prepared, using rooster plasma and extract from chick embryos incubated for 7 days. The supernatant for roller tube cultures consisted of 10 parts of ascitic fluid, 10 parts of Tyrode's solution and 1 part of embryonic extract. Observations were made both on living cells and of preparations stained with hematoxylin-eosin-azure.

Upon comparison, cultures from a patient with a diagnosis of scirrhous carcinoma of the stomach were found to resemble those obtained from a patient with cirrhosis of the liver in that both started with a collection of polymorphonuclear leucocytes, lymphocytes, monocytes, polyblasts and mesothelial cells. After 10 days both resulted in pure cultures of spindle cells resembling fibroblasts.

In contrast, a patient with a diagnosis of intra- and extra-cystic papillary cystadenoma of the ovary and another with an adenocarcinoma of the ovary with metastasis to the peritoneum both showed organized epithelial tissue bits initially and these developed into well-formed cysts, suggesting a definite tendency to gland formation. In these 2 cases no spindle cells were seen.

Work in progress is attempting to define the effect of gonadotropins, sex hormones, anti-organ immune sera and x-rays on such cellular aggregates *in vitro.*

126. *Intrasplenic injections of alloxan in the rat.*[1] Ralph G. JANES, Department of Anatomy, State University of Iowa.

Alloxan is apparently effective in damaging the pancreatic islands for a few minutes only following its injection. In an attempt to determine whether the liver was responsible for the conjugation or destruction of alloxan, this substance was injected intrasplenically. Of 16 non-fasted rats which lived following intrasplenic injections of alloxan (50–150 mg/kg), 5 became severely diabetic. Of 10 rats which received similar amounts of alloxan intraperitoneally, 2 became severely diabetic. Alloxan was most effective in producing a severe diabetes when the rats were fasted 3 days before they were injected. Unless alloxan was injected very slowly (1/100 cm^3/min.) the rats failed to survive intrasplenic administration of the drug. When injected rapidly, infarcts in the liver were common or the liver cells showed a marked vacuolation, caused in part by a hydropic condition. The data indicate that alloxan was just as effective in producing diabetes when given intrasplenically as when given intraperitoneally. However, it is difficult to compare the effectiveness of intrasplenic and intraperitoneal injections, since it has been shown that the intravascular administration of alloxan is more effective than other methods and because intrasplenic injections might be considered comparable to intravascular injections. Nevertheless, it may be concluded that alloxan is not inactivated to any great extent by the liver.

[1] This work was supported in part by the Hoffmann-LaRoche Co.

94. *Effect of intermittent exposure to 30,000 feet simulated altitude on brain structure.* Arthur V. JENSEN and William F. WINDLE, Departments of Anatomy, University of North Carolina and University of Washington.

Young adult male guinea pigs were subjected to conditions simulating an altitude of 30,000 feet in a decompression chamber 6 hours daily and 6 days weekly for 100, 150, 200 or 250 hours. At appropriate times, the experimental animals and controls were killed by perfusion through the vascular system with a formaldehyde fixing solution to preserve the nervous system and other organs in situ. Histological technique was carefully controlled.

The animals responded to reduced barometric pressure by maintaining a quiescent state, usually with episodes of physical distress, collapse and, apparently, unconsciousness. After removal from the chamber, comparison with controls rarely revealed significant behavioral or physical differences. No permanent neurological defects could be observed. Some weight loss occurred.

Focal areas of degeneration occurred in the vermis of the cerebellum of all animals of the 200- and 250-hour groups and in 3 of the 7 animals studied in the 100- and 150-hour groups. Elsewhere in the brain, small areas of cell shrinkage or (less commonly) impaired staining occurred, fortuitously, it seemed. For the most part, brain tissues appeared to be normal and most regions looked exactly like comparable regions in the controls. The focal areas of degeneration or other structural change may have resulted from vascular stasis; marked leucostasis was encountered in the 250-hour group.

50. *A study of mitochondria in dry smears of embryonic blood.*[1] Oliver P. JONES, Department of Anatomy, University of Buffalo.

Studies on the megaloblast-normoblast problem have been made for a number of years (Jones, '34, '35, '38). During the course of these investigations it was necessary to study material similar to that which Doan et al. used when formulating their theory for erythropoiesis. Dry smears of blood from chick blastoderms revealed that both the cytoplasm and nuclei were not as devoid of structure as described by Doan et al. A comparative study of dry smears and supravital preparations showed a striking correlation between the pale or yellowish areas of hyaloplasm in the basophilic cytoplasm and the distribution of mitochondria. Because of this it was tentatively concluded that the so-called hyaloplasm represented negative images of mitochondria. More extensive studies were conducted on primitive erythroblasts from 11-day rat embryos by means of various cytological, cytochemical and optical techniques. It was found that freshly dried smears of embryonic blood cells preserve cytoplasmic detail to an extraordinary degree. Light areas in basophilic cytoplasm previously described as hyaloplasm or paraplasm represent, for the most part, the negative images of underlying mitochondria. Cytochemical techniques reveal that the light areas (mitochondria) are relatively rich in lipoid material and contain a carotenoid pigment — probably vitamin A. Cytoplasmic basophilia is due to ribonucleoprotein. Phase microscopy makes it possible to study structures in unstained blood cells which are usually hidden in routine preparations.

[1] Aided by The Dr. Grant T. Fisher Fund.

68. *Observations on the differentiation and connections of specialized conducting tissue in the human heart.* Cornelius T. KAYLOR and Jane S. ROBB, Departments of Anatomy and Pharmacology, Syracuse University College of Medicine.

Serial sections of 4 human fetal hearts aged 15½ weeks, 4½, 5½ and 8 months, were studied histologically to determine the differentiation and connections of A-V nodal and bundle tissue.

Nodal and bundle cells could be recognized in the youngest heart mainly by differential staining reactions. The topography of the definitive node and bundle is discernible at this early time. In the older hearts, conducting tissue showed distinct cytological and staining characteristics. The location of the A-V node and the bundle is similar to that described for rodents, carnivors, ungulates, and primates including the adult human. In the auricular septum, nodal cells merge imperceptibly with auricular muscle cells. In the ventricular septum, bundle cells merge with myocardial cells. Numerous septal connections, not in accord with usual descriptions, were observed in all hearts. The absolute size of the A-V node and bundle seemed to be maintained throughout development. No myocardial tissues were observed to be contributing to the specialized conducting tissues.

Sections of 2 4½-month fetal hearts were studied for nerve supply to A-V node. Preliminary examinations show nerve fibers to cardiac vessels and structures above the coronary sulcus, few if any nerve endings in conducting tissue.

37. Another approach to the problem of efferent fibers in dorsal roots. Donald L. KIMMEL and Elizabeth K. MOYER, Department of Anatomy, Temple University School of Medicine.

In 12 Wistar rats, the central segments of sectioned dorsal roots were anastomosed end-to-end with arterial sleeves to prevent the ingrowth of regenerating fibers. At 2 to 18 weeks after the operation, the cord with the anastomosed roots as well as the peripheral stumps of the sectioned dorsal roots with their ganglia were removed under ether anesthesia. Small sections were taken from the anastomosed roots for osmic fixation. The remaining tissue was fixed in formol-acetic-alcohol and impregnated by Bodian's protargol method.

In studying the serial sections, the anastomosed roots of 6 animals were found to be contaminated by aberrantly located ganglion cells, intradural anastomoses, or regenerating fibers from the surrounding roots. In all the uncontaminated anastomosed roots of the remaining animals a few, fine, normal-appearing fibers were found. The intact fibers were sparse and scattered throughout the roots. At the cord-rootlet junction they lay in a position corresponding to the small fibers of a normal rootlet.

11. Anatomical relations of possible significance in the production of the scalenus anticus syndrome. Homer D. KIRGIS, Department of Anatomy, Tulane University, School of Medicine, and Division of Neurosurgery, Ochsner Clinic, and Adrian F. REED, Department of Anatomy, Tulane University, School of Medicine.

One hundred and twelve dissections have been made to determine anatomical relations of possible significance in production of the scalenus anticus syndrome. The etiological significance of a cervical rib or an elongated cervical transverse process is well-known but the majority of cases do not present these anatomical variants. Additional anatomical features which seem important include the relations of the scalenus anticus muscle at its origin, the presence of a scalenus minimus muscle or a vertebro-costal ligament, the relations of the scalenus medius muscle, and the sites of insertions of the scalene muscles. Each subject revealed muscle fibers joining the upper portion of the scalenus anticus from an origin along the inferior and posterior aspects of the intervertebral canal. These, upon contracting, may exert pressure on the nerve located immediately inferior to their points of origin. Gross and microscopic examination demonstrated many muscle fibers to have their origins from the sheaths of adjacent nerves. The scalenus minimus muscle was present in 55.4% of the cases and the vertebro-costal ligament in 25.9%. The widths of the insertions of the scalenus anticus, minimus (or vertebro-costal ligament) and medius muscles and the intervals between the insertions of these structures were measured. Although certain anatomical relations may favor the development of the scalenus anticus syndrome, it is apparent that no variation from the normal anatomy need be present for its occurrence.

49. Intraepithelial granular cells in urinary tracts of rats infested with Trichosomoides crassicauda. Hadley KIRKMAN, Department of Anatomy, Stanford University.

The cells are absent in the transitional epithelium of uninfested rats. Their origin and homologies remain uncertain. On the basis of staining reactions and other tests applied to fresh and fixed material it is believed that they are not mast cells, Russell body cells, serous cells, mucous cells, plasma cells, fibrocytes, macrophages or hemophages.

They exhibit certain similarities to eosinophilic myelocytes, globular leucocytes (Schollenleukocyten), and ''secretory leucocytes'' of fishes. Unlike the myelocyte (but like the mast cell) they exhibit no mitochondria with Janus green, anilin acid fuchsin or iron hematoxylin. Their granules stain well, but not metachromatically, with alkaline toluidine blue and thionine. With eosin they stain but lightly. Pale lavender with Wright's stain they color brilliantly with azocarmine. They are negative to indulin and to Nadi reagent. On the other hand, like eosinophilic myelocytes their granules assume a pale yellow coloration with neutral red, are highly refractile and give a variable peroxidase reaction.

Unlike certain descriptions of ''globular'' and ''secretory'' leucocytes they contain no iron or hemoglobin and exhibit no siderophilia. Their granules are less labile and less eosinophilic.

In view of lack of precise data concerning the nature and significance of ''globular leucocytes'' it is concluded tentatively that the 2 cells may be related and that the possibility of a causal relationship between roundworm infestation and Schollenleukocyten might well be investigated.

51. In vivo acceleration of fibroblastic mitosis by injection of extract of inflamed tissue. Fred KOLOUCH, Department of Surgery, University of Minnesota Medical School.

The usual fibroblastic mitotic activity commencing 24 hours following induction of sterile inflammation by the subcutaneous injection of egg white in the rabbit led the author to investigate the effect of the subcutaneous introduction of a saline extract of connective tissue inflamed for 24 hours. The extract was made from connective tissue removed from a rabbit 24 hours following subcutaneous injection of egg white. 0.1 cm^3 of the extract was injected subcutaneously at multiple points on the back of normal rabbits. Biopsies prepared as dry tissue spreads (stained with Wright-Giemsa) were studied at 2, 4, 8, 12, 16, 24, 48, 72, and 96 hours following injection. An acceleration of the entire inflammatory cycle occurred as compared with the response to egg white or an extract of normal rabbit connective tissue. An increased number of polymorphonuclear leukocytes as well as an earlier appearance of hemic lymphocytes followed by a rapid transformation of lymphocytes to macrophages occurred. The fibroblasts showed an earlier nuclear hyperchromasia and cytoplasmic basophily. By 24 hours mitotic activity of the fibroblasts was considerably increased over that of the controls.

86. Gigantism produced in normal female rats by chronic treatment with pure pituitary growth hormone. III. Pituitary-cytological changes. Alexei A. KONEFF and Choh Hao LI, Division of Anatomy and Institute of Experimental Biology, University of California.

Normal plateaued female rats were injected for 437 days with pure growth hormone. Non-injected rats of the same age were used as controls. The average weight of the pituitaries was increased, but the increase was not proportional to the body weight gain. Pituitaries were examined in 4 micra celloidin sections stained with Mallory-Azan. Acidophils were markedly decreased in number. Planimetric measurements showed a significant decrease in size of acidophils (mean of injected group 156 ± 6, control group 189 ± 7). There was considerable depletion of granules in these cells. The extent of the changes in the acidophils paralleled the body weight increase, being most pronounced in the animals showing the greatest gain. The findings must be interpreted as indicating decreased functional activity of acidophils — the cells believed to be the site of growth hormone production. The basophils of six of eight pituitaries did not manifest any significant deviation from the normal picture; castration-like changes were present in the basophils of the remaining two pituitaries. No changes were noted in the intermediate or posterior lobes of the pituitaries.

100. New apparatus for the study of cardiac and circulatory ballistics. Vernon E. KRAHL, Department of Anatomy, Wayne University. (Introduced by F. Gayor Evans.)

Two vertical ballistocardiographs are described. One is a simple appartus consisting of a bath scale, writing lever and kymograph. It does not give quantitative information concerning heart output, but produces the characteristic ballistic waves and records changes in the size of impacts. It has proved satisfactory in the teaching laboratory as a means of demonstrating the effects of exercise and certain drugs upon cardiac output. In the second and more sensitive instrument, the recoils of the body produce deformations in resistance strain gages which are arranged as a Wheatstone bridge. The unbalance of the bridge is amplified and the signal fed into a cathode ray or other oscillograph. Records obtained with each ballistocardiograph are illustrated and discussed.

31. General conclusions from an experimental study of the cerebral cortex of the albino rat. Wendell J. S. KRIEG, Institute of Neurology, Northwestern University Medical School.

One hundred minute lesions were made in the cerebral cortex and thalamus of the albino rat, the brains prepared by the Marchi method, and the revealed connections stereographically reconstructed. Findings were then synthesized and described on the basis of anatomical cortical areas, permitting the following general conclusions:

Applications of the Marchi method after cortical lesions does not distinguish between thalamocortical and corticothalamic fibers. Every thalamic nucleus studied sends fibers to the cortex and none studied sends fibers elsewhere. Internuclear connections are very scant.

Nearly every area sends nonthalammic projection fibers to the cerebral peduncle or further. These pass to nearly every main division of the brain, except striatum and hypothalamus. Most areas have homeotopic callosal connections systematically arranged but no heterotopic connections were observed.

Association fibers are few, and are mostly intracortical, but the concept of an undifferentiated feltwork is not sustained. Many associational connections are directed caudomedially so that the retrohippocampal region receives from a widespread expanse.

All categories of connection indicate a high degree of point for point specificity and the fibers are arranged discretely and orderly in the fiber accumulations. Exceptions exist. Destruction of the outer layers of the cortex or undermining does not produce substantial cell loss of the remaining layers, due perhaps to the numerous intact collaterals. Retrograde degeneration is not a fruitful method for cortical investigations.

96. *Neuroectodermal neoplasms of borderline dysplastic character. A survey of different types and their relationship to normal histogenesis.* Hartwig KUHLENBECK and Webb HAYMAKER, Army Institute of Pathology, Washington, D. C. and Department of Anatomy, Woman's Medical College of Pennsylvania.

In a survey of the tumor material available at the Army Institute of Pathology, the following 5 types of neuroectodermal neoplasms were found to be of special interest as examples of disturbed normal ontogenetic processes.

1. Atypical subependymal spongionueroblastoma associated with tuberous sclerosis (Bourneville's disease).

2. Neoplastic transformation of hyperplastic corpus pontobulbare and subependymal cell plate in the wall of the fourth ventricle (spongioblastoma subependymale diffusum).

3. Dysplastic, persistent transitory embryonic superficial granular layer of the cerebellum associated with cerebellar neuroblastoma or spongioneuroblastoma containing anaplastic Purkinje and other neuronal elements.

4. Transformation of dysplastic superficial granular layer of the cerebellum, associated with ectopic Purkinje cells, into glioneuroma with neuronal elements of Purkinje cell type.

5. Transformation of the margin of a dysplastic optic cup into a glioneuroma with neuronal elements of pyramidal cell type, filling part of the anterior half of the eye bulb.

The significance of the subependymal cell plate as a potential source of diverse typical neuroectodermal neoplasms has already been pointed out in a previous communication. In addition, the present series shows that this structure may be involved in dysplastic processes of borderline neoplastic character. Other embryonic structures which may undergo similar transformations are the corpus pontobulbare, the transitory superficial granular layer of the cerebellum, and the margin of the optic cup.

91. Reactions of interstitial and capsule cells in autonomic ganglia to various stimuli.[1] Albert KUNTZ and Normal M. SULKIN, Department of Anatomy, St. Louis University School of Medicine.

Hyperplasia of the interstitial tissue, including the capsule cells, in autonomic ganglia has been induced in experimental animals (cats and dogs) by prolonged faradic stimulation of preganglionic fibers and by periganglionic application of alcohol. It is also a common phenomenon associated with various pathologic states in man. The cells in question undergo amitotic division.

Under conditions of hyperplasia, some interstitial cells assume spheroid forms and appear to be devoid of processes. Within the ganglion cell capsules some cells become separated from the capsular wall. Some of these gradually become spheroid and encroach upon the ganglion cells. At the points of contact of the spheroid cells with a ganglion cell concavities arise in its peripheral cytoplasm. These concavities gradually become deeper and the spheroid cells sink into the ganglion cell body. This involves dissolution of some of the cytoplasm of the ganglion cell apparently due to a cytolytic process.

[1] This work has been supported by a grant from the Life Insurance Medical Research Fund.

33. Some morphological features of the human cerebellum. O. LARSELL and W. A. STOTLER, Department of Anatomy, University of Oregon Medical School.

Two adult human cerebella, 1 with unilateral agenesis of one cerebellar hemisphere and the other with greatly reduced anterior lobe, and several normal child cerebella, show clearly the spino-cerebellar and trigemino-cerebellar tracts in simple relations approaching those of lower vertebrates. The 2 tracts lie close together and form a commissure corresponding to the cerebellar commissure of lower forms. In the case on one-sided agenesis the tracts are shown to be chiefly crossed. The dorsal nucleus of Clarke, the dorsal spino-cerebellar tracts, lateral cuneate nucleus and direct cuneo-cerebellar tract appear only on the side homolateral to the intact hemisphere. A band of fibers apparently corresponding to the lateral commissure of lower vertebrates extends medially from the base of the floccular peduncle and decussates across the midplane. The arciform nucleus of the medulla oblongata is connected with the contralateral side of the cerebellum by fibers which course near the olivo-cerebellar tract. A small homolateral connection also is present. The stria medullares of the fourth ventricle floor connect the arciform nucleus with the contralateral corpus pontobulbare. No ventral external arcuate fibers, as usually described, are present. Olivo-cerebellar and ponto-cerebellar connections appear to be predominantly contralateral, while reticulo-cerebellar connections are homolateral.

109. The effects of partial starvation on somatotype. Gabriel W. LASKER, Department of Anatomy, Wayne University, and Laboratory of Physiological Hygiene, University of Minnesota. (Introduced by Ernest D. Gardner.)

Under the direction of Dr. Ansel Keys and his co-workers at the University of Minnesota, 34 volunteers were limited to a diet averaging 1,760 calories daily for 24 weeks. During this period they lost an average of 24% of their body weight.

Standardized nude photographs taken at the beginning and end of this period were made available to the author for study. Somatotyping techniques have been applied to these photographs to see whether so-called ''constitutional'' traits were influenced by changed dietary conditions.

Though partially starved individuals do not look identical to habitually lineal individuals, application of somatotype criteria proved practicable. Using 2 different codifications of Sheldon's technique of somatotyping, the majority of traits are shown to change in most of the individuals. Every individual shows a noticeable change in somatotype. Only in the wrists and ankles is there ordinarily little change. Similarly, preliminary studies by 5 somatotypers working under the direction of Dr. James Andrews in Professor Hooton's laboratory at Harvard confirm the findings of changed somatotype in every individual. Furthermore, 17 measurements taken by Sheldon as indicative of somatotype all show a tendency to decrease in the present series, most of them in every individual. The technique of somatotyping as applied in this study would thus seem to be a better measure of nutritional state than of inherent constitution.

128. *Protection against alloxan diabetes.* Arnold LAZAROW, Department of Anatomy, Western Reserve University.

Glutathione, cysteine, or thioglycolic acid previously have been reported as protecting rats against diabetes when these sulfhydryl compounds were administered intravenously prior to a diabetogenic dose of alloxan.

In the present study British Anti-Lewisite (BAL) was tested for protective action against alloxan diabetes. BAL is a disulfhydryl compound which can reactivate sulfhydryl groups of enzymes, previously inactivated by heavy metals. When BAL was injected in doses of 0.36 mM/kg immediately preceding a diabetogenic dose of alloxan, all rats were protected. However, when doses of 0.75 mM/kg of BAL were administered 5 minutes after alloxan, all rats developed diabetes. The diabetogenic action of alloxan may be due to its inactivation of essential sulfhydryl groups of enzymes within the beta cells. The failure of BAL to protect against diabetes when it was administered 5 minutes after alloxan may be due to an irreversible inactivation of sulfhydryl groups by alloxan.

Methionine, in doses of 2.9 mM/kg did not protect against alloxan. Thiourea, in doses of 7.5 mM/kg followed by alloxan, 40 mg/kg, proved acutely toxic. When nicotinamide (7.5 mM/kg) was injected intravenously immediately preceding alloxan (40 mg/kg), 5 of 18 rats developed diabetes. When alloxan alone was administered intravenously (40 mg/kg), 20 of 21 rats developed diabetes.

122. *Factors influencing steroid hormone action in rats and mice.* James H. LEATHEM, Department of Zoology, Rutgers University.

The route of administration is a determining factor in the response of 22-day-old rats to a single injection of 0.25 mg of testosterone propionate. After 72 hours the seminal vesicles are increased 3 fold when the intramuscular and subcutaneous routes are used but only slight stimulation follows intraperitoneal injection. In the mouse, a similar difference in injection routes is shown when the adrenal X zone is the criterion. Furthermore the volume of oil does not influence the response to a single injection.

Extending the injection period to 10 days it was found that the daily subcutaneous injection of 0.3 cm^3 sesame oil into 22-day-old mice would modestly suppress seminal vesicle weight. Therefore, the 0.25 mg total dose of testosterone proprionate was retested by dividing in into 10 daily subcutaneous injections given in 0.1 and 0.4 cm^3 of oil. The increase in seminal vesicle weight was roughly 3 fold in both groups, thus simulating the single injection in degree of response and in an absence of the oil effect. Unlike the single injection in which the intraperitoneal route was not effective, the intraperitoneal injection of 0.025 mg daily for 10 days caused a 250% increase in seminal vesicle weight and testis weight was only slightly influenced in comparison with the subcutaneous route. Therefore, the mode of administration and experimental period are effective agents in influencing results.

120. Detached respiratory squames from the lung of the albino mouse. Charles C. MACKLIN, The University of Western Ontario Faculty of Medicine, London, Canada. (See demonstration.)

If through a drop of physiological saline the fresh collapsed albino mouse lung be gashed repeatedly and the exuding mixture spread on a slide, dried and stained as for blood with Wright or Giemsa, there appear numerous thin cell derivatives which are identified as the desquamated respiratory squames or plates of Kölliker. The appearance is widely varied, and one cannot say that the average squame is either definitely nucleated or unnucleated; but rather that the nucleus has undergone profound change, tending to merge with the cytoplasm and to become extremely thin and pale-staining. Often its position is indicated, if at all, only by a less weakly stained central area. Frequently a sunburst form is presented, the nuclear substance seeming to be continued as radiations into the cytoplasm. Often an arc of the nuclear circumference forms part of the boundary of the squame. With ordinary section stains, the squames show little or no color. In fresh mounts, with weak illumination, they may be described as ghostlike. Trypan blue granules, in the chronically stained mouse, were seen in both fresh and fixed squames, so confirming the finding of John L. Bremer. Failure to see squames in ordinary sections is due to their extreme thinness and their frequently having been exfoliated, to appear only as debris. In the albino rat analogous, usually completely anuclear, squames were seen.

28. Sulfhydril groups, lithium, and the motility of human spermatozoa. John MACLEOD, Department of Anatomy, Cornell University Medical College, New York, N. Y.

Evidence is presented that the motility of human spermatozoa is dependent upon the integrity and lability of sulfhydril groups in certain enzyme systems. The addition of inhibitors such as Cu and arsenicals which are known to affect the integrity of sulfhydril groups, results in destruction of motility. Cysteine, glutathione and BAL protect the motile activity of the spermatozoa against these inhibitors and in certain circumstances, may reverse the loss of motility induced by the previous addition of the inhibitors.

The Li ion in very low concentration also has a profound effect in destroying the motility of human spermatozoa. The nature of this action is still under investigation. In all cases of loss of motility (Cu, arsenicals, or Li), a marked depression of glycolysis is also apparent.

14. The relation of separately innervated areas of muscle to amount of shortening and strength of contraction. J. E. MARKEE, W. W. THOMPSON, and S. O. THORNE, Jr. Department of Anatomy, Duke University School of Medicine.

We have previously reported that areas of a muscle innervated by separate branches of a nerve contract separately on electrical stimulation of individual branches, and have called attention to the pattern of branching of the nerves supplying long limb muscles of humans.

The present report deals with the long muscles of the thigh acting over both the knee and hip joints. In the human we have investigated the relation of muscle fascicles to each other and to the fibrous skeleton of the muscle. In the dog we recorded the contraction of separately innervated parts of these individual muscles.

It appears that these long thigh muscles are divided into 2 fundamental groups: those in which fibers are arranged in parallel e.g. rectus femoris, semimembranosus, in which stimulation of additional separate branches increases primarily the strength of contraction without affecting materially the amount of shortening; and those in which the fascicles are arranged essentially in series, and in which stimulation of additional separately innervated parts increases primarily the amount of shortening rather than the strength of contraction. The semitendinosus is a muscle of this type resembling structurally and functionally a two-bellied muscle, the two superficial bellies attaching to an intervening tendinosus inception. Functionally the sartorius behaves similarly; separately innervated areas produce additional shortening by contracting in series. However, in the sartorius there are no macroscopic septa dividing these areas.

106. On the embryology of levator ani muscle. Cecil P. MARTIN and Richard M. H. POWER, Department of Anatomy, McGill University.

A study of comparative anatomy, human embryology and some human malformations suggests that the pubo-rectalis should be interpreted as the caudal continuation of the ventral longitudinal muscle band which forms the depressor muscles of the hyoid bone, the sternalis and the rectus abdominis muscles. The ano-vaginal cleft is the lower end of the Linea Alba separated from the abdominal part by the development of the pubis. This interpretation contributes to an understanding of some anomalies.

102. Basic types of the arterial blood-supply of the supramesocolonic organs.[1] Nicholas A. MICHELS, Daniel Baugh Institute of Anatomy, Jefferson Medical College.

Complexity of vascularization of liver and gallbladder is due primarily to site of origin, secondarily to mode of terminal (extra-hepatic) branching of the 3 major arteries to the liver, viz. right, left and middle hepatic (latter for quadrate lobe). Patterns varied to such an extent that in the 200 bodies investigated no 2 are the same. Analysis shows that any sample no matter how complex falls into one of the following 10 basic types.

The celiacal hepatic supplies: (1) Right, left and middle (textbook type). (2) Right and middle hepatic — left replaced from left gastric. (3) Left and middle hepatic — right replaced from superior mesenteric. (4) Middle hepatic — right replaced from superior mesenteric, left from left gastric. (5) Right, left, middle — left hepatic small, hence an accessory left hepatic from left gastric. (6) Right, left, middle — right hepatic small, hence an accessory right hepatic from superior mesenteric. (7) Right, left, middle — all 3 being small there is an accessory right from superior mesenteric and an accessory left from left gastric. (8) Combination pattern of an accessory right hepatic with a replaced left hepatic or an accessory left hepatic with a replaced right hepatic. (9) Celiacal hepatic absent — entire hepatic trunk from superior mesenteric. (10) Celiacal hepatic absent — entire hepatic trunk from left gastric.

Basic arterial patterns will be presented of the supraduodenal, retroduodenal (posterior superior pancreaticoduodenal), dorsal pancreatic (superior pancreatic) and transverse pancreatic (inferior pancreatic).

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[1] Aided by a grant from the American Philosophical Society.

18. Histochemistry of the sebaceous glands of the rat. William MONTAGNA, Department of Anatomy, Long Island College of Medicine, Brooklyn, New York. (Introduced by James B. Hamilton.)

Rectal and skin sebaceous glands were used in this investigation. The rectal glands are larger, more complex, and more active than those of the skin, but histochemically they are essentially alike. The studies discussed here apply to both types of glands.

Alkaline phosphatase is abundant in sebaceous glands, especially along their peripheries. Ribonucleic acid is present especially in the immature sebaceous cells.

Lipids are visualized by the application of several techniques. (a) Phospholipins along the periphery of the glands are shown by the Smith-Dietrich method and by the use of Sudan black (Baker, '44). (b) Nile blue sulphate reveals large amounts of unsaturated triglycerides in the larger acinar cells and sebum. (c) Birefringent spherocrystals are found in the sebum and cells undergoing sebaceous degeneration. These crystals are removed by solvents and do not form digitonides. Since the anisotropic material is weakly Schultz positive, perhaps it represents cholesteryl esters. (d) In addition to these lipids the glands contain a substance which is stained with Sudan black-B after prolonged immersion in solvents at 60°C. This substance may represent the saturated, monobasic, high molecular alcohol (eicosyl), suggested by Ameseder ('07) and Melczer and Deme ('42).

24. *The distribution of alkaline phosphatase in the developing teeth of the 4-day rat.* Anna MORSE and Roy O. GREEP, Harvard University School of Dental Medicine.

We find here favorable material for studying the precise localization and distribution, and the relative concentration of this enzyme in tissues specialized for the purpose of mineral deposition. The substrate used was sodium glycero-phosphate plus 0.0015 M magnesium. The pH was varied by half units from 7 to 11. Incubation time varied from $7\frac{1}{2}$ minutes to 24 hours.

Alkaline phosphatase was demonstrated in all the dental tissues except the basal formative end of the enamel-organ and the cytoplasm of the pre-ameloblasts. However, its apparent distribution and concentration was greatly influenced by the pH of the substrate, period of incubation, and stage of development and differentiation of the tissue. Optimal circumstances for different tissues were not uniform. For instance, the stratum intermedium after an incubation period of only 1 hour was positive to a variable degree over a fairly wide range of pH and strongly positive over a much narrower range. The stellate reticulum near the outer enamel epithelium remained negative after much longer periods of incubation becoming positive only after 4 or more hours and within a narrow pH range. The cytoplasm of the secreting ameloblasts was never more than weakly positive under any circumstances.

From serial sections through longitudinal and cross sectional planes, the localization and concentration of the enzyme in areas of comparable differentiation was found uniform irrespective of mesial, distal, buccal, or lingual position.

25. *Tissue cultures of adult human sympathetic ganglion cells.* Margaret R. MURRAY and Arthur P. STOUT, Department of Surgical Pathology, Columbia University. (Introduced by Raymond C. Truex.)

Portions of 22 sympathetic ganglia obtained surgically from 19 adult human individuals were cultivated in lying-drop cultures for a period of several months. Though many cells remained anchored in the explant, a considerable number migrated away into the culture medium for a distance of a few millimeters. These migratory cells resembled fetal and regenerating ganglion-cells in many respects. They maintained a conspicuous polarity in the absence of normal functioning. Their content of Nissl substance, nuclear chromatin, pigment, neurofibrils and Golgi bodies was variable and often atypical. The emigrated ganglion-cells did not organize themselves into a tissue or form synaptic relationships, although they might establish chance associations with one another or with satellite cells or capillaries.

Ganglion cells remaining in the explant habitually produced branching nerve fibers which extended for an average of 2 mm into the medium, and survived for months. They were often closely invested by Schwann cells which migrated and divided in a manner similar to that which they adopt in the developing animal. When the newly grown nerve fibers were severed in vitro, many of them regenerated. The latent period before the beginning of neurite growth and ganglion cell migration varied significantly among ganglia from different individuals.

58. *Histopathological effects of plutonium, fission products, fast and slow neutrons, and x-rays on the thymus.*[1] **Raymond G. MURRAY, Department of Anatomy, Tufts College Medical School. (Introduced by William Bloom.)**

The qualitative effects of x-rays on the thymus were roughly comparable in several hundred mice, rats, rabbits, guinea pigs, and chickens, agreeing essentially with the reports in the literature. The extreme sensitivity of thymocytes to doses near the LD50/30 days paralleled the sensitivity of lymphocytes in the spleen; epithelium, prominently revealed in both cortex and medulla, was insensitive. Regeneration, nearly complete a month after LD50/30 days, was primarily from medullary reticular cells and surviving lymphocytes. Evidence that epithelium participated in lymphocyte formation was suggestive but inconclusive. Changes resembling Hassall's body formation occurred in cortical epithelial cells which possibly had been phagocytes. Mice had small but otherwise normal thymuses after 20 daily 80 r treatments (single LD50/30 days about 500 r). The effects of fast and slow neutrons resembled those of x-rays.

Beta emanation from strontium^{89} and yttrium^{91} (2–5 $\mu c/g$), concentrated in the sternum after parenteral administration, depleted adjacent thymus lobules. Inhaled cerium-praseodymium (300-day) and yttrium^{91} (200 μc/rat) caused more damage in adjacent thymus than in the heavily contaminated lung. Following zirconium^{95}–columbium^{95} and barium^{140}–lanthanum^{140} (3–5 $\mu c/g$), the rat thymus was completely depleted of lymphocytes.

Plutonium was deposited irregularly in the thymus. Areas containing plutonium were depleted and adjacent uncontaminated areas were relatively normal.

[1] Work done under contract between University of Chicago and Manhattan District, Corps of Engineers, War Department.

32. *Structural aspects of inhibitory and facilitatory reticulospinal connections.* W. T. NIEMER and H. W. MAGOUN, Department of Anatomy, Northwestern University Medical School.

Information concerning reticulo-spinal connections has been obtained by exciting brain stem components recently found to exert inhibitory or facilitatory influences upon spinal motor activity.

Inhibition of cortically or reflexly induced movement, elicited from the ventromedial bulbar reticular formation, results from activating intrinsically bulbar elements for it is obtained unimpaired after chronic hemisection of the midbrain or pons. Reticulo-spinal connections mediating inhibition descend through the ventrolateral portion of the cord, being most concentrated in the ventral third of the lateral funiculus. Though their influence is predominantly ipsilateral, some spinal crossing occurs.

Facilitation of movement is obtained by exciting the basal diencephalon, midbrain and pontile tegmentum and bulbar reticular formation, outside the inhibitory field. Facilitatory effects elicited from the tegmentum below chronic hemisections of the midbrain or pons indicate the existence of intrinsic facilitatory components at successive levels down the brain stem. Bilateral facilitation of movement is evoked from one side of the brain stem, the crossed effect involving both brain stem and spinal decussations. In the cord, facilitatory reticulo-spinal connections overlap those mediating inhibition but in general are more dorsally situated, being most concentrated in the middle third of the lateral funiculus.

In spite of their overlap in the cord, facilitatory and inhibitory paths possess sufficiently distinct concentrations to permit the selective elimination of either facilitation or inhibition by appropriately placed spinal lesions.

3. Histochemical comparison of the ''resting'' and ''exhausted'' cells of the pancreas and the salivary glands of the mouse and rat. Charles R. NOBACK, Department of Anatomy, Long Island College of Medicine.

The ''resting'' pancreatic acinar cell (apical zone filled with zymogen granules) shows strong basophilia in the basal zone, strong lipase activity in the apical zone and slight alkaline and acid phosphatase activities throughout the cytoplasm. Lipase activity in the basal zone was noted but may be due to the limitation of the technique.

The ''exhausted'' acinar cell (induced by pilocarpine) shows more intense basophilia in the basal zone than in the ''resting'' cell. Traces of basophilia were identified apically. All basophilia, identified as ribose nucleic acid, was abolished in all acinar cells by ribonuclease. Either slight or no lipase activity was observed in the cells. This loss of lipase activity was corroborated by chemical determinations. No changes were noted in the activity of phosphatases.

Basophilia in ''resting'' and exhausted submaxillary and parotid gland will be discussed.

Phosphatases are abundant in the submaxillary gland, less abundant in the parotid gland, present in slight amounts in the pancreas and practically absent in the sublingual glands. Alkaline phosphatase activity is most intense in sections incubated at pH 8 in the submaxillary gland and at pH 8.5 in the parotid gland. In the submaxillary glands ''serozymogenic'' cells contain more of the phosphatases than ''special serous'' cells.

125. The effects of exposure to beta rays on the cerebral cortex of the cat. Rosalind NOVICK, Department of Anatomy, University of Minnesota Medical School. (Introduced by Fred Kolouch.)

Changes induced in the cerebral cortex by exposure to beta rays have been investigated in a series of cats. In these animals capillary tubes of radium emanation were applied directly to the surface of the cranial dura over the cerebral cortex, to give dosages of 60 to 2200 millicurie minutes. Sharply circumscribed lesions, consisting of concentric zones varying in the severity of the alterations, are produced. From the center outwards are found zones of necrotic tissue, then shrunken and hyperchromic neurons, followed by a thin layer of acutely swollen nerve cells, and, finally, the unaltered brain substance. The most severely injured neurons show an early disintegration of cytoplasm, accompanied by an aggregation of nuclear chromatin into large round clumps, and, ultimately, the disintegration of the nucleus. The astrocytes in the zone of the lesion become swollen and fragmented. The oligodendroglia undergo changes which result in the appearance of cytoplasm, the rounding-up of the cell, and the progression towards a darker, more closely packed nucleus. Microglia also show similar tendencies, and the 2 cell types become indistinguishable.

27. *The morphology and behavior of mast cells obtained from mastocytomas and cultivated in vitro.* George H. PAFF, Frank BLOOM, and Christopher REILLY, Departments of Anatomy and Pathology, Long Island College of Medicine.

Mast cells from 2 mast cell tumors of dogs have been cultivated in vitro for several months. Observations were made on living unstained cells kept in roller tubes and in hanging drops; on surviving toluidine blue stained cells; on fixed cells stained with iron hematoxylin. Direct observations were supplemented by time-lapse cinematographic studies.

In some of the roller tube cultures mast cells grew as epithelioid sheets. In hanging drops cells grew as irregular spindle and multi-processed cells. All cells which grew from the original fragment contained cytoplasmic granules which stained metrachromatically with toluidin blue despite the fact that the original tumor contained other cell types. It is suggested that mast cells may have a growth inhibitory function on other cells possibly by elaboration of heparin.

Although many cells were mature as judged by the presence of many large cytoplasmic granules, other cells had granules varying from a dust-like to a coarse size. In many cells there was a perinuclear area devoid of granules.

Cinematographic studies for periods of 24 hours revealed that division occurred in some cells by amitosis, although fixed, stained cultures showed mitotic figures. Ameboid movement occurred in some cells while others in the vicinity failed to retract a process or change shape for as long as 6 hours.

The relationship of these mast cells to heparin formation will be briefly discussed.

110. *An x-ray motion picture of the thorax.* Paul R. PATEK and Irving REHMAN, Department of Anatomy, The University of Southern California.

The x-ray motion picture illustrates the radiographic anatomy of the living human thorax. These x-ray motion pictures demonstrate the movements of the heart during the resting state and immediately following mild and violent exercise. The excursions of the diaphragm are visualized during shallow, normal, and deep breathing as well as during thoracic and abdominal respiration. The respiratory cycle of the lungs during normal and forced and also during abdominal and thoracic types of respiration are shown.

67. *Local overgrowth as a possible factor in embryological malformations of the brain.* Bradley M. PATTEN, Department of Anatomy, University of Michigan Medical School.

Evidence suggesting the possibility that local overgrowth of the neural plate in early stages of its development might be a factor in the genesis of myeloschisis and spina bifida was presented at the 1946 Anatomists' meetings. During the past year extension of this study to cover the other parts of the central nervous system has brought to light several instances of disturbances in brain development in which local overgrowth appears to be an underlying factor. Photomicrographs are presented showing examples of such areas of abnormal growth from specimens in the Michigan collection. In these embryos, ranging from 7.5 to 30 mm, local portions of the walls of the neural tube are richly plicated as if they had grown too large for the space in the head which they were destined to occupy. In some cases the redundant brain walls have caused the head to become grotesquely high-crowned; in others the overgrown brain protrudes as an encephalocoele. A preliminary survey of the Carnegie collection shows the availability there of similar cases with which the series can be extended. These findings suggest that a causative mechanism entirely different from an ''arrest of development'' may lie behind certain types of embryological disturbances. If this concept is borne out it would indicate the desirability of giving increased attention to growth accelerating factors as possible causative agents in abnormal growth.

36. *Crossed fibers in the facial nerve in human embryos and fetuses.* Anthony A. PEARSON, Department of Anatomy, University of Oregon Medical School, Portland, Oregon.

At the pontine flexure of the brain stem there is a layer of fibers just beneath the floor of the fourth ventricle. These fibers pass toward the midline and many of them cross to the opposite side. The rostral part of this layer includes fibers of the trigeminal nerve. Among the more caudal fibers of this layer are fibers of the facial nerve.

Only part of the motor fibers of the facial nerve follow the course of its internal genu. There are fibers from the rostral end of the facial nucleus which pass dorsally to enter the root of VII without taking the course around the abducens nucleus. Some of these fibers continue toward the midline and are thought to join the facial nerve on the opposite side. This is more clearly shown in the brain stem of an anencephalic baby. In this specimen the facial nucleus is divided into 2 main groups. A caudal part sends its axons through the path of the internal genu. Fibers from the rostral part of the nucleus enter the facial root more directly. Many of them pass toward the midline and are thought to cross the raphe.

Some of the fibers of nervus intermedius enter the fasciculus solitarius, and others continue toward the midline and join the layer of fibers beneath the floor of the fourth ventricle.

73. *A study of the factors controlling the differentiation of Mauthner's cell in Amblystoma.* Jean PIATT, Department of Anatomy, University of Pennsylvania.

Mauthner's cell in Amblystoma occurs at the level of the VIIIth nerve roots and is in close anatomical and functional relationship with these roots, also the VIIth and Xth lateral-line roots. The consistent association between this giant neuron and the vestibular and lateral-line roots suggests the possibility that this relationship may be one of cause and effect, i.e., Mauthner's cell may differentiate under the influence of VIIIth or lateral-line nerve roots. A greatly varied series of experiments was carried out on Amblystoma punctatum embryos to test this thesis.

Mauthner's cell failed to differentiate in about one-third of those cases in which the VIIIth roots were prevented from developing. A supernumerary Mauthner's cell developed in a few cases in which experimentally induced ectopic VIIth, Xth lateral-line and VIIIth roots entered the brain at heterotopic levels, more frequently in association with the VIIIth. Neither consistent nor positive results accrued in many cases but the data, on the whole, indicate that the differentiation of Mauthner's cell is influenced to some extent by the presence of VIIIth, chiefly, and perhaps also VIIth and Xth lateral-line nerve roots. A total of 212 experimental animals were individually studied and form the basis for these tentative conclusions.

58. Myelinated fibers in grey rami communicantes. Joseph PICK, Department of Anatomy, New York University College of Medicine.

The source and direction of fine myelinated fibers in grey rami communicantes to the fifth, sixth, seventh lumbar and first sacral spinal nerves were studied by secondary degeneration in 17 cats on osmic acid preparations. The lumbo-sacral sympathetic of 12 sides in normal cats were used as material for comparison. Four series of experiments were performed. Series I included simple division of the sympathetic trunk or isolation of the sixth lumbar sympathetic ganglion by sectioning its internodal connections. Series II consisted in removing the spinal roots and dorsal rootganglia of the nerves to the lumbo-sacral plexus. Series III was a combined operation as performed in series I and II. In series IV the trunk was cut at the second lumbar segment followed by the removal of the spinal roots and dorsal rootganglia of the first lumbar to the second sacral nerves.

In none of these experiments was there apparent degeneration of the main bulk of myelinated fibers (measuring 2–6 μ) within the grey rami. The majority of these myelinated elements were interpreted as postganglionic to the lumbo-sacral plexus. The few large myelinated fibers (measuring 10–12 μ) occasionally seen in the lumbo-sacral sympathetic were apparently somatic to the psoas muscle and to the ligaments of the spine.

59. Effects of experimental cryptorchidism upon the thymus and mediastinal lymph nodes in the rat. James C. PLAGGE, Department of Anatomy, University of Illinois, College of Medicine.

Abdominal confinement of the testes of the rat from the fiftieth to the eightieth day of life did not increase significantly the weights of the thymus or mediastinal lymph nodes. Observations on the seminal vesicles, coagulating glands and ventral prostates of these animals showed that only the latter were significantly lower in weight.

A marked increase in the weights of the thymus and the lymph nodes was noted in male rats after castration for a corresponding 30-day period.

Body weight was definitely decreased by castration but not by cryptorchidism.

41. The vascular pattern of the endometrium of the pregnant rhesus monkey. Elizabeth M. RAMSEY, Carnegie Embryological Laboratory. (Introduced by Robert K. Burns.)

A series of injected maternal placentas shows that connection between the intervillous space (IVS) and the spiral arteries is established about the eighteenth day. Continued arterial growth and progressive narrowing of the endometrium by pressure of the conceptus produce increased coiling and back and forth looping of the arteries. The number of arteries communicating with the IVS is increased by development of small branches into independent stems. Subsequently the ends of adjacent vessels coalesce to enter by a common mouthpiece and finally the number of communications is again reduced by obliteration of many vessels through occlusive intimal proliferation. In the final trimester, as the uterine wall stretches, the coils of the arteries are ''paid out'' and their course becomes circumferential and gently undulating. Endothelial proliferation early appears in the terminal portion of the artery, increasing in amount and proximal extent until at about day 50 it reaches nearly to the muscularis; thereafter diminishing. The terminal mouthpiece lacks a continuous media and adventitia; is lax and distended. The distal portion maintains its tone.

The veins draining the IVS decrease in number and become greatly distended. Exit from the IVS occurs in later stages only at the margins of cotyledons and at the placental margin.

A separate, relatively inconspicuous vascular system which retains most of its preimplantation characteristics supplies the endometrial wall per se.

34. The course of the fibers from nucleus cuneatus and nucleus gracilis through the human brain stem. A. T. RASMUSSEN, Department of Anatomy and W. T. PEYTON, Department of Surgery, University of Minnesota.

Swank-Davenport preparations of the entire brain stem and thalamus of a 41-year-old man that died 9 days after surgical tractotomy show the following features that are at variance with the generally accepted notion concerning this tract: looping of internal arcuate fibers (in narrow sense) around the solitary fasciculus; invasion of the region of the tectospinal fasciculus, extending the dorsal limits of the medial lemniscus; absence of degeneration in the ventral external arcuate fibers; invasion of the ventral two-thirds of the lateral lemniscus throughout the midbrain. Termination in a restricted region of the posterolateral ventral nucleus of the thalamus conforms to experimental work on higher animals.

42. Influence of shape and size of a conceptus on maternal blood-flow through the uterus of the rabbit: vascular readjustments during uterine accommodation. S. R. M. REYNOLDS, Department of Embryology, Carnegie Institution of Washington.

The shape of the rabbit conceptus is spheroidal after implantation. The spheroid increases in size until the twenty-second day, whereupon it undergoes rapid conversion to a cylindrical form. Measurements of the rate of maternal blood-flow about the spheroidal conceptus show that there is progressively impaired flow of maternal blood through the lateral collecting vein in the uterine wall. Upon attainment of maximum spheroidal size there is sudden and profound ischemia. After conversion to the cylindrical shape, there is a quick restoration of the local blood-flow about the conceptus. This is maintained throughout most of the period of rapid fetal growth.

Injection of a diffusible dye through the vascular tree under 110 mm of mercury pressure reveals an inability of the dye to reach the uterine wall in appreciable concentration about the conceptus at the time of local ischemia. Injection of the venous system of the uterus with vinylite and digesting away the tissues shows that during the phase of spheroid shape of the conceptus the collecting veins about the conceptus are placed under increasing tension and are widely separated. This condition is most pronounced at the time of uterine ischemia. After conversion, the collecting veins return toward each other, as some of the veins which were crowded into interconceptus sites take a new position over the growing fetus.

47. *The effect of sulfaguanidine on reproduction in the rat.* Philip V. ROGERS, Department of Biology, Hamilton College.

A group of young female rats and the males known to have sired their previous litters received a diet containing 2.5% sulfaguanidine. About 30 days later 2 of the animals were discarded because of abnormal estrous cycles. Lack of sperm in the vaginal smears and the continuance of normal cycles indicated that 3 of the remaining 23 rats failed to copulate when placed in separate cages with the males for several estrous periods. An additional 3 copulated but failed to conceive. The others became pregnant and had an average of 3.2 offspring compared with 9.9 in the litters born before they had received the sulfa drug. Six of these produced no visible offspring even though post midterm bleeding was observed in each case.

A second group of animals was given sulfaguanidine and mated in the same manner. A laparotomy was performed on each female 8 days after pregnancy started. One animal showed no implantation sites. With minor exceptions the number of offspring born to the remaining 7 corresponded exactly with the number of embryos observed during the exploratory operation. The average litter of the group was 3.2. Virtually the same results were obtained on twenty second-generation sulfa rats. Laparotomy on the 10 females of this group revealed an average of 3.3 embryos. With 1 exception all were born.

9. *Changes in the human skull associated with muscle atrophy resulting from anterior poliomyelitis and other causes.* William M. ROGERS, Department of Anatomy, College of Physicians and Surgeons, Columbia University.

Alterations in the form of certain bones, subsequent to atrophy of muscles innervated by the fifth cranial nerve, have been observed in 6 patients and 1 cadaver.

These studies included 3 cases of anterior poliomyelitis, 1 of atrophy of masseter, internal pterygoid and temporalis, 1 of atrophy of masseter associated with scleroderma, 1 of atrophy of masseter and of both pterygoids with hypertrophy of the temporalis, and 1 adult cadaver.

In each case the atrophy was unilateral with marked asymmetry of the face.

Conventional roentgen films and laminagrams showed appreciable reduction in size of those parts of bones to which atrophic muscles attached.

In 7 cases of atrophy of masseter and internal pterygoid, reduction of bone at the angle of the mandible and a decrease in the vertical dimension of the zygomatic arch occurred. In 4 cases atrophy of the temporalis presented a smaller coronoid process and indistinct temporal lines varying with degree of muscle atrophy.

In 5 cases regressive changes in the condyle were associated with decreased function of the external pterygoid.

On the opposite side of the head, there was a compensatory hypertrophy of muscles and of those parts of bones to which these enlarged muscles were attached. The increased function of the hypertrophic muscles also caused an enlargement of the condyle.

121. *Retardation of development in the kidneys of mentally deficient patients.* Edward C. ROOSEN-RUNGE, Department of Anatomy, University of Louisville.

The kidneys of patients with mental deficiency (99 cases of prenatal or natal origin) were found to be underweight in about 70% of the cases; below 12 years of age 85% were underweight. The kidneys were small even in relation to the relatively low body length of the patients and the smallness could not be explained by shrinkage following pathological lesions.

Histological examinations revealed a retardation in development on the basis of the following criteria: (1) The histological and cellular differentiation of the glomeruli was often delayed. (2) The average diameter of the glomeruli increased at a retarded rate and rarely reached normal values in the adult. (3) The uriniferous tubules remained below normal in diameter and did not expand in length at the normal rate as indicated by the relatively poor development of the cortex and particularly of the "cortex corticis" (Hyrtl).

It was concluded that damage to the central nervous system which occurs early in life, before or around the time of birth, almost invariably causes a retardation of differentiation and growth of the kidneys. In later life, the underdeveloped kidneys become very susceptible to disease which most often takes the form of nephrosclerosis.

127. *Changes in islets of Langerhans in living mice after alloxan administration.* Chanai RUANGSIRI, Department of Anatomy, Washington University. (Introduced by E. V. Cowdry.)

Approximately 100 young Swiss mice were studied at intervals after intraperitoneal injection with alloxan. The pancreas was exposed by Covell's method ('28) and then stained with neutral red and janus green in normal saline solution.

Less than 30 minutes after alloxan injection, B cells formed clefts of fluid in cytoplasm, which gradually enlarged and fused with fluids in neighboring cells. Later, the fluid escaped from the cells into the intercellular spaces and accumulated at pericapillary spaces. Fluid gradually drained into capillaries.

The B granules disappeared where fluid was accumulating. Around the fluid clefts in the cytoplasm, the granules formed clumps. Elsewhere they appeared normal.

The nuclei decreased in size in about an hour, later became distorted. At 8–9 hours they appeared to crack and fragment. After 10 hours, only fragments of nuclei were seen.

Contrary to general belief, A cells also showed signs of degeneration by degranulation, vacuolation and disintegration of cells.

In alloxan-treated animals, endothelial cells of the capillaries were swollen. Leukocytes stuck to capillary walls and capillaries were blocked off.

One hour after alloxan injection, increase in wandering cells was generalized. Later, islets were infiltrated with wandering cells and macrophages. Mitoses were commonly seen both in macrophages and mononuclear wandering cells.

119. The vascular channels of human bone. Elbert B. RUTH, Department of Anatomy, University of Cincinnati.

The vascular channels of human bone are described, from early fetal to late adult life, as seen by the Colquhoun mercuric sulfide and the Ruth thick section techniques. On the basis of this study it is suggested that the vascular channels be designated as pericortical, with reference to the channels of vessels grooving the external surface of the cortex and giving rise to the vessels that enter the bone through the external canals of Volkmann; and endocortical, with reference to all of the canals and canaliculi found in the cortex of compact bone, and including the following: nutrient (medullary) canal, external canals of Volkmann, internal canals of Volkmann, Haversian canals, inter-Haversian canals, translamellar canals, longitudinal interlamellar canals, circumferential interlamellar canals, Haversian canaliculi, and lacunar canaliculi. These channels, with their relationships to each other, and to the bone as a whole, are described at different ages from the 3 months fetus to the age of 93 years — with special reference to the bone picture from birth to maturity. It is suggested that these vascular channels of bone form an hydrokinetic system which offers an explanation of various phenomena of bone development and physiology.

72. Cholinergic drugs and the hypophysis in salamander larvae. Charles H. SAWYER, Department of Anatomy, Duke University School of Medicine.

Larvae of Amblystoma punctatum reared in solutions of various cholinergic drugs develop certain characteristics in common with Blount's ('32) triple-pituitary animals. For instance, in physostigmine the larvae become intensely dark and their gills and dorsal fins are much reduced. These observations led the author to investigate the cholinergic-hypophyseal relationships, of which only the pigmentation effects are considered in the present report. In the responsiveness of pigment cells to stimulation, 2 developmental stages may be distinguished. In the first, or *pre-pituitary* stage (before the hypophysis has become functional), dermal melanophores attain maximal expansion directly in response to cholinergic drugs, and atropine induces pigment cell contraction. In the second, or *pituitary* stage, the melanophores give very little or no response to cholinergic drugs directly (in hypophysectomized animals) but expand maximally in intact larvae cultured in acetylcholine, mecholyl, physostigmine or prostigmine. The latter effects have been blocked by neither atropine nor nicotine. The results are interpreted as indicating that secretion of the melanophore-expanding hormone of the hypophysis is stimulated by cholinergic drugs.

15. *Combined injection and dissection studies to demonstrate variations of the bronchopulmonary segments of the left upper lobe.* J. Gordon SCANNELL, Department of Anatomy, University of Minnesota. (Introduced by E. A. Boyden.)

A series of 13 left upper lobes were studied by a combined injection and dissection technic in order to correlate variations of the surface projections of bronchopulmonary segments with variations of bronchial pattern. Unembalmed, "normal" lungs were used. After preliminary dissection, the various segmental and subsegmental bronchi were differentially injected with colored gelatin in the manner of Brock. After formalin fixation detailed dissection of the bronchi was carried out and camera lucida records made.

Variants of hilar bronchial anatomy were interpreted in the light of recent studies by Boyden and Hartmann. Typical and atypical specimens were selected for demonstration of the following major variants: (1) the development of an accessory anterior bronchus, $BX^{2}b$, with downward displacement of the anterior segmental bronchus; (2) such variations in segmental representation along the interlobar fissure as absence of the posterior ramus of the anterior segmental bronchus, $B^{2}a$, with the appearance of an accessory bronchus or the compensatory expansion of superior lingular and posterior segments; (3) variations of bronchial supply to the apex.

Attention is directed to 3 characteristic ridges on the medial aspect of the left upper lobe, herein named the subclavian, lingular, and posterior crests.

93. *Effects of pamaquine on the central nervous system.* Ida G. SCHMIDT, Department of Anatomy, University of Cincinnati.

As reported previously, plasmocid, an 8-aminoquinoline with antimalarial properties, produced a complex neurological syndrome when administered to the rhesus monkey. This syndrome was associated with extreme degenerative lesions in the nuclei of nerves III, IV, VI and VIII, cerebellar nuclei and in most nuclear groups associated with their pathways, and with less severe and extensive lesions in other areas.

Pamaquine, although closely related to plasmocid in chemical structure, affected the heart, circulation, and formed elements of the peripheral blood and bone marrow and gave no outward evidence of inducing central nervous system injury. However, it did produce microscopic lesions in the brain stem. Small doses given over a 21-day period gave rise to mild swelling and chromatolysis and/or moderate pyknosis of limited numbers of cells in the nuclei of nerves III, IV and VI.

Scattered cells in some or all of the following nuclei were similarly affected: supraspinal, ambiguus, lateral reticular, lateral cuneate, dorsal motor, vestibular, cerebellar and large-celled nucleus ruber.

Larger doses given repeatedly produced marked swelling and chromatolysis in the nuclei of nerves III, IV and VI, sometimes in the interstitial nucleus of Cajal, and to a lesser and more variabe extent in the other nuclei mentioned above. These changes were in all respects much less than those induced by plasmocid.

132. Histological studies concerning the endocrine kidney. Hans SELYE and Helen STONE, Institut de Médecine et de Chirurgie expérimentales, Université de Montréal, Montréal, Canada.

The excretory function of 1 kidney has been selectively abolished in the rat, using a method described in a previous communication [Journal of Urology *56,* 399 (1946)], by decreasing the blood pressure to a level at which filtration becomes impossible, but the nutrition of the renal tissue is not seriously affected.

Most animals so treated develop a rapidly fatal hypertension, periarteritis nodosa, cardiac infarcts and sometimes hemorrhages into the brain. However, some of them survive the acute critical phase and then develop a certain degree of resistance to the activity of the purely "endocrine kidney."

The juxta-glomerular apparatus is often excessively developed, but usually, it is not visible in "endocrine kidneys" at the height of the hypertensive reaction.

Methyl-testosterone stimulates epithelial growth, even in "purely endocrine" kidneys, a fact which supports the view that this steroid has a direct renotrophic effect independent of its action upon urine formation.

Bilateral adrenalectomy inhibits the normal, rapid, post-operative decrease in the size of the kidney, and thus interferes with the "endocrine transformation" of the organ.

(This work was performed with the aid of a grant of the Commonwealth Fund.)

84. Gigantism produced in normal female rats by chronic treatment with pure pituitary growth hormone. I. Growth and organ changes. Miriam E. SIMPSON and Choh Hao LI, Institute of Experimental Biology, University of California.

Pure growth hormone caused continuous growth in rats when injected chronically from 211 to 647 days of age. The daily dose was gradually increased from 0.4 to 2.0 mg, administered intraperitoneally. The initial growth rate, 22 gm per 10 days, had slowed to a fifth this rate by the end of the experiment. The average final weight of the 8 injected animals was 550 gm, of the controls, 300 gm. The length of the injected animals exceeded that of the controls by 4.5 cm. The liver, kidneys, heart, stomach and intestine increased in weight in proportion to body weight. Weights of endocrine and reproductive organs bore no relation to body weight. The average absolute weights of ovaries, uteri, preputials, adrenals and pituitaries were slightly increased above those of controls. The weight increases of the adrenals and pituitaries were statistically barely significant; that of the preputials was highly significant. Thyroid weights were slightly decreased. Lobular-alveolar development of the mammary tree occurred and was in contrast with the simpler tree of the controls.

5. *Hyaluronidase and the growth of malignant epithelial tumors.* W. L. SIMPSON and A. R. Gopal AYENGAR, Barnard Free Skin and Cancer Hospital and Department of Anatomy, Washington University School of Medicine.

The association of ''spreading factors'' with the growth of malignant epithelial tumors has been recognized for some years, although the nature of this relationship has never been elucidated. Suggestions on the possible nature of this association were made by Cramer and Simpson in 1943. The present report describes preliminary experimental efforts to test these hypotheses. The effects of a ''spreading factor'' from the testis (hyaluronidase) on the growth and invasive capacities of a mouse transplantable squamous cell carcinoma are described, as well as the relationship between the injection of hyaluronidase and the development of malignant skin tumors in response to 20-methylcholanthrene.

Numerous examples of enhancement of invasive growth, with the destruction of muscle and bone, demonstrate the striking effect of the local injection of the hyaluronidase about the transplantable carcinoma. In a few cases distant metastatic lesions have developed very soon after this type of treatment. On the basis of these preliminary experiments further studies have been instituted into the mechanism of this relationship.

23. *The responses of syncytium and fibrinoid of the human placenta to acid and basic dyes under controlled conditions of staining.* Marcus SINGER and George B. WISLOCKI, Department of Anatomy, Harvard Medical School.

Employing a technique of histochemical study in which tissue sections are stained in solutions of varying pH but in which other conditions such as dye concentration, ionic strength and temperature are held constant, we have investigated the staining characteristics of fibrinoid and placental syncytium. When the light absorption of these placental structures, measured by photometric means, is plotted against the pH of staining, curves are obtained which characterize the constituent proteins. The syncytium of the early placenta possesses 3 areas distinguished by their responses to acid and basic dyes: (1) an outer brush border, having a low affinity for both of these dyes; (2) an outer zone notably acidophilic in character; and (3) an inner or nuclear zone the cytoplasm of which is relatively basophilic. Comparison of the staining indices of placental syncytium at various ages shows that there are well-defined changes in affinity for acid and basic dyes which occur especially during the first few months of development. These changes involve a rapid increase in the capacity to bind acid dyes and a gradual decrease in affinity for basic dyes.

The staining characteristics of placental fibrin and fibrinoid are similar to one another and show little variation with age. The staining indices of both resemble closely that of a sectioned plasma clot.

79. *Mitosis and regeneration in the corneal epithelium of rats as affected by hypo- and hyperthyroidism.* George K. SMELSER, Columbia University.

The rate of epithelization of standard thermal burns of the rat cornea was determined in thyroidectomized, in normal and in hyperthyroid rats. The burns of the thyroidectomized animals healed at a normal rate, but those of the hyperthyroid were slower in healing. The number of mitotic figures in the regenerating epithelium was significantly greater than that in the intact cornea of the left eye in 62.5% of normal and 83% of the thyroidectomized rats but in only 37.5% of the hyperthyroid group.

The number of mitotic figures in the intact corneal epithelium was the same in normal, hypo- and hyperthyroid rats, however, in colchicine treated rats the number found in the hyperthyroid animals was 40% less than that in control corneas. Colchicine effectively arrested mitotic division in metaphase in all rats.

The data suggest that excess of thyroid hormone reduces the speed of cell movement in regenerating corneal epithelium and lengthens the time required in a mitotic division. Ablation of the thyroid, resulting in serious impairment of body growth, did not interfere with these processes.

75. *The adrenal glands of normal and castrated mice.* Fern W. SMITH, Department of Anatomy, Yale University. (Introduced by William U. Gardner.)

The lipid content of adrenal cortical cells in mature (10–12 mo.) male and female mice of several inbred strains (A, JK, L, CHI, I and N) and hybrids (A × C_3H and NHO) were studied. Frozen and paraffin sections of formalin and Bouin's fixed tissue were examined when stained with haematoxylin and eosin, Sudan IV and Sudan Black B. Additional sections were studied following the Schultz test for cholesterol and with polarized light. Hyperplasia of the adrenal cortex occurs subsequent to castration (Woolley, '39; Smith, '45). These hyperplastic areas in the adrenal cortices of castrated mice, studied by the same methods, were compared with the normal adrenal glands. The Sudan stains indicated a concentration of lipids in the outer zona fasciculata of intact animals. The extent of involvement of the zona glomerulosa and juxta-medullary areas was variable. In the castrated animals the distribution of lipids in the fascicular layers was similar to that found in control mice of the same strain. In the hyperplastic areas of the cortex in the castrates, the enlarged, vesicular, lightly staining cells (haem. and eosin) were observed to be highly sudanophilic, and to contain birefringent globules and cholesterol. Dustlike sudanophilic particles were found in the small round subcapsular type of cell but consistent results have not been obtained with polarized light or with the Schultz test.

80. *The relation between structure and function in the cerebral cortex.*[1] Wilbur K. SMITH, Department of Anatomy, University of Rochester, School of Medicine and Dentistry.

Largely as a result of studies on the precentral cortex in primates and the corresponding cortical region in other mammals, there has arisen the general concept that a specific relation exists between cytoarchitecture and response elicited by electrical excitation. Evidence against this view is furnished by studies on the frontal eye fields, the cingular cortex and the pyriform lobe.

In the frontal cortex the electrically excitable region yielding ocular responses is not limited to a single cytoarchitectural area. Furthermore, within the eye fields responses similar in character have been obtained from cortex differing in neuronal arrangement.

The rostral cingular cortex, designated as area 24 by Brodmann from studies on Cercopithecus, is a unique region of the cortex in that it yields a number of divergent responses. If this cortical region possessed cytoarchitectural uniformity, it would offer substantial evidence against the above concept. However, the cortex included by Brodmann in area 24 varys so widely in cellular arrangements that it can hardly be considered as constituting a single area. Studies on this region have thus far failed to yield a constant relation between the structure of the responsive focus and the response elicited.

Investigations on the pyriform lobe and on other parts of the cerebral cortex yield evidence supporting the view that the responses elicited depend mostly, perhaps entirely, upon the connections of the cortical cells with lower levels rather than upon cytoarchitecture.

[1] Aided by a grant from The John and Mary R. Markle Foundation.

117. *Multisinusal versus nonsinusal spleens.* Theodore SNOOK, Department of Anatomy, Tulane University.

It was recently suggested (MacKenzie et al., '41) that the term "sinus" (venous sinus) is not an apt designation for the primary venules in the splenic pulp of mice, rats, rabbits, guinea pigs and cats. "Capillary veins" or "primordial veins" were favored as more applicable terms.

Examination of sections of spleens (human, monkey, guinea pig, dog, rabbit, rat, squirrel, skunk, cat, cow, horse, pig, mole, bat, weasel) impregnated for reticulum has shown that the suggestions given above hold for the spleens of the cat, cow, horse, pig, mole, bat and weasel. Here sinuses are lacking, and the finest venous tributaries are continuous with the pulp reticular spaces. However, in the spleens of the other animals listed (e.g., human, monkey, guinea pig, dog, rabbit, rat, squirrel, skunk) true venous sinuses (Milzsinusen) with characteristic lining cells and reticular fibrous support are present and occupy a large portion of the red pulp. The course followed by arterial blood to the sinuses appears to be distinctive for each animal studied.

It seems desirable to classify mammalian spleens in 2 groups: those whose spleens are multisinusal; and those which are nonsinusal.

89. *Reinnervation phenomena as revealed by prolonged observations of vagus nerve stumps and associated sense organs.* Carl Caskey SPEIDEL, School of Anatomy, University of Virginia.

Multiple successive operations on vagus nerves in frog tadpoles have yielded clearcut data on trophic and tropic effects. Individual nerve stumps and associated lateral-line sense organs have been kept under observation for many months.

Compensatory reinnervation progresses readily along distal stumps of freshly cut nerves but not along old distal stumps. Loop or plexus formation between vagus branches of right and left sides sometimes occurs. In such tadpoles, a whole zone of sense organs may be alternately shifted experimentally from an innervation by fibers of one side to those of the other, then back again, etc. An interesting variety of reversed (distal to proximal) innervation is also possible.

All stages in individual sense organ atrophy and degeneration, together with recovery, have been watched, and correlated with periods of denervation and re-innervation. Complete degeneration and failure of recovery have also been noted. Sense organs are not induced by nerves from indifferent epithelium. Denervated organs of a few months' standing have less capacity for regeneration and readjustment than freshly denervated or innervated organs.

Frustrated nerves without organs, as well as organs without nerves, waste away and die. Stranded sheath cells on aneural stumps atrophy and ultimately disappear. Old aneural stumps of more than 6 weeks' standing display marked loss of ability to penetrate repair zones as compared with fresh distal stumps.

4. *Histological studies on nerve elements and their endings at the epithelial cells of the gastric mucosa.* Rosette SPOERRI, Department of Biology, Graduate School of Arts and Science, New York University. (Introduced by Harry A. Charipper.)

The presence of "interstitial neurons" in the lamina propria of the gastric mucosa in guinea pig, cat and man, and their neural nature have been demonstrated using a tribasic staining method and checked by silver techniques. The most significant features in favor of the neural character of these cells are their occurrence as isolated morphological units, their impregnation by silver methods, the presence of a nucleolus in a centrally placed, clear, spherical or ovoid nucleus, of occasional neurofibrils and chromophil substance, the varicose appearance of some processes, and the manner of their termination. The protoplasmic extensions have been observed establishing connections with all types of gastric epithelial cells reaching not only the gastric glands but also the surface epithelium; their terminations could be demonstrated as bulb-like, knob-like and club-like structures or arborizations and could be traced directly to an epithelial cell, sometimes penetrating the glandular epithelium or the surface epithelium thus establishing a closest contact with the epithelial cells themselves. The existence of the interstitial neurons gives a new value to physiological experiments which demonstrate a nervous control of gastric secretion; they may represent an important constitutent and terminal link of the enteric nervous system. The gastro-intestinal innervation thereby may be effected through 3 systems: (1) through the sympathetic and parasympathetic systems; (2) through the intramural ganglion-cell apparatus; and (3) through the interstitial neurons.

132a. *The inductive effect of the intact posterior chorda-mesodermal axis on competent prospective ectoderm in Amblystoma.* Walter R. SPOFFORD, Department of Anatomy, Vanderbilt University.

While it is generally agreed that the neural plate in the salamander originates by means of an inductive event mediated by the underlying chorda-mesoderm, it is by no means apparent from inspection whether the mesoderm arising in the posterior part of the neural plate is the result of the "neural" induction, or whether, like the orthodox mesoderm with which it is continuous around the dorsal lip of the blastopore, it arises by an autogenous self-differentiative process. To determine the nature of the inductive influence of that part of the chorda-mesodermal axis underlying the posterior neural plate mesoderm, the posterior part of the neural plate was removed from 12 neurulae, and replaced by vitally stained competent prospective ectoderm from stage 11 gastrulae. After delayed closure of the posterior neural folds, the graft material was later found to be disposed as somites in the variously deformed tails of the hosts.

In a second series wherein a larger part of the neural plate was removed and replaced by competent test material, each graft formed spinal cord in its anterior part, and somites in its posterior part.

Although it is well known that there are ''regional'' differences in the ''organizer,'' this experiment demonstrates a sharp break in the inductive properties of the chorda-mesodermal axis in its posterior region, with neural-inducing capacity anteriorly, and mesodermal-inducting posteriorly.

87. Localization of motor cells in the thoracic spinal cord of the rhesus monkey. James M. SPRAGUE, Department of Anatomy, Johns Hopkins University.

It was shown by the author that the motor cells of the dorsal primary rami (axial muscles), and ventral primary rami (limb muscles) are not segregated into medial and lateral nuclei in the brachial cord of the fetal sheep (J. Comp. Neur., *85:* 127–140). The cells are indiscriminately mixed and show no indication of the localization frequently described in adult mammals. The following report concerns sub-adult rhesus monkeys, in which ventral rami were cut on one side of the animal, and dorsal rami of the same segments on the other. $T_{7,\ 8,\ 9}$ were cut in this manner in 1 animal, and $T_{10,\ 11,\ 12}$, L_1 in another. In a third both rami were cut on the same side in $T_{5,\ 6,\ 7,\ 8,\ 9}$. The chromatolytic cells were plotted by means of a calibrated ocular grid. No medial and lateral columns are revealed. Motor cells of both dorsal and ventral rami are distributed throughout the ventral horn. The cells of the dorsal rami extend at least one full segment above and below their nerves of exit. At the upper and lower ends of this distribution the cells are limited to the anterior tip of the ventral horn. In the segments of which the nerves were cut, these cells spread throughout the horn. Ventral rami cells are mostly restricted to the levels of their nerves of exit. They are found everywhere but the anterior tip of the horn.

114. Rate of renewal of the cells of the intestinal epithelium in the rat. C. E. STEVENS and C. P. LEBLOND, Department of Anatomy, McGill University.

The rate of renewal of cells may be estimated by counting the percentage of cells undergoing mitosis. Mitotic counts carried out on the ileum of the rat revealed that the half life of the cells of the epithelium is quite short, approximately 1 to 2 days. In the duodenum, the half life of the epithelial cells is somewhat longer.

The cells of the intestinal epithelium originate from mitoses taking place throughout the length of the Lieberkuhn glands. The cells are then gradually pushed up along the sides of the villi and seem to be eliminated at the tip of the villi where a characteristic structure formed by the dying cells is usually visible.

97. *Primitive bloodvessels of the spinal medulla of the rabbit injected while alive.* Leon H. STRONG, Department of Anatomy, University of Michigan.

The vascularization of the dorsal and ventral halves of the spinal cord is enough different to suggest a functional significance. The medial ramus of the intersegmental artery forks into a ventral and lateral radical for the ventral and dorsolateral surfaces. The lateral radical ascends to the upper limit of the segmental ganglion, forks caudally and rostrally establishing a longitudinal trunk by anastomosis, but continues its dorsal course. At the ganglion, it sends medially, perpendicular to the lateral surface, 2 parallel capillaries which reach the ependyma. One turns dorsally, the other ventrally, subependymally. The upper capillary branches dorsally about its midpoint. This branch reaches nearly to the surface, parallel to the subependymal capillary before it anastomoses medially with it. Between the 2 dorsal capillaries, transverse parallel capillaries develop, forming a dense subependymal plexus that fills the area between them and connects intersegmentally. The lateral half of the dorsal area of the cord is left nonvascular.

The ventral radical reaches just lateral to the midline ventrally, forks caudally and rostrally, establishing a longitudinal trunk close to its fellow but nonanastomotic with it. From this trunk a subependymal capillary arises dorsally which joins the lower of the 2 medial capillaries from the lateral longitudinal trunk. From the quadrangle of vessels thus formed, long centrally directed capillaries form a central loose plexus.

78. *The factor of nutritional interruption in the transplantation of adrenal medullae in the mouse.* C. Donnell TURNER, Department of Zoology, Northwestern University.

Unlike the cells of the adrenal cortex, those of the medulla apparently continue their secretory capacity for an indefinite length of time and there is no provision for their replacement. Adrenal medullae persist more frequently in the mouse, when whole glands are transplanted, than in other mammals studied; this is believed to be due to the fact that the medulla of the mouse touches the capsule at the hilus and hence is not completely surrounded by cortical tissue. This makes it possible for nutritional materials of the host to reach the medulla directly without first diffusing through the cortex.

When whole adrenals from young donors are inserted below the kidney capsule of appropriate hosts, the persistence of medullae is correlated with the position of the hilus with reference to the kidney capsule. If the hilus is caused to project through a pore in the kidney capsule, medullae persist in approximately 2% of the cases; medullae persist in about 80% of the grafts when the hilus is firmly pressed into the substance of the kidney cortex, or when whole glands are introduced into the anterior chamber of the eye. The incidence of medullary persistence is increased by sectioning the donor adrenals so as to expose a greater medullary surface. It is concluded that nutritional interruption is an important factor in determining the survival of medullary cells in adrenal grafts.

71. *Vascular changes in the gonopodium of poeciliid fishes during the metamorphosis of the anal fin.* C. L. TURNER, Department of Zoology, Northwestern University.

In the males of poeciliid fishes various structures, including some of the fins, become profoundly modified in the male with the onset of sexual maturity. Experiments indicate clearly that the modifications of these target organs occur only under the influence of hormone stimulation while other parts of the body are not affected. The same modifications can be induced in females by hormone treatment.

During the process of metamorphosis in the anal fin of the normal males or in females treated with androgenic hormones the vascular systems of the susceptible tissues undergo changes which resemble those of the early stages of inflammation. The changes are marked by the following stages: (1) Dilatation of capillaries and venules and changes in the permeability of the endothelium occur. Leakage of fluid from the blood produces edema in the surrounding tissues. (2) The flow of the blood becomes sluggish. The endothelium becomes sticky and traps masses of blood cells. (3) Stasis occurs and capillaries and venules become blockaded by plugs of blood cells. (4) The endothelium breaks down and the masses of blood cells are digested by fibrocytes or free macrophages. (5) Circulation is reestablished in affected tissues by the formation of new blood vessels.

Ethynyl testosterone in a concentration of 1 mg/20,000,000 cm^3 of water is sufficient to bring about vascular changes in the newly regenerated anal fin of a female.

61. *Colchicine influence during embryonic development in rats.* John H. VAN DYKE and M. G. RITCHEY, Department of Anatomy, Washington University School of Medicine.

Embryos of pregnant Wistar rats given a single intraperitoneal dose of 0.06 mg of colchicine per 100 gm of body weight on the sixth, eighth, tenth, twelfth or fourteenth day of gestation and sacrificed *3 days later* have been studied.

Embryos thus subjected on day 6 and 8 were dead and partially resorbed. Embryos of females treated on day 10–14 were living but showed general developmental retardation. Three embryos (10 day series) possessed *gross abnormalities.* Two of these specimens developed a transverse crease on the dorsal surface near the arm pads, apparently representing transient growth inhibition resulting from colchicine. Diencephalic hydrocephalus occurred also in one, while the cephalic portion of the other, although enlarged, appeared normal. Below the level of this crease the embryos were bulbous and *necrotic.* The site of maximal mitoses (using neural tube as an index), as well as of aberrant mitotic figures (typical colchicine effect), occurred at the cephalic end of these specimens. A kinky tail characterized a third embryo.

Our initial study suggests that colchicine retards development of advanced embryos but has a lethal effect on young ones. While its influence lacks uniformity, colchicine may temporarily inhibit development, or cause death of a portion differentiating subsequent to its injection (caudal part), but the precocious region (cephalic end) appears more resistant and may survive for a limited period.

10. *The relation of the temporal muscle to the form of the skull.* S. L. WASHBURN, Anatomy Department, Columbia University.

The effect of the temporal muscle on the developing skull has been debated for many years. In order to determine the relation of the muscle to the form of the skull, unilateral removal of the temporal muscle was performed on day-old rats. In 10 of 22 animals the long neck muscles of the same side were separated from the nuchal crest. The animals were sacrificed from 3 to 5 months later. On the operated side no temporal line formed, the coronoid process of the mandible failed to develop, and the nuchal crest was absent when both temporalis and the nuchal muscles had been removed. The brain case was normal in shape when temporalis alone was absent. After the double operation, a change in internal shape, as shown by endocasts, was discernible. Although the temporal muscle of the rat is large relative to the brain case, effects of its removal are limited to bony crests and processes immediately associated with its origin and insertion. Even after the additional removal of muscles from the nuchal crest, only very minor changes in the form of the endocast can be detected.

76. *Studies of regenerating adrenal cortex in the ears of rats, and the ability of such grafts to maintain life.* R. G. WILLIAMS, Department of Anatomy, University of Pennsylvania.

Group 1. 28 rats were adrenalectomized. The capsule and zona glomerulosa from 1 gland in each animal were separated as 1 piece and implanted in the left ear; the remainder of the gland was implanted in the right ear.

Group 2. 20 rats were adrenalectomized; 1 gland was split as in the previous group but the pieces were replaced in the original adrenal bed.

Group 3. 10 rats were adrenalectomized and no grafts made.

All animals in Group 3 died within 105 days. Accessory cortical tissue, if present, did not maintain rats in the absence of the chief glands. 10 animals of Group 2 and 12 of Group 1 died within 3 months with symptoms of adrenal deficiency. Grafts at the lower temperature and under the other environmental differences in the ear maintained life as effectively as those in the abdomen. In all animals the original fasciculata and reticularis cells disappeared completely and quickly. Regeneration occurred in all animals but was very limited in those that died. Regeneration was from the glomerulosa, and possibly from the capsule, by centripetal proliferation. Well-defined fasciculata and reticularis were present 20 months after operation. Holocrine secretion appeared to be the fate of some reticularis cells resulting in large spaces which became vascularized by sinusoids.

63. *Anomalies of the genito-urinary tract induced by maternal vitamin A deficiency in fetal rats.* James G. WILSON, Department of Anatomy, University of Rochester, and Josef WARKANY, Department of Pediatrics, University of Cincinnati.

Fetal rats from mothers maintained in chronic vitamin A deficiency exhibited 2 types of anomalies in the genito-urinary tract. In both sexes epithelia that composed, or were derived from, the urogenital sinus underwent keratinizing metaplasia. Abnormal keratinization first appeared in the urethral plate on the eighteenth day of gestation and by the twentieth day it was present in all epithelia of sinusal origin distal to the juncture of the genital ducts with the sinus. In older fetuses and newborns keratinization was greater in degree but still largely limited in extent to the genito-urinary tract distal to, and around, the termination of the genital ducts. The metaplastic changes first appeared, and were always more pronounced, on the dorsal wall of both the urogenital sinus and structures derived from it.

Other anomalies consisted of hypoplasias and frank malformations. Relative to development in other regions, the urogenital sinus and its associated structures were retarded. This general retardation was exemplified by such conditions as: retention of the primitive juncture of ureters with the mesonephric ducts; persistence of both sets of genital ducts, Mullerian and mesonephric, in either sex for several days beyond the time regression normally occurs in one set; and failure of the Mullerian ducts to unite distally to form a utero-vaginal canal, even in mature female fetuses. Hypospadias and horseshoe kidneys were commonly occurring malformations.

82. *Cytological studies on spontaneously occurring adenomata of the anterior hypophysis of the rat.* J. M. WOLFE and A. W. WRIGHT, Departments of Anatomy and Pathology, Albany Medical College.

Three types of spontaneous adenomata have recently been described in the anterior lobes of old female rats; namely, adenomatous nodules, hemorrhagic adenomata and chromophobe adenomata, the last 2 being larger and more discrete than the first (Wolfe, Bryan and Wright, '38). Lesions of the first 2 types often contained acidophiles as well as chromophobes, the latter only chromophobes. Forty-four additional lesions have been studied, including all of the above varieties. Adenomatous cells tended to be enlarged. Enlargement of the nuclei occurred in cells of most lesions while hypertrophy of the nucleoli was found almost constantly. The nuclei were often characterized by wrinkled membranes, lobulation and evidence of nuclear budding; the nucleoli often contained vacuoles. The degree of enlargement of cells, nuclei and nucleoli varied within wide limits and was often extreme. In many adenomatous acidophiles and chromophobes hypertrophy of the Golgi apparatus was pronounced; this was not constant. The mitochondrial picture varied; often they were more numerous than normal; just as frequently only normal numbers were present. They were occasionally quite hypertrophied. The adenomatous acidophiles were invariably packed with granules; there was no cytologic evidence indicating that they were releasing hormone. The cytoplasm of some of the chromophobes was scant and stained light blue; in other cells the cytoplasm appeared abundant, dense and stained deep blue.

105. *Changes with age in the weight of the liver in Wistar albino rats.*[1] Eleanor H. YEAKEL and Edmond J. FARRIS, The Wistar Institute of Anatomy and Biology.

The weight of the liver was determined in 164 male and female albino rats 21–1000 days of age. The average liver weight increased rapidly from weaning to

75 days, and thereafter maintained a uniform weight to 500 days (approximately 11 grams in males, and 8 grams in females).

In the old age range of 600–900 days, no change in absolute or relative weight of the liver occurred in the male rats, but many of the livers of the females were significantly heavier. At 1000 days, the weight fell in the males, paralleling a loss in body weight, to an average of 9.68 gm. The weight remained high in the females (11.46 gm).

The increase in liver weight in a number of old female rats appeared to be correlated with the presence of tumors elsewhere in the body, specifically in the pituitary, mammary glands, and uterus, rather than with body weight changes. The average weight of the liver in 14 old females with no tumors was 8.91 gm, and in 27 tumor-bearing animals was 12.00 gm. Average relative weights were also greater in the latter group. The weights of mammary gland tumors in 12 old rats compared with the corresponding liver weights suggested that spontaneous mammary growths influenced liver weight when the tumors exceeded a critical weight of approximately 30 gm.

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[1] This investigation was aided by a grant from the Samuel S. Fels Fund.

8. Anatomical factors in senile alopecia; experimental production of baldness.
M. Wharton YOUNG, Department of Anatomy, Howard University.

The constant pattern of senile alopecia suggests that there is a definite anatomical basis for this condition. The retention of the marginal zone of scalp hair (also other body hair including the beard) indicates that local factors within the scalp are more important etiologically than systemic deficiencies.

The tela subcutanea contains the hair bulbs embedded in fat, also the superficial muscles and the larger vascular networks. Injection studies reveal a richer vascular supply in the marginal (muscular) zone of the scalp than in the central (aponeurotic) zone. Likewise, senile alopecia is confined to the aponeurotic area, whereas, hair persists in the marginal area overlying the occipital and auricular muscles. Consequently, a sharp line of demarcation is established between these 2 contrasting regions. The bald area shows a loss of hair follicles and fat with increased fibrosis in the subcutaneous layer.

Beginning alopecia is associated with tension zones in the scalp resulting in diminished blood supply in these areas. Scalp tension may result from contraction of muscles attached to the galea, from continued growth of the cranium, or from external pressure. Some victims of the atomic explosion developed a "senile-type" of alopecia. Tension areas were produced experimentally in rhesus monkeys by suturing the free scalp edges after excising elliptical segments, thus reducing the circumference. These operations resulted in persistent baldness closely resembling the human types.

46. The ovarian follicular cyst and the pattern of sexual behavior in the female guinea pig.
William C. YOUNG, Wesley A. INNES, Richard C. WEBSTER and Paul G. ROOFE, Department of Anatomy, University of Kansas.

In our experience during which the ovaries of several hundred guinea pigs have been examined microscopically we have only recently encountered an animal in which a follicular cyst, to be distinguished from the rete ovarii cyst, was present.

Each ovary was largely medulla partially capped by a relatively inactive cortex. No vesicular and only 15 primary follicles, most undergoing atresia, were found. A hemorrhagic cyst containing strongly eosinophilic luteinized granulosa cells surrounding a degenerating primary oocyte and a corpus luteum-like structure were also present. An enlarged uterus seen only macroscopically was suggestive of pyometra caused by constant estrogenic stimulation; metaplastic epithelium in the rete ovarii was similarly suggestive.

Prior to ovariectomy the animal displayed male-like behavior at cyclic intervals but was not found in heat although after ovariectomy short periods of heat followed the injection of estradiol benzoate and progesterone. The history of the animal considered with other data to be reviewed suggested that the amount of estrogenic stimulation is correlated with the appearance of male-like behavior in female guinea pigs and possibly certain other mammals. In a test of this hypothesis, reduction in the quantity of injected estrogen was followed by a much greater decrease in male-like behavior than in mating responses. The implications for the problem of the physiological and somatic basis of sexual behavior patterns will be discussed.

ABSTRACTS READ BY TITLE

136. The effect of splenectomy and inanition on the bone marrow of plethodontid urodeles. William C. BARRETT, Jr., Department of Anatomy, Western Reserve University.

Splenectomy, either alone or combined with inanition, affords a means of testing the competence of the lymphogranulo-poietic bone marrow of plethodontid salamanders. Fifty-eight specimens of *Desmognathus fuscus* and *Eurycea bislineata* were splenectomized and the blood and bone marrow was studied at intervals for a period of 10 months. Removal of the spleen, the sole erythropoietic organ, did not stimulate erythropoiesis in the bone marrow or in the hepatic or mesenephric hemiopoietic loci. Splenectomy, combined with inanition for 1 to 3 months, and followed by feeding likewise failed to stimulate erythropoiesis. Following simple splenectomy erythropoiesis was confined to the peripheral blood. The number of erythroblasts and hemoblasts in the blood stream increased temporarily, but returned to normal after 4 to 6 weeks. Splenectomy followed by inanition produced a marked anemia and a high mortality rate. Subsequent feeding produced an increase in the hemoblasts and immature erythrocytes accompanied by increased mitotic activity. The bone marrow of the starved and splenectomized animals showed signs of depletion. It is concluded that the bone marrow of plethodontids, normally lymphogranulopoietic, is not a favorable site for erythropoiesis. Also, the spleen, usually considered the amphibian erythropoietic organ, plays a minor role in erythropoiesis compared with the circulating blood. The spleen as well as the lymphogranulopoietic organs contribute undifferentiated elements to the blood where they differentiate heteroplastically into erythroblasts.

137. *The mechanism of menstruation.* G. W. BARTELMEZ, Department of Anatomy, The University of Chicago.

It has been suggested (Schlegel, '45) that the superficial ischaemia and necrobiosis of menstruation are due, not to constriction of the coiled arteries (Daron, '36) but rather to the opening of arterio-venous anastomoses and the shunting of blood from the surface to such an extent that a superficial ischaemia is produced. If this were true it should be possible to inject particles like starch grains into the arteries and find them in the veins of the endo- and myo-metrium. Certain of our injected macaque uteri provide evidence on this score. They were prepared by filling the entire vascular bed after vasodilatation with carmine gelatin or India ink via the arteries and then injecting starch grains (9–14 μ), which had been colored by reduced silver or glygocen methods. The uteri were serially sectioned at 100–500 μ. One ischaemic (pre-menstrual) case and 5 menstruating uteri (days 1 to 6) were surveyed. Even arteries which contained starch grains in the superficial zone usually showed basal constrictions except in the sixth day case. No starch was found in any vein either in the endometrium or myometrium. It would seem that in the macaque uterine arterio-venous anastomoses are either not patent during menstruation or they are not abundant enough to be significant in the production of the menstrual ischaemia.

138. *Stratification of the human erythrocyte by ultracentrifuging.* H. W. BEAMS, Department of Zoology, State University of Iowa.

Ultracentrifuged human erythrocytes are stratified into 2 or more layers which may be sharply stained, after fixation in Champy's fluid, by Heidenhain's hematoxylin, or Mallory's triple stain. Such results do not support the earlier views that the interior of the erythrocyte is chiefly of a homogeneous nature. Similar observations have been reported for the erythrocytes of the rat (Beams and Hines, Anat. Rec., *90*).

Many of the leucocytes likewise show a shifting of their components following ultracentrifugation.

139. *Evidence for 2 types of thyrotropic activity in Amphibia.* Raymond F. BLOUNT and Isabel H. BLOUNT, Department of Anatomy, University of Texas.

The type of activity produced by the thyrotropic hormones of the pituitary at different ages during the initiation of the pituitary function has been studied. Rana pipiens tadpoles were stages within "paddle," premetamorphic and metamorphic groups by the system of Taylor and Kollros ('46). During short intervals within these age brackets animals were placed in thiouracil solution (0.05%) in order that the chemical removal of pituitary inhibition would result in overproduction of the thyrotropic hormone characteristic of each age. Representative test and control animals were prepared for microscopic study. The treated animals showed an increase in the storage of colloid during the premetamorphic stages. The amount of colloid was increased relative to the epithelium. The cells showed no increase in height as compared with the controls such as would be expected if a discharge hormone were present. This phase is of course followed during metamorphosis by the production of the thyrotropic hormone causing discharge of colloid. These results support the evidence from additional pituitary transplants in the salamander and from heteroplastic transplantations between a neotenous form, in which storage hormone alone exists, and a metamorphosing form in which a discharge hormone also is present.

140. Vascularization of the cornea of the rat resulting from deficiency of B vitamins. Lester L. BOWLES, Department of Microscopic Anatomy, University of Georgia School of Medicine. (Introduced by Lane H. Allen.)

In the investigation of corneal changes resulting from B vitamin deficiencies, a control diet was used consisting of: ''vitamin free'' casein 20 gm, cod liver oil 2 gm, cottonseed oil 3 gm, salt mixture 4 gm, sucrose 71 gm, choline chloride 200 mg, calcium pantothenate 2 mg, thiamine chloride 0.4 mg, pyridoxine hydrochloride 0.4 mg and riboflavin 1.6 mg. To produce deficiency of pantothenate, pyridoxine or riboflavin, diets the same as the control were used, but with the respective vitamin omitted.

Nine of 23 rats (6 litters) placed on pyridoxine-deficient diet at 48–57 days of age developed definite corneal vascularization. Six older rats showed no corneal changes, 6 younger rats died too soon for corneal vessels to appear. Thirteen of 22 rats (5 litters) placed on pantothenate-deficient diet at 41–52 days developed extensive corneal vascularization. Five of the 22 and 5 younger rats died too soon for vessels to appear. The corneal capillaries in these deficiencies were numerous and large with prominent anastomoses, as contrasted with the fewer vessels in fine networks resulting from riboflavin deficiency. Correcting the deficiency of pyridoxine, pantothenate or riboflavin in the respective diet resulted in regression of the vascularization.

Litter-mate controls, rats with symptoms of nicotinic acid deficiency, starved rats and rats on a caloric-deficient diet showed no significant corneal changes.

This study was aided by grants from the John and Mary R. Markle Foundation.

141. An analysis of variations in the bronchial pattern of the right upper lobe of 50 lungs. Edward A. BOYDEN, Department of Anatomy, University of Minnesota.

With the cooperation of Dr. J. Gordon Scannell, 50 lobes have been dissected and pictorial records made of the distribution of arteries, veins and bronchi in the 3 bronchopulmonary segments — apical (B^1), anterior (B^2) and posterior (B^3).

In contrast to the left upper lobe, in which the key to variations is the outgrowth from the apical bronchus of an accessory anterior ramus (BX^2b) and the downward displacement of B^2 to form a trifurcate pattern (26%), the striking modification on the right side is the outgrowth from the anterior bronchus of an accessory apical ramus (BX^1b) which takes over the territory usually supplied by B^1b. In half of these 28 cases, not only the whole mediastinal surface of the lobe but also part of the apex is supplied by B^2. Also the prevailing pattern is trifurcate (46%); for although 54% of specimens are bifurcate 10 different combinations of bronchi occur.

Of further interest is the tendency for the clinically important posterior bronchus that borders upon the interlobar fissure, to separate into independent halves; then its upper ramus (B^3a) arises with the apical bronchus, its lower ramus (B^3b) with the anterior bronchus.

Finally, that ramus of the anterior segment which borders on the secondary fissure (B^2a) occurs in 98% of specimens; yet on the left side the corresponding bronchus occurs in only 60% of specimens.

142. Re-growth of nerve fibers in the spinal cord. James O. BROWN, Department of Anatomy, Woman's Medical College of Pennsylvania.

Transected spinal cords in a series of cats and dogs showed evidence of regrowth of nerve fibers. By the end of 24 hours some of the cut fibers already showed signs of regeneration in the cat. These abortive attempts at regeneration appeared as fine fibers growing in various directions. Their disorientation permitted the influx of connective tissue the abundance and rapid growth of which formed a barrier that impeded the slower growing nerve fibers from bridging the gap. This is suggestive of the abortive regeneration in peripheral nerves observed by Perroncito ('07).

Strips of omentum, gall bladder, amnion and aorta were used to wrap around the region of transection in an attempt at preventing the invasion of connective tissue from the dura. None was successful. This corresponds to the already observed interference in nerve regeneration due to the ingrowth of connective tissue following a break in the wall of an arterial sleeve used for peripheral nerve splicing (Weiss, '43).

Failure of functional restoration appears to be due not so much to the complete lack of regenerative cell outgrowths as to the presence of insurmountable obstacles which prevent regenerating nerve fibers from becoming oriented and from growing forward to reach their original terminations.

143. The cerebral cortex of the 3-month infant. J. LeRoy CONEL, Department of Anatomy, Boston University School of Medicine.

The results of this investigation will be published in the third volume of a series of monographs in which the postnatal development of the cerebral cortex is described. Nine criteria are used for evaluating the progress of development. During the time between the first and third months of postnatal life development of the cortex has been fairly uniform in rate in all areas, but growth has been especially active in area FAγ (4) particularly in the region of the hand. In the parietal lobe development has advanced further in area PB than in other areas. In the occipital lobe area OC leads the other areas in development. Area TC is more advanced than other areas in the temporal lobe. Growth has not been as active in areas OC and TC as it has in areas FAγ and PB. Of the horizontal layers of cells, layers V and VI are the most advanced in development.

144. The growth of the brain in the fetal dog. Robert L. CORDER and Homer B. LATIMER. Department of Anatomy, University of Kansas.

Dog fetuses weighing from 6 to 410 gm were used. The method of removal and weighing the brain was the same as that used in similar studies from this laboratory. The brain weight increases from 0.3 to 10 gm, and the cord, from 0.03 to 0.6 gm. The brain weight plotted on body weight forms a straight line curve and the cord weight, a curve very slightly convex superiorly.

The brain was divided into 4 parts and each part was weighed separately. When plotted on body weight, the prosencephalon forms a straight line, and the cerebellum, 2 straight lines, the second rising more rapidly. The mesencephalon also increases as an arithmetical function of the body weight for the first half of the growth period and then remains at a constant level until birth. The medulla forms a curve slightly convex superiorly throughout.

The prosencephalon increases in its percentage of the total brain weight throughout the entire fetal period, but more rapidly at first. The cerebellum likewise increases in its percentage of the brain weight but its curve is concave superiorly, or its increase is more rapid toward the latter part of the fetal period. The mesencephalon and the medulla both decrease in relative weight and both curves are concave superiorly, or the relative changes occur most rapidly in the early part of the fetal period.

145. Changes induced in the anterior pituitary gland of the white rat by prolonged treatment with thiouracil. Alden B. DAWSON, The Biological Laboratories, Harvard University.

The pituitary glands used in this study were obtained through the courtesy of Mr. W. L. Money who has been studying the effects on the thyroid gland of chronic treatment with thiouracil (Money, '46). Adult male rats were maintained on 1.0% thiouracil (in drinking water) for periods up to 16 months. Pituitary glands were taken at 4- to 8-week intervals beginning with the fourth month. At this time ordinary basophiles and thyroidectomy cells were numerous and the acidophiles were but slightly reduced in number. Thereafter, there was ordinarily a rather rapid decrease in unmodified basophiles and a more gradual decrease in thyroidectomy cells. The acidophiles also slowly decreased in number. Maximum reduction of unmodified basophiles and acidophils usually occurred after 10 to 12 months. In late stages of treatment, practically all granular cells disappear and even thyroidectomy cells become rare, approaching the condition found at the end of 40 days, in the glands of rats thyroidectomized at birth (Severinghaus, '42). The deleterious effects of thiouracil are sometimes repaired by the development of a certain type of thyroid adenoma. Single or periodic injections of dibenzanthracene appeared to accelerate the rate of degranulation of the chromophiles and disappearance of thyroidectomy cells. Prolonged treatment with thiouracil may produce effects in the anterior pituitary gland as extensive as those found after total thyroidectomy.

146. *The ovary of the pika.*[1] Kenneth L. DUKE, Department of Anatomy, Duke University School of Medicine.

The Lagomorpha includes 2 families: Ochotonidae (pikas) and Leporidae (rabbits and hares). Of the genera making up this order the pika is the most primitive. Twenty-six reproductive tracts of *Ochotona princeps uinta* were obtained for study. These animals were trapped during the months of July, August, and September.

The germinal epithelium is cuboidal near the root of the ovary but may become flattened as it approaches the ventral aspect. There are few cellular proliferations from the germinal epithelium into the cortex. The cortical stroma is a compact connective tissue containing many fibroblast nuclei. Some oocytes lie immediately beneath the germinal epithelium, while others are deeper in the cortex. Corpora lutea and large vesicular follicles are present in July. Luteal remnants and numerous growing and small vesicular follicles are common in September. Atresia is widespread during the September period. Call-Exner bodies are seen in the granulosa of the follicles in immature animals. The theca interna, even of vesicular follicles, is thin and fibrous. There is but slight hypertrophy of the theca interna, even in atretic follicles. Interstitial cells are inconspicuous; however, the stroma of immature females contains large epithelial or epithelioid cells. The medullary stroma consists of fibrous connective tissue with few nuclei visible as compared with the cortical stroma.

[1] Aided by a grant from the Duke University Research Council.

147. *Some observations on growth of limbs with deficient nerve supply.* Donald DUNCAN, Department of Anatomy, University of Texas.

The sciatic nerves of 8 young rats, 10–12 days old, were severely damaged but not completely interrupted by application of tantalum clips. The animals were observed for 140–150 days before sacrifice. During this interval sensibility was present but diminished in all animals and movements of the foot and toes were very feeble or absent. Weights of the leg segments with skin removed were 41–68% less than those on the side with intact nerve supply. The tendons of the undersized muscles were of the same size as those of the opposite side and tibial lengths were not appreciably different except in 1 animal. In this one the tibia of the partially denervated leg was 2 mm (6%) shorter than its mate. The experiments indicate that severe but incomplete denervation of a limb during the post-natal growth period of the species examined affects neither the length of the bones nor the size of the tendons.

148. *An histological study of duck limb primordia transplanted to embryonic chick hosts.* Herbert L. EASTLICK and N. Karle MOTTET, Department of Zoology, State College of Washington.

Anterior and posterior limb primordia of 26–38 somite duck embryos were transplanted to the flank of 2–3 day chick embryos. This report is based on a study of serial sections of 41 non- or poorly innervated grafts, 5–21 days old.

Premuscle masses are formed in 7 day grafts and subnormal amounts of faintly striated fibrillae appear in 8 day specimens. In 11–18 day grafts the muscle fibers appear to be edematous. Macrophages are numerous between the fibers and sarcolysis and active phagocytosis is taking place. Venous congestion is moderate with some erythrocyte extravasation. Fat organs are developing in the loose connective tissue and most of the fat cells are multivacuolated. In 19 day grafts desquamation of the epidermis and dermal hemorrhage occur. The venous congestion is especially prominent in the subdermal vessels. Fat lobules and loose connective tissue are predominant and little or no muscle remains. In 21 day specimens the fat lobules are composed primarily of univaculate cells. Venous congestion is extensive, edema is prominent, and many erythrocytes are extravasated into the connective tissue between the fat lobules.

149. *Distribution of phosphatase and glycogen in the amphibian embryo.* Herbert ELFTMAN and W. M. COPENHAVER, Department of Anatomy, College of Physicians and Surgeons, Columbia University.

The localization of alkaline phosphatase, acid phosphatase and glycogen has been studied in Amblystoma punctatum embryos ranging from late blastulae to feeding stages (Harrison's stages 8 to 46).

In late blastulae, alkaline phosphatase is present in the yolk granules of both animal and vegetal poles. Acid phosphatase is much more abundant in the vegetal pole, contrasting with glycogen, which is more abundant in the animal pole.

During the development of the heart there is a marked increase in the alkaline phosphatase of the endothelial nuclei. Glycogen is present in the anlage of the heart; at later stages it is more striking in the ventricle than in the atrium. The yolk granules in the heart wall decrease in size and number in the course of development but continue to give a positive phosphatase reaction.

The cells of the retina exhibit a polarity in the distribution of alkaline phosphatase and of glycogen before stage 40. Alkaline phosphatase is associated with developing nerve fibers; in the central nervous system it is differentially distributed. Acid phosphatase is more abundant in the ventral portion of the nerve cord than in the dorsal in the later stages.

The pronephric tubules have large amounts of alkaline phosphatase in their proximal portions, the concentration decreasing distally. Alkaline phosphatase is also associated with the development of the skeletal elements, teeth and the digestive tract.

150. *Factors affecting the estimation of concentration of sperm in ram's semen by the photoelectrometric method.* L. Otis EMIK and George M. SIDWELL, Bureau of Animal Industry, Agricultural Research Administration, U. S. Department of Agriculture. (Introduced by Clair E. Terrill.)

Ejaculates from Rambouillet, Columbia and Targhee rams were used to measure the effects of various factors on estimation of concentration with a colorimeter.

Chlorazene was the diluter of choice, despite its low opacity and change upon prolonged standing, because it alone dissolved clumps which were prevalent in ram's semen. A dilution of 1: 200 gave readings in the optimum range of the colorimeter's scale, and a lower error of estimate.

Scores for estimating visible debris in the samples effected highly significant improvement in estimation of concentration. Residual turbidity, after centrifugation of diluted samples, was not a suitable estimate of debris. Semen with the debris removed gave a regression line through the point of origin.

Two brands of diluting pipettes exhibited significant difference in volume and ability to drain; however the variability of individual pipettes was similar.

The optimum technique (chlorazene at 1: 200 with scoring of debris) gave a standard error of estimate of 37.8 ($\times 10^7$ sperm per ml). The error due to workers' and instruments' variability was negligible. The sampling and dilution errors for the colorimeter and hemacytometer were 21.3 and 21.8, respectively, but the error for the hemacytometer at the working mean was increased to 26.7 due to the Poisson distributional error of the small volume measured. The error due to non-scorable debris and heterogeneity of sample was 34.2.

151. Development and chromosome number of the offspring of a tetraploid axolotl female mated with a diploid male. G. FANKHAUSER, Department of Biology, Princeton University, and R. R. HUMPHREY, Department of Anatomy, University of Buffalo.

A 2-year-old tetraploid female axolotl, mated with a diploid male, spawned 120 eggs, of which 98 were fertile. Three eggs died before or during gastrulation; 4 made little progress beyond stage 25. The majority of the remaining embryos showed early abnormalities in the circulatory system, particularly a failure of the blood island to drain into the general circulation. In some of these embryos, however, normal circulation was subsequently established. At stage 40, 49 larvae of fairly normal type were still living. A majority of these developed stasis, hematomata, ascites, etc., or simply refused to start feeding. Of 16 still alive (age 2 months), 2 exhibit extreme edema.

The chromosome number has been determined to date in amputated tailtips of 19 larvae, in which it ranges from 32 to 44, with only 2 larvae showing the expected triploid number, 42. In 5 larvae the number is 41, in 3, 40. The preponderance of unbalanced chromosome complements explains the high percentage of abnormal or poorly viable larvae. It also indicates that irregularities in the pairing of the 4 homologous chromosomes at meiosis are frequent enough in tetraploid oocytes to produce a majority of hypo- and hyper-diploid eggs, and of hypo- and hyper-triploid offspring. The further analysis of this spawning will have considerable theoretical interest since this represents the first breeding test of a tetraploid vertebrate.

152. The influence of cosmic radiation showers arising from lead plates on the rate of carcinogenesis. Frank H. J. FIGGE, Department of Anatomy, University of Maryland School of Medicine.

The ultraviolet luminescence spectra of a number of carcinogenic compounds suggested the hypothesis that the action of these substances may be related to their conversion of some form of penetrating radiation such as cosmic radiation to a form of energy capable of inducing intracellular malignant transformations.

To test this hypothesis, 182 inbred mice were injected with 0.5 mg of methylcholanthrene. These mice were divided into 2 groups. The group of 69 control mice in 3 aluminum cages received normal, unmodified sealevel cosmic radiation. The 113 mice in 5 aluminum cages with lead plate covers, were subjected to the normal sealevel cosmic radiation plus the showers of radiation which result from passing cosmic radiation through 1 or 2 lead plates 1 cm thick. The average latent period for the malignant transformations in the controls was 80 days. The average latent period for induction of sarcomas in the 113 experimental mice receiving the intensified cosmic radiation was only 60 days, or three-fourths that of the controls. A repetition of this experiment gave the same results. While these results are significant, experiments to test this hypothesis in a more conclusive manner are in progress.

153. Pigmentation of adipose tissue as influenced by unsaturated fatty acids in vitamin E deficient rats. Lloyd J. FILER, Ruth E. RUMERY and Karl E. MASON, Department of Anatomy, University of Rochester School of Medicine and Dentistry.

Accumulations of acid-fast pigment in adipose tissue cells and associated foreign-body reactions previously observed in rats fed vitamin E deficient diets high in cod liver oil (20%), or containing the equivalent of this as the most highly unsaturated fraction of the oil, are not dependent upon the presence of C_{20}–C_{22} fatty acids of fish oils. Comparable feeding of the methyl esters of linseed, soy bean, and corn oils resulted in marked, moderate, and slight cellular reactions in the fat depots, respectively. These experiments indicate that C_{18} fatty acids with 3 double bonds (linolenic acid) are highly effective in producing these histological changes, while C_{18} fatty acids with 2 double bonds (linoleic acid) exert much lesser, but similar, effects if fed at sufficiently high levels for long enough periods of time. Compared to lard, the unsaturated fats mentioned markedly accelerate dystrophic changes in skeletal muscles and kidney nephrosis.

In rats fed E-deficient diets containing 10% methyl esters of linseed oil and given graded, oral doses of purified natural tocopherols, from 30 to 75 days of age, the daily intake of alpha tocopherol (0.2 mg) and of gamma tocopherol (2.0 mg) just sufficient to protect against the alterations in adipose tissue cells also proved to be critical levels for preventing muscle and testis injury. Other antioxidants (ascorbic acids, gallic acid, propyl gallate, nordihydroguaiaretic acid) proved ineffective.

154. The motor and sensory fibers in the rootlets of the seventh cranial nerve of the cat. James O. FOLEY, Department of Anatomy, Medical College of Alabama.

The roots of the right seventh cranial nerve were sectioned intracranially in 10 cats and, after allowing 14 days for degeneration of the motor axons, counts of the fibers in the right nerve proximal to the geniculate ganglion were made and compared with similar analyses of the left seventh nerve of the same animals.

The enumerations showed that there were from 6902 to 10276 special visceral efferent axons, 890 to 2551 general visceral motor axons and 1363 to 3175 sensory nerve fibers in the proximal part of the seventh cranial nerve of the 10 cats. Expressed in terms of percentages, the special visceral efferent axons ranged from 57.16 to 79.13, the general visceral efferent fibers from 8.18 to 20.93, and the sensory axons from 12.51 to 25.81.

When the separate values for the different functional components of all nerves were averaged, it was found that the special visceral efferent component was represented by 8506 axons (68.86%), the general visceral efferent component by 1854 fibers (15.01%), and the sensory component by 1994 axons (16.13%).

155. *Ceroid pigment in the testis of mice treated with estrogens.* Marthella FRANTZ, Department of Anatomy, University of Minnesota. (Introduced by Arthur Kirschbaum.)

Brown pigment has been reported to appear in the testes of mice treated with estrogenic hormone (Hooker and Pfeiffer, '42) and of vitamin E deficient rats (Mason and Emmel, '44). Pigment of this type has also been reported to occur spontaneously in the testes of many species of mammals, especially in animals of relatively advanced age.

Lillie, Daft and Sebrell ('41) observed a brown pigment in hepatic cirrhosis of rats induced by dietary means, which they designated as "ceroid." In hematoxylin-eosin sections the testicular pigment of mice treated with estrogens appeared to be similar. In order to establish the ceroid character of this testicular pigment, sections of testis of estrogen-treated Bagg albino mice were stained with the methods of Endicott and Lillie ('44). The pigment's ceroid nature is indicated by its acid-fastness, sudanophilia, stainability in dilute aqueous solutions of basic aniline dyes (basic fuchsin, methyl and crystal violet, methyl green, safranin O, methylene and toluidin blue), its lack of stainability with similar solutions of acid aniline dyes (orange G, Congo red, trypan blue, acid fuchsin, eosin Y and B, and Phloxine B), and its insolubility in common fat solvents.

156. *The effect of amputation of the dorsal fin upon the development of Sd melanomas in hybrid fishes.* E. D. GOLDSMITH, Myron GORDON and Ross F. NIGRELLI, Department of Anatomy, New York University College of Dentistry and New York Aquarium, New York Zoological Society.

Hybrid fishes (Platypoecilus — Xiphophorus) carrying the Sd gene develop melanomas of the dorsal fin.

The dorsal fins of 8 6-week-old, 6 3-week-old, 8 2-week-old fish were amputated at the base. At these ages the fins had few or no macromelanophores. The fins regenerated completely within 2 weeks and were removed again at that time and at 2-week intervals for a total of 10 amputations. In the 2-week interval few or no macromelanophores developed in the regenerated fins, although a number did develop on other parts of the body. Within a month following the last amputation, most of the dorsal fins developed some pigmentation. Continuous observations for 5 additional months disclosed additional pigmentation in these fins. However, it was far less than in the unoperated individuals. In fact, the dorsal fins of unoperated control fish, 5 months of age, had more pigment than the regenerated fins of 11-month-old animals.

The explanation for the difference might lie in the fact that the tissues comprising the regenerated fins are chronologically younger than that in the controls. Further work is in progress.

157. *The excretion of fluorescein by the liver under normal and abnormal conditions, observed in vivo with the fluorescence microscope.* Allan L. GRAFFLIN,[1] Department of Anatomy, Harvard Medical School, and Department of Medicine, College of Physicians and Surgeons, Columbia University.

All observations were made upon male specimens of *Rana pipiens,* receiving fluorescein (uranine) in standard dosage of 30 mgm per kilogram. After injection the dye appears promptly in the plasma, then in the hepatic cells and the bile capillaries. The peak of excretion (all bile capillaries sharply delineated, high intensity; all hepatic cells diffusely stained, low intensity) is reached in approximately 1 hour at ca. 15°C. The sequence of events from the peak of excretion to complete elimination is essentially a progressive diminution in fluorescent intensity of the dye to non-detectable levels, achieved first in the hepatic cells, then (much later) in the bile capillaries. Elimination is usually complete in 24 hours or less. The present observations upon the normal mechanism of fluorescein excretion have failed to confirm the findings of Hirt, Ansorge and Markstahler (Z. f. Anat. u. Entw., *109:* 1, 1939) in the later stages of excretion (extensive sprouting of fluorescein-containing vacuoles from bile capillaries, widespread intracellular vacuole formation, discontinuity of bile capillaries, ''lakes'' of fluorescein). However, the whole range of these findings has been observed in animals subjected to highly abnormal conditions, with anoxia probably the most important single factor. A refractory period of at least several hours' duration, as insisted upon by Hirt et al., does not occur, with respect to fluorescein excretion, in the livers of frogs maintained under normal conditions.

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[1] Senior Fellow in Cancer Research, American Cancer Society, recommended by the Committee on Growth, National Research Council.

158. *Histogenesis of Mauthner's neurone in Amblystoma.* Walter F. GREENE, Syracuse University and Osborn Laboratory, Yale University.

The differentiation of this intercalated neurone can be correlated with the morphologic stages of Harrison and the motility phases described by Coghill. The early location of this neuroblast is constant and definite among the marginal tracts of the medulla. Axone growth and decussation; Nissl bodies; pigment migration; and dendritic field become apparent in synchrony with the total reaction pattern. The dendritic field becomes elaborate as the sensory stimulation (acousticolateralis inflow) is perfected. Axonal growth extends caudad along the brain stem in advance of ganglionic and receptor maturity of the labyrinth and lateral line organs. Experimental extirpation of its synaptic primary sensory neurones can fractionally deplete the lateral dendritic field of Mauthner's cell. In such cases the relations to the motor pattern of the medulla remain intact. Several cases are recorded in which this intercalated neurone was absent from the operated side of the medulla. Decussation of the remaining axone was normal in most of these cases.

159. The acid phosphatase reaction in degenerating axons of certain sensory tracts. Walter L. HARD [1] and Bernard FERRARA, Department of Anatomy, Medical College of the State of South Carolina.

The presence of acid P-ase in the axons of both sensory and interstitial neurons of the CNA is reported. The sensitivity to injury of pyramidal tracts fibers in the cat and monkey has been identified by the disappearance of P-ase from the injured axon (Hard and Lassek, '46; Lassek and Hard, '46). The present study extends these observations to fibers of the trigeminal, spinocerebellar, and gracili fasciculi.

Axonal injury was produced in a total of 33 cats by appropriately placed lesions in the medulla (15), low thoracic cord (10), and dorsal roots (8). Phosphatase disappeared from fibers of the descending trigeminal tract within a 32-hour postoperative time interval. In hemisected animals, as well as dorsal root sections, maximal loss of P-ase occurred in the posterior funiculus by a 3-day postoperative time. However in these hemisections the spinocerebellar tracts showed little or no (4 animals) degeneration at 3 days, extensive degeneration at 4, and maximal loss at 5 days postoperative times. In hemisections the degeneration of cortico-spinal fibers during a 3-day postoperative interval confirms earlier work. These observations reflect a distinct variation in susceptibility to injury of diverse neurons. Studies are in progress to determine if variations in enzyme concentrations of the diverse neurons is alone sufficient to explain this type of response.

[1] Present address, Department of Anatomy, School of Medicine, University of South Dakota, Vermillion.

160. Changes in mitochondrial content of motor nerve cell bodies following axone section. J. Francis HARTMANN, Department of Anatomy, University of Minnesota.

In this experiment, the right sciatic nerve was cut in a series of 19 rats, after which animals were killed at daily intervals from the second through the tenth postoperative day, perfused with Regaud's fluid, and the appropriate segments of the spinal cord removed for study. Alternate 3 μ cross sections of the cord were stained for Nissl substance with carbon-thionin and for mitochondria with the basic fuchsin technique of Fain and Wolfe. Beginning with the seventh postoperative day, a definite increase in fuchsinophilic granulation is evident in the chromatolyzed cells in the ventral horn of the operated side as contrasted with the control side. Counts of mitochondria in affected and normal cells show an average calculated number of 170.4 million per mm^3 on the operated side as opposed to 90.2 million on the control side on the tenth postoperative day. Direct comparison of Nissl-stained sections of individual cells with adjacent sections through the same cells stained for mitochondria reveals that the increase in fuchsinophilic granulation follows approximately the same time course as does chromatolysis. The increased red granulation in the affected cells is thought to be due to increased numbers of mitochondria rather than to degraded Nissl material staining with basic fuchsin because the granulation is as intense in the axone hillock as elsewhere in such cells.

161. The plasmal reaction. E. Russell HAYES,[1] Department of Anatomy, The Ohio State University. (Introduced by R. A. Knouff.)

The plasmal reaction (PR) of Feulgen and Voit ('24), if applied rigorously, can be used for the strict demonstration of acetal-lipids. The bases for a rigorous PR are: (1) obtaining a negative control section from the same block of tissue, which allows precise definition of the unmasking mechanism; (2) restricting treatment of the fresh tissue sections to 2–5 minutes in $HgCl_2$; this, by decreasing the time element, minimizes the unmasking effects of acid and of oxidation and thus takes full advantage of the high specificity of $HgCl_2$ for the acetal bond. Formol fixation rapidly diminishes and finally destroys the "true" PR and progressively develops a secondary reaction with fuchsin-sulfurous acid that is independent of $HgCl_2$. In normal rat adrenal and liver these 2 carbonyl-lipid reactions are so markedly different in distribution that it seems clear that formol fixation (oxidation?) is not equivalent to $HgCl_2$-unmasking and probably demonstrates carbonyl-lipids other than acetals. Observations indicate that this "true" PR is rather stable in the rat adrenal under stresses that markedly alter the Sudan and Schultz reactions.

[1] Now at the University of Buffalo.

162. Some observations on the human nervus vertebralis. Edward A. HOLYOKE and Alister I. FINLAYSON, Departments of Anatomy and Surgery, University of Nebraska College of Medicine.

Although the nervus vertebralis is frequently a well-defined feature of the sympathetic system in the cervical region it has been entirely ignored in most standard works on anatomy and has received scant attention in the literature. There is little agreement as to its frequency, connections, or significance.

The vertebral nerve has been observed in a series of 12 out of 20 adult cadavers during the present routine course in gross dissection. It was found on one side, usually the right, in 8 cases and on both sides in 4 (making 16 nerves observed). It is a nerve of variable size passing superiorly from the inferior or occasionally (6 instances) the middle cervical ganglia to perforate the prevertebral fascia and longus capitis and cervicis muscles. In accord with Broek (except 1 case) it crosses ventral to the vertebral artery from lateral to medial to communicate in most cases with the fifth, sixth, and seventh, or sixth and seventh cervical nerves. Two cases communicated with the fifth and sixth and one with the fourth and fifth nerves. In one instance there were 2 separate nerves communicating respectively with the fourth and sixth cervicals.

We do not yet have evidence as to the functional significance of this nerve. It does not seem to connect extensively with the sympathetic plexus of the vertebral artery.

163. Temperature influence during embryonic development in rats. C. Y. HSU, Department of Anatomy, Washington University School of Medicine, St. Louis, Mo. (Introduced by John H. Van Dyke.)

Pregnant rats (Wistar) were incubated 0.5–10.5 days after insemination at 39°C. for 1, 2 or 4 hours. Rectal temperature, following treatment, averaged 5–6°F. above normal.

Six females treated for 1 hour 0.5–1.5 days after insemination and 6 animals subjected for 2 hours during the same gestation periods failed to produce young, with the exception of 2 cases. Delivery in one was normal; in the other, 2 offspring were dead. Parturition in animals treated similarly during advanced stages of pregnancy was normal.

However, with the exception of the following individuals, a third group of females (12) subjected for 4 hours 0.5–10.5 days after insemination gave positive results. Three animals (4.5, 5.5 and 6.5 days pregnant) delivered no offspring. Another 6.5 day specimen produced a litter of 10 (2 twins, 2 within fetal membranes, dead; the remaining apparently normal). Uteri of a third 6.5 day pregnancy were examined immediately before delivery (5 were completely resorbed, 3 partially resorbed and 3 apparently normal).

Examination of corpora lutea during pregnancy in this preliminary study suggests that, despite certain individual variations, increase in body temperature for a relatively short period may terminate early pregnancies, that the period immediately before, during and just after implantation is more susceptible to severe heat treatment, and that later stages of gestation are most resistant to temperature changes.

164. The suprahyoid hypoglossal ansa (of Hyrtl): a nerve crossing the midline of the body. Robert M. JAMESON and William O. REINHARDT, Department of Anatomy, University of California Medical School.

Instances of anastomosis across the midline of the body of peripheral or cranial nerves in man are of sufficient rarity to warrant attention. The present example is one in which the 2 hypoglossal nerves were continuous across the midline in relationship to the antero-superior aspect of the hyoid bone. This loop or ansa (found in a 49-year-old male cadaver) was approximately one-third the diameter of either of the main roots of the hypoglossal nerves. The loop maintained a uniform diameter across the midline, gave off a few fine twigs to the geniohyoid muscles on either side of the neck, and traversed in its course the deep cervical fascia in immediate relationship to the antero-superior aspect of the hyoid bone. Attention has been directed to this variation by Hyrtl, who named it the suprahyoid hypoglossal ansa, distinguishing it, thereby, from the infrahyoid cervical ansa of the hypoglossal nerve. The ansa was first described by Bach in 1835, has been seen by Arnold (1851), and by Hyrtl (1865). This suprahyoid loop or ansa of the hypoglossal nerve has not been studied in any previously cited case as to the origin of its contained fibers. It is possible that the suprahyoid ansa may be of cervical rather than hypoglossal origin, and might, therefore, be more properly designated as a suprahyoid cervical ansa. Such extra-cranial anastomoses across the midline of the body have been noted more constantly in other forms, i.e., the chiasmatic anastomosis of the internal laryngeal nerves in turtles, and the hypoglossal anastomosis in the suprahyoid region of the alligator and the ostrich. The incidence of this variation has been quoted at 10% on the basis of Bach's original findings (1835) of 3 cases in 32 subjects examined. In view of the paucity of further reports of instances of this loop formation, a frequency of 10% is certainly too high.

165. The Golgi element in primitive erythroblasts of the 11-day rat embryo. Oliver P. JONES, Department of Anatomy, University of Buffalo.

Preliminary observations indicate that structures similar to the Golgi element described by J. R. Baker ('44) are also preserved in dry smears of embryonic blood. These may be stained supravitally by immersing the yolk sac in an isotonic salt solution of dahlia violet. Portions of these structures have a strong affinity for Sudan black after dry smears have been fixed in formol-calcium. With phase microscopy some of these forms are true discs or rings which may appear rod-like, elliptical or circular depending on their orientation. The centers, which are sudanophobic, correspond to neutral red vacuoles. Fetal livers from 15-day embryos are also suitable for similar studies on the definitive generation of erythroblasts.

166. On the nature of Auer rods.[1] Raymond S. KIBLER, Department of Medicine, University of Buffalo and the Buffalo General Hospital. (Introduced by R. R. Humphrey.)

Early myeloid cells of 5 cases of acute myeloblastic leukemia have been studied by means of a combined Sudan black-Romanowsky technique. Auer rods exhibited an intense sudanophilia; the largest rods possessed a sudanophobic core. It was observed that Auer rods were present in each case and that they occurred in a much greater percentage of the promyelocytes than could be demonstrated by a standard Romanowsky stain (May-Grünwald-Giemsa). In many instances within the same promyelocyte there were demonstrated all stages of Auer rod development from an intensely sudanophilic granule to the largest rod. Phase microscopy appeared to confirm the observation that the largest Auer rods possessed a well-defined cortex in which the lipoid was concentrated, and that the presence of a sudanophilic cortex is not an optifact. The Kurloff bodies of the guinea pig and lymphoid azure granulation to which the Auer rods have been likened were shown to be constantly sudanophobic. Differential extractions indicated that the contained lipoid was a phospholipid. It is suggested that the rod-like structures reported in a variety of disorders of the blood and/or bone marrow resemble Auer rods only in their morphology and differ from the genuine Auer rods of myeloid leukemias in chemical composition.

[1] Aided in part by a grant from the Ciba Pharmaceutical Company.

167. *Morphological evidence suggesting that human extrinsic eye muscles are phylogenetically relatively primitive.* Hadley KIRKMAN, Department of Anatomy, Stanford University.

In a total of 30 cadavers examined 73% showed macroscopically visible nodular swellings near the middle of the bellies of some or all of the extrinsic eye muscles. No similar swellings were observed in any other skeletal muscles examined.

From a second series of 35 cadavers a total of 37 extrinsic eye muscles and a total of 70 extra-ocular skeletal muscles, including lumbricales of index fingers, posterior venters of digastrics, tensor tympanis, and one levator palpebrae superioris, were examined microscopically. In 36 of the eye muscles the swellings were due to accumulations of localized contraction nodes in single muscle fibers. Similar nodes were observed in but one of the extra-ocular muscles (one of 12 tensor tympani muscles).

The nodes were independent of the state of rigor since the specimens were embalmed from 6 to 168 hours post mortem.

Localized contraction nodes are characteristic of smooth muscle, are common in skeletal muscle of insects and lower vertebrates, and occur in denervated mammalian skeletal muscle. In striated muscle they are generally interpreted as a relapse to a smooth muscle type of contraction.

It is concluded that the morphological observations reported here furnish data, in addition to those already available in the literature, in support of the hypothesis that mammalian extrinsic eye muscles are phylogenetically more primitive than other mammalian skeletal muscles.

168. *Topography of fibers in the medullary center of the human cerebellum.* Wendell J. S. KRIEG, Institute of Neurology, Northwestern University Medical School.

Fiber-stained sections are inadequate to show the complete course of fibers of the medullary center of the human cerebellum and Marchi experiments are impracticable. However, fiber dissection is quite feasible on the decorticated cerebellum after certain treatments.

Brachium pontis is shown to distribute to lateral lobe generally but in diminished numbers to its inferior part. Deep rostral fibers make a sharp bend forward and medially to supply anterior lobe, fanning out medially and laterally. Fibers to lateral part of superior semilunar lobule form superficial stratum of pons, converge in region of trigeminal root. Those which reach medial part of lobule run medially anterior to dentate nucleus, then turn backward within their folia. Fibers to inferolateral part of lateral lobe are superficial and caudal in pons, while those to inferomedial part are deep but not necessarily caudal.

Restiform body can be traced as a unit throughout medullary center. As soon as it has cleared brachium pontis deeply, fibers turn backward to pass to superior semilunar lobule. This tract continues medially just in front of hilus of dentate, sending fibers backward to middle lobe and forward to anterior lobe. Component to middle vermis curves backward around medial edge of dentate.

(For remainder of connections see abstract of demonstration.)

169. The lateral infracostal artery. Benjamin N. KROPP, Department of Anatomy, Queen's University, Kingston, Ontario.

In a series of 82 dissection room subjects, the frequency of occurrence and relations of the lateral infracostal artery were noted. Typically this artery is a branch of the internal mammary and descends obliquely and laterally close to the mid-axillary line deep to the parietal pleura. It arises near the origin of the internal mammary, rarely from the subclavian, and may traverse 6 intercostal spaces. A vein accompanies the artery, and both communicate with the corresponding intercostal vessels of the spaces they cross.

The artery has been mentioned under a variety of names, A. thoracica interna, Hodges, 1858; A. mammaria interna lateralis, Henle, 1868; and others. Adachi, 1928, found it on the right side alone in 8.1% and on the left side alone in 14% of all cases. In the present series, the origin and course of the vessel conformed in general to earlier descriptions. The distribution, however, was found to be variable and the incidence markedly higher than previously reported. In 42.6% of all cases, the artery was present on one or both sides; in 14.6% on the right side alone, in 18.2% on the left side alone, and in 9.7% on both sides. When present it reached the fifth intercostal space in 30% of cases. The occurrence of the artery has no apparent relation to the sex or age of the subject.

170. Components of supporting tissue and ganglion cell capsules in autonomic ganglia.[1] Albert KUNTZ and Norman M. SULKIN, Department of Anatomy, St. Louis University School of Medicine.

Sections of autonomic ganglia of the cat and of man have been prepared by various methods including Masson's trichromic technic, Quade's ferric gallate stain, Cajal's gold chloride sublimate method for astrocytes and the methods of Del Rio Hortega, Penfield and Grino for oligodendroglia and microglia. The data obtained in a study of these preparations support the following conclusions: The supporting tissue within the ganglia normally includes but few collagenous elements. The ganglion cell capsule typically consists of a single layer of flat cells with few cytoplasmic processes. Most of the cellular components of the interstitial tissue are similar in cytologic structure, size and staining reactions to the capsule cells. On the basis of these characteristics which they possess in common with oligodendroglia the capsule cells and the related interstitial cells appear to be homologous with the oligodendroglia in the central nervous system.

[1] This work has been supported by a grant from the Life Insurance Medical Research Fund.

171. Acid phosphatase in peripheral degenerating and regenerating neurons. A. M. LASSEK and E. D. BUEKER, Department of Anatomy, Medical College of the State of South Carolina.

The reaction of acid phosphatase in motor cells and sciatic nerve axons has been studied in 53 animals (carnivora, aves and amphibia) following either high crushing or sectioning. The following was found in a series of cats: The enzyme disappears from the distal portion of the severed axons rather completely between the third and fourth postoperative days whereas it remains intact in the central stump during the entire degenerative and regenerative cycles. Acid phosphatase accompanies the growing nerve and is quantitatively reduced during the maturation of the growing axons. Its speed of advance peripherally is much faster in the crushed than sectioned nerves. We have detected an increase of acid phosphatase in affected anterior horn cells as early as the first and as late as the eighty-fourth days (the latest sacrifice time to date). Maximal results are obtained in the cells between the twelfth and eighteenth days. Injections of commercial formalin or gentian violet into the central stump at the time of section has not reduced the early heightened acid phosphatase activity in the involved motor cells in preliminary experiments. The reaction is also present in chickens and frogs under similar experimental conditions. It is concluded that the enzyme, acid phosphatase, is present in all stages of nerve regeneration in the animals studied.

172. Electrophoretic study of liver cytoplasmic proteins. Arnold LAZAROW and Jack BERMAN, Department of Anatomy, Western Reserve University.

Guinea pig livers were freed of blood by perfusion through the inferior vena cava. The liver cells were fragmented by squeezing the liver through bolting silk, and the resulting liver brei was suspended in 1.5 volumes of 0.85% saline. The nuclei, mitochondria, and submicroscopic cytoplasmic components were removed by differential centrifugation, and final clarification was accomplished at 18,000 × g. for 60 minutes, using the multispeed attachment for the International centrifuge. The resulting supernatant, a clear amber liquid, showing practically no opalescence, was dialyzed against veronal buffer pH 8.6 and ionic strength of 0.1. The electrophoretic pattern was measured in the Tiselius apparatus. Control studies were carried out on the serum obtained from the same guinea pig.

A number of distinct components were observed in the electrophoretic pattern of the liver proteins. On comparing the serum and the liver patterns, it was noted that the fastest liver component migrated at about 65% of the rate of serum albumen. Purified guinea pig serum albumen was added to the liver protein solution before electrophoresis (20% albumen to 80% liver proteins). The pattern of the mixture revealed that the added albumen migrated at its normal rate, and the fastest liver component migrated at 64% the rate of the added albumen. Studies are being carried out to determine whether the fastest moving liver component is a complex of albumen.

173. Gonadotrophic hormone content of rat pituitaries following thiouracil and testosterone propionate. James H. LEATHEM, Department of Zoology and Bureau of Biological Research, Rutgers University.

Eight adult male rats were fed 0.5% thiouracil in the stock diet for 20 days and served as the control for food intake. Three additional groups of 8 rats each were pair fed and included rats given thiouracil and 0.5 mg testosterone propionate (perandren, Ciba) daily; normal rats and normal rats injected with the androgen. Pituitaries were macerated in a centrifuge tube to which was added distilled water and a drop of 2% NaOH. Littermate recipients were 22-day-old female rats which were injected twice daily for 4 days and killed 24 hours later. Ovarian weight increase was the criterion of response; each rat received but 1 pituitary. Ovarian weight in recipient rats was increased from 13 mg to 70 mg by 1 pituitary from thiouracil-fed rats and to 65 mg by a pair fed control pituitary indicating the absence of an effect. Rats fed thiouracil and injected with androgen increased ovarian weight to only 28 mg with 1 pituitary but a similar reduction in hormone content occurred with pair fed androgen injected normals whose pituitaries increased ovarian weight to 31 mg. Since the pituitary weights were essentially the same, thiouracil did not alter the gonadotrophic hormone content or the effect of the androgen when compared with pair fed controls.

174. The epicytes of mammalian lungs. Charles C. MACKLIN, The University of Western Ontario Faculty of Medicine, London, Canada.

These scattered entodermal epithelial cells, also known by other names, as septal cells, were found universally distributed in the alveolar walls of healthy lungs of 10 mammalian species including man, in such numbers as to suggest that they would make a substantial mass if merged. The form is much altered by fixation methods. Most favorably displayed in material fixed by vascular perfusion, the profile resembles a carafe, with the larger end containing the nucleus, around which is the most conspicuous feature, a zone of clear bodies or vacuoloids. When epicytes show phagocytosis the ingested dust particles or dye molecules are found not in the vacuoloids but in the cytoplasm between them. Epicytes which extend through the partitional alveolar walls have 2 free surfaces, a larger and a smaller; and both of these in silver-washed material may be surrounded by silvered rings and sprinkled with silvered granules. They lie between larger areas often regarded as containing epithelial squames, and occupy sockets bordered by capillaries which frequently indent their sides. They have been found apparently being released from their sockets to become free dust or foam cells. Epicytes resemble the epithelial cells of the respiratory bronchioles, where cell diversification is apparently proceeding. After fixation by bronchial injection they are flattened, and after immersion fixation they are usually so buried in the surrounding tissue as to be overlooked.

175. Failure of the theory of partial sex-linkage to explain known facts of inheritance of xeroderma pigmentosum. Madge T. MACKLIN, Department of Zoology-Entomology, The Ohio State University.

If a recessive gene is partially sex-linked, at least 1 crossover must occur in 75% of families producing affected sons when parents are related, and 3 crossovers must occur in 25% of families. The crossovers must occur at specific places somewhere between the ancestor with the affected gene and the affected son. In similar matings, producing affected females, 62.5% require 2 crossovers, 6.25% 4 crossovers, and 31.25% require none. Haldane sets the preferred crossover value for xeroderma at 14, mentioning 35 as an upper estimate. With 14 as the value, 25% of consanguineous matings should produce affected males, 75% affected females. Of 53 recorded consaguineous matings, 29 produced only affected males, 24 only affected females. At a value of 14, χ^2 is 23; at 35, χ^2 is 3.46. The first excludes the hypothesis of sex-linkage, the second not quite. Tests for linkage with sex in families in which the affected paternal grandparent can be identified on this hypothesis, give χ^2 of 0.5. The recorded data do not support the hypothesis of partial sex-linkage for the gene causing xeroderma pigmentosum. The family regarded by Haldane who did not know the relationship of the parents as most strongly supporting partial sex-linkage requires 7 crossovers to produce the 7 affected males.

176. An adrenergic link in the ovulatory mechanism of the rabbit. J. E. MARKEE, Charles H. SAWYER and W. Henry HOLLINSHEAD, Department of Anatomy, Duke University School of Medicine.

The results of our previous experiments with electrical stimulation to determine the nature of the control of the nervous system over the hypophysis have indicated that the hypothalamico-pituitary pathway probably contains a non-nervous link. In the present work the known neurohumoral agents, acetylcholine and adrenaline, have been administered to mature female rabbits both intravascularly and by direct instillation into the anterior hypophysis. Intravenous injection of acetylcholine did not induce ovulation in any of 7 animals. Ovulation also failed to occur in 22 of 23 animals in which acetylcholine was instilled into the hypothesis; it is therefore unlikely that acetylcholine is concerned with the release of luteinizing hormone in the rabbit.

Intravascular administration of adrenaline in 20 animals was not followed by ovulation. However, direct instillation of adrenaline into the hypophysis produced ovulation in 12 of 51 animals in all dilutions used, and 5 of 10 with 1/1000 dilution. There was no correlation between the amount of hypophyseal damage observed and the occurrence of ovulation. The effect of adrenaline could not be attributed to lowered tissue oxidation, for neither systemic anoxemia nor injection of potassium cyanide into the hypophysis produced ovulation.

The present experiments are interpreted to indicate that the humoral link in the hypothalamico-pituitary path involved in ovulation is of an adrenergic nature.

177. The scalenus minimus muscle: its occurrence and morphology in Brazilian Whites and Negroes. J. P. P. MELLO,[1] Medical School, University of Brazil, Rio de Janeiro. (Introduced by N. L. Hoerr.)

Forty-six individuals (26 males and 20 females), totalling 92 sides of Brazilian Whites (21), Mulattoes (16) and Negroes (9), were dissected. The scalenus minimus was found in 63% of the individuals and in 45.6% of the sides considered. The total number of muscles was 38, out of which 69% were unilateral (70% on the right, 30% on the left), and 31% bilaterally.

There are strong indications that the occurrence of the scalenus minimus is influenced by racial factors, the Whites appearing to have the highest ratio (70%), the Negroes the lowest (40%), with the Mulattoes in an intermediate position (55%).

No indication of sex variation was seen.

All of the 38 muscles studied originated on the seventh cervical vertebra. In 70.2% of the cases, this was the only origin. In 25.43%, an additional origin on the sixth cervical vertebra was noted, and in only 4.25% another slip came from the neck of the first rib.

The insertion was in all cases the inner margin of the first rib. Sometimes the tendon of insertion was observed to spread out upon the pleural dome, but careful dissection always succeeded in separating this tendinous expansion from the endothoracic fascia. In cases of a well developed muscle, the first rib showed a small spine on the inner border, corresponding to the muscle insertion.

Concerning relations, the muscle lay always between the subclavian artery and the lowest trunk of the brachial plexus.

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[1] Commonwealth Fellow in Anatomy, Department of Anatomy, Western Reserve University, School of Medicine.

178. *Effects of thiouracil, estradiol and testosterone propionate on the time of appearance of the ossification centers of the rat.* Charles R. NOBACK, Department of Anatomy, James C. BARNETT, Department of Radiology, Long Island College of Medicine and Herbert S. KUPPERMAN, Department of Experimental Medicine, University of Georgia, School of Medicine.

A delay of from 1 to 5 days in the time of the appearance of the epiphyseal ossification centers of the extremities of the rat results after the daily injection of thiouracil (deracil). Rats received daily subcutaneous injections of 1–1.25 mg of thiourea for 10 days after birth and of 2.5 mg of thiourea thereafter. Roentgenographic studies were made of the ossification centers which appear normally 8 to 10 days after birth. The 5 litters examined were divided into 10 control and 16 experimental animals. Effects of thiourea were indicated by the poor growth of the animals and by cellular hyperplasia of the thyroid glands noted after the animals were sacrificed.

The daily subcutaneous injection for the first 9–10 days of life of 12.5 μ of estradiol in sesame oil or of 125 μ of testosterone propionate in sesame oil into either male or female rats does not influence the time of appearance of the same centers studied above. A total of 30 litters consisting of 93 control and 87 experimental animals were studied. The animals were sacrificed at 9–10 days of age, cleared and stained with alizarin red for the examination of the skeleton. Wet weights of the uteri and seminal vesicles were used to determine the activities of the hormones.

Littermate animals of the same sex served as controls in all experiments.

179. *A third pressoreceptor area of the body: afferent nerve endings in the coeliac and superior mesenteric arteries.* José F. NONIDEZ, Department of Anatomy, Cornell University Medical College.

The experiments of Heymans, Bouckaert, Farber and Hsu[1] in the dog point to the existence of pressoreceptors for the splanchnic circulation the function of which can be demonstrated after afferent denervation of the carotid sinus and the arch of the aorta. The work here reported shows that these pressoreceptors occur at the bases of the coeliac and superior mesenteric arteries, respectively; they are terminations of medium-sized nerve fibers branching in the externa as well as in the outer half of the media of these arteries. The branches of arborization are longer than in the corresponding terminations of the carotid sinus and the aorta, and lack the pronounced reticulated swellings found in those areas; since they are so extensive they are seldom included within 1 section. Arborizations of this type do not occur in the descending aorta, nor are they present in the renal, inferior mesenteric and spermatic arteries.

The origin of the splanchnic pressoreceptor fibers has not been determined. They probably reach their destination by way of the splanchnic nerves via the coeliac plexus because, as shown by Heymans and coworkers, the splanchnic vascular reflexes are abolished after acute destruction of the spinal cord. The silver nitrate method of Cajal (chloral hydrate fixation) was used.

[1] Spinal vasomotor reflexes associated with variations in blood pressure. Am. J. Physiol., *117:* 619–626, 1936.

180. Malformations produced by sodium thiocyanate in the early development of the chick embryo. Wiktor W. NOWINSKI [1] and José PANDRA. From the Institute of General Anatomy and Embryology, Faculty of Medicine, University of Buenos Aires. (Introduced by Charles M. Pomerat.)

Sodium thiocyanate in concentrations ranging from 0.4–24.2 mg was injected into the vitellin sack of unincubated hen-eggs and then incubated for 40–48 hours. In this way a series of malformations was obtained of which the following types occurred regularly: (1) Omphalocephaly, accompanied in the great majority of cases by a non-individuation of the cell material in the region of the forebrain and midbrain; (2) The neural plate in the region of the metencephalon did not form the neural tube. This malformation is accompanied by a stretching of the notochord in the vertical direction; (3) Due, in all probability, to the change in growth rate of the neural folds, a superposition of the 2 neural folds was formed. In this case, the fold which grew inside the neural canal, continues to grow in 3 directions: towards the bottom of the neural tube and along its lumen. The slowly growing fold, once in contact with that of the other side, stops growing. From this fact the conclusion is drawn that the factor which regulates the closure of the neural tube is simply a mechanical one: when the 2 borders touch each other, their further growth is inhibited.

The above mentioned malformations occurred either alone, or in combinations, the third one being the most frequent.

[1] Now in the Department of Anatomy, University of Texas Medical Branch, Galveston, Texas.

181. The physiology of the tubal sac of the oviduct of the rat. D. Louise ODOR and Richard J. BLANDAU, Department of Anatomy, University of Rochester, School of Medicine and Dentistry.

The dilatation and muscular activity of the tubal sac of the oviduct of the rat during the period of behavioral estrus has been investigated. It was found that dilatation of the sac begins at the onset of heat and continues at a gradual rate until maximal dilatation was attained 6 to 8 hours later. A definite constriction forms at the distal end of the tubal sac as the dilatation proceeds. Muscular activity of this portion of the oviduct, at the onset of heat, consists of pronounced pendulum-like contractions which occur at the rate of 10 to 15 per minute. Four to 6 hours later these pendulum-like contractions subside and are replaced by strong peristaltic waves which sweep along the tubal sac and end in the region of the distal constriction. Particulate matter or freshly ovulated ova inserted into the periovarial space are propelled into the proximal end of the tubal sac by the ciliary activity of the fimbria. Their further progression into the distal part of the sac is accomplished by strong peristaltic waves which carry the material in the lumen forward at a rapid rate. Once the particulate matter has reached the distal end of the sac it remains more or less stationary, the distal constriction preventing any further forward progression of the material in the lumen.

182. Cell counts in the anterior pituitary of the cat. Clinton M. OSBORN, Department of Anatomy,[1] The Ohio State University.

Hypophyses of 35 cats representing controls, inanition, and acute and chronic adrenal insufficiencies were fixed by the method of Dawson and Friedgood, '37. Serial sections cut frontally or sagittally at 5 micra were stained with Mallory-Azan or Masson. Outline drawings of every twentieth section were made by projection on millimeter graph paper and squares labeled to coincide with ocular micrometer squares. Using O.I. objective with 10 × ocular, total counts of cells displaying nucleoli and within the limits of the micrometer field were recorded in appropriate squares of the graph paper. By such a "checkerboard" recording, a quantitative picture showing cell type distribution over the entire section or any part of a section may be obtained. Data concerning cell type distribution in 3 dimensions are available by combining graphic records of appropriate consecutively counted sections.

Counts averaging 30,000 cells per gland indicate for acidophils, basophils and chromophobes respectively 42%, 16.5%, 41.5% (normal female sacrificed 2/27/40); 42.2%, 19.8%, 38% (normal male sacrificed 3/18/40); and 36.3%, 10.8%, 52.9% (adrenalectomized female sacrificed 3/18/40 in acute insufficiency 11 days after operation). Qualitatively the basophils in the adrenalectomized animal showed marked degranulation.

Chromophobes are most numerous in the medial third; basophils, more evenly distributed, reach peaks at either edge of the medial third; acidophils are most numerous laterally.

[1] In cooperation with the Department of Physiology, The Ohio State University.

183. Inclusion bodies voided from nerve cells and embedded in walls of blood vessels. James W. PAPEZ, Zoology Department, Cornell University.

Neurons affected by inclusion bodies tend to degenerate and release the inclusion bodies in clusters into the intercellular substance. The inclusions migrate to small blood vessel (veins) and become embedded in cytoplasm of connective tissue of outer wall of vessel forming vascular cuffs. Here they undergo degeneration, release their lipid, and a sclerotic patch is formed in the wall of the vessel encroaching on the lumen (see demonstration D 2). When inclusion bodies migrate along outer walls of larger cerebral vessels, numerous sclerotic patches may be formed on the vessels, typical of cerebral arteriosclerosis.

In some instances the inclusion bodies enter the small blood vessels and become attached to the red blood corpuscles which swell to a spherical shape and stain darkly with cresyl violet.

184. Protargol used with the silver gelatin stain for nerve fibers. Anthony A. PEARSON, Department of Anatomy, University of Oregon Medical School.

The silver gelatin stain is often improved by a preliminary treatment with protargol. The method is outlined as follows:

Fix tissues in formalin, Bouin's fluid, or in any of the usual fixatives for silver stains. Imbed, cut, and remove the paraffin in the usual manner.

Place sections in a .5% solution of protargol for 8 to 10 hours at 37°C. Add copper as in the Bodian stain.

Water adjusted approximately to a pH of 4 to 4.5 should be used for the solutions in the following steps until the sections are placed in gold. This adjusted water can usually be prepared by adding 1 to 5 cm^3 of 1% citric acid to 1 liter of distilled water.

Drain off the protargol solution and place the sections in 2% silver nitrate for about 45 minutes in a dark oven at 55°C. This step may sometimes be omitted.

Place slides in a solution containing 10 cm^3 of 2% silver nitrate and 40 cm^3 of 3% gelatin. Place in a warm bath (about 55°C.) and add 4 cm^3 of 1% hydroquinone. Agitate this solution gently until the sections turn a tobacco brown.

Wash in warm water. Tone lightly in 1% gold chloride. Wash and place in 1% oxalic acid until the sections turn slightly purple.

Wash, dehydrate, clear, and mount sections in the usual manner.

185. Luteinization of theca interna cells in the ovaries of albino rats. E. Stanley PEDERSON and John S. LATTA, Department of Anatomy, College of Medicine, University of Nebraska.

In histological examination of rat ovaries in various stages of the sex cycle several examples of thecal luteinization in the walls of unruptured vesicular follicles were found. Two of these follicles were classified as simple follicular cysts, the remainder as atypical examples of atresia.

The hypertrophied thecal cells formed quite compact masses in juxtaposition to the stratum granulosum, which was in various stages of disintegration. Usually such masses were found only in one portion of the theca, however in some instances the masses had completely encircled the follicle.

The hypertrophic cells are histologically identical with lutein cells of the rat ovary. Transitional stages between typical theca cells and ''theca lutein'' cells were obvious.

In some instances, in which the stratum granulosum had entirely disappeared, interstitial cells (theca) were found which were apparently in the process of transformation into ''theca lutein'' cells.

From the material observed thus far it seems probable that this phenomenon occurs in unruptured follicles early in diestrum.

186. *Some observations on the surface of some epithelial cells.* Paul H. RALPH, Department of Anatomy, The Ohio State University. (Introduced by R. A. Knouff.)

By the use of the phase-difference microscope, it can be seen that the surface of desquamating cells from some stratified squamous epithelia (cheek, heel, tongue et al.) exhibits a complex pattern of tiny ridges or thickenings. These ridges exhibit variety and individuality equal to that of human finger prints. At the point where 2 epithelial cells are in contact the pattern of ridges of 1 fits into the grooves between the ridges on the surface of its neighbor. They may be seen only with the phase-difference microscope. There appear occasionally suggestions that they may be related in some way to the tonofibrils found in cells deeper within the epithelium. However, the resemblance of these patterns to striae developed in drying protein and collodion films suggests the possibility that this is merely an effect of desiccation.

187. *Bulbar structures involved in swallowing.* R. RHINES, Department of Anatomy, Northwestern University Medical School.

In cats under light nembutal anesthesia, the medulla was explored for swallowing responses by the Horsley-Clarke method. Rhythmic swallowing was elicited during stimulation of the dorsal part of the medulla in, and immediately adjacent to, the tractus and nucleus solitarius. Stimulation was not effective elsewhere in the bulb except along the intramedullary course of the glossopharyngeal and vagus nerves. Caudally, the excitable areas converged to the midline in the region of the commissural nucleus. No rhythmic swallowing was produced by stimulation caudal to this level.

Because of the close proximity of the tractus and nucleus solitarius and the adjacent gray, it was impossible to place precise enough lesions to designate the specific structure responsible for rhythmic swallowing. However, the extension of excitable points to the level of the commissural nucleus, where the tractus solitarius has almost disappeared, suggests that the response obtained was not solely due to activation of primary afferents.

Dysphagia is commonly observed clinically in instances of bulbar involvement, and has usually been attributed to injury to the nucleus ambiguus. The demonstration of supranuclear structures concerned with this activity suggests that they should be scrutinized also in determining the neuropathological basis for impairment of the swallowing reflex in affected cases.

188. *Effects of pressure on the uterine eggs of Rana pipiens.* Roberts RUGH and Douglas MARSLAND, New York University.

Previously the authors found that the mature spermatozoön of Rana pipiens was not appreciably affected by moderate hydrostatic pressure of 8,000 lbs./sq. in. for periods up to 30 minutes. Similar tests have been made recently on the uterine eggs, all of which are known to be in metaphase of the second maturation division. In each experiment 1 egg-filled uterus of an ovulating female was used as a control while the other similar uterus was subjected to hydrostatic pressure of

8,000 lbs./sq. in. for 3, 10, and 30 minutes. Immediately after compression the eggs from both uteri were fertilized with normal spermatozoa.

Some entire batches of eggs appeared to be undamaged by the pressure. They fertilized and developed normally. However, other batches of eggs displayed widespread cytolysis, some immediately after decompression. Other batches showed a low percentage of cleavage, death at gastrulation, and a high percentage of abnormalities during later development. These results indicate a varying susceptibility to compression damage in the eggs from different females. It appears that in this egg, as in previous observations, the cells are more susceptible to compression damage during periods when movements (e.g., mitotic) are occurring. Moreover, a preliminary test indicates that the germinal vesicle of the ovarian egg is quite resistant to pressure damage.

188a. Persistent pupillary membrane and hyaloid artery in the postnatal rat and mouse. Meredith N. RUNNER,[1] Roscoe B. Jackson Memorial Laboratory. (Introduced by Katharine P. Hummel.)

Although persistence of the pupillary membrane and hyaloid artery is common in man there is practically no evidence of an hereditary basis for this defect. The eyes from 6 strains of inbred mice and 2 strains of inbred rats were examined prior to opening of the eyelids. The hyaloid artery, vascular lenticular capsule and pupillary membrane were present in all eyes examined. Their major portions disappeared about the time that the eyes opened. A series of vascular injected, cleared and mounted eyes depicts the embryological basis for one of the most common anomalies in the eye of man.

One strain of Jackson Laboratory rats showed 100% persistence of an incomplete pupillary membrane throughout life and 100% persistence of the hyaloid artery for the first week after the eyelids opened. The pupillary membrane in the newly opened eyes changed from an almost complete, vascularized membrane to a few vessels at the age of 2 months. A similar but less pronounced situation was found in one strain of mice (Bagg alb. C). Here both the extent of the membrane and frequency of persistence was less. In contrast to the situation commonly seen in man, the persistent pupillary membranes in rats and mice are vascularized structures. They consist of one of twelve vascular loops derived from the second and third arcades of the anterior vascular capsule.

[1] Finney-Howell Fellow.

189. Some observations on the maturation of the histomorphology of human bone. Elbert B. RUTH, Department of Anatomy, University of Cincinnati, and Emil W. HAURY, Department of Physical Anthropology, University of Arizona.

The histomorphology of the femurs of a series of human fetuses, infants, children, young adults, and adults is described. The ages of the subjects range from the 3-month fetus to the 93-year-old adult. Most of the femurs are from the collection of American Indian skeletons in the Arizona State Museum. Ages given are only approximately as estimated by the physical anthropologist, with a few exceptions where the age is known. The development of the laminar type of bone characteristic of the fetus and infant is described, followed by the origin and development of Haversian canals between the first and second years. The characteristic circumferential and longitudinal vascular spaces of the fetus and infant are followed through their metamorphosis into the Haversian type of vascular channels, and these in turn are described as the histomorphology of the femur begins to approximate that of the adult at about the eighth year. The circumferential interlamellar canals and the relatively wide band of external circumferential lamellae that appears to be charcteristic of the bones of children and young adults apparently remains until the eighteenth year, becoming less complete or absent in older bones.

190. *An analysis of variations in the arteries and veins of the bronchopulmonary segments of 50 right upper lobes.* J. Gordon SCANNELL, Department of Anatomy, University of Minnesota. (Introduced by Edward A. Boyden.)

The 3 segmental arteries — apical (A^1), anterior (A^2) and posterior (A^3) — arise exclusively from the anterior side of the hilum in only 8% of specimens. In the remainder, the lobe is supplied from both the anterior and interlobar sides. Anterior arteries are grouped into a commonly bifurcate *truncus anterior* (88%) or into a *truncus superior* and a *truncus inferior* (12%). A^1 is always anterior in origin. A^2 and A^3 may be either anterior or interlobar (the "recurrent" and "ascending" arteries of Appleton) or both. A^2 is wholly recurrent in 58%, mixed in 42%; A^3 is wholly recurrent in 18%, ascending in 12% and mixed in 70%, but its ramus A^3a is wholly recurrent in 60%, and A^3b is wholly ascending in 56%.

Of the 3 veins — Apical-anterior (V^1), inferior (V^2) and posterior (V^3), the last is the most variable. In 50%. it is formed by confluence of interlobar and central divisions, but the posterior vein may be wholly central in position (28%) or wholly interlobar (16%). In the remaining 6%, the 2 divisions empty separately — the interlobar coursing either posterior to the pedicle of the middle lobe, to enter the superior pulmonary, or posterior to the pedicle of the inferior lobe to reach the inferior pulmonary vein. In 52% of cases the interlobar division receives a tributary from the superior segment of the lower lobe.

191. *The fundamental transverse arrangement of cross striae in myofibrils of striated muscle.* Carl Caskey SPEIDEL, School of Anatomy, University of Virginia.

Recently the "actin-myosin screw theory" of myofibril structure has been proposed by Szent-Gyorgyi and Gerendas ('45). It is claimed that as a striated muscle fibril is rotated, the cross striae shift in spiral fashion like the thread of a screw, the shift being equal to one sarcomere length as a fibril is rotated 180 degrees. Szent-Gyorgyi has elaborated a detailed theory of muscle contraction based on this concept.

To check this screw theory, I have examined an exceptionally favorable type of striated muscle from the pharynx of the sea-spider, Anoplodactylus lentus. A typical fiber is a short, slender, single-celled structure containing about a dozen coarse myofibrils. The sarcomeres are of large size, each measuring about 9 micra in length in resting condition.

A simple technique was devised which permitted very satisfactory observations on whole fibers as these were rotated various amounts from 0 to 360 degrees. The muscle nucleus afforded a convenient fixed landmark. Dozens of fibers were carefully examined, attention being focussed in each case on individual myofibrils as rotation proceeded.

These observations clearly showed that there was no spiral or screw-like shift of cross striae within individual myofibrils. On the contrary, throughout the rotation the transverse arrangement of Q, J, and Z discs within each myofibril was maintained at the same level.

Similar results were obtained with shrimp (Palaemonetes) muscle.

192. *Response of germinal epithelium of the mouse to thyroxin in the ovarian capsule.* Kathryn STEIN and Jewel QUIMBY, Department of Zoology, Mount Holyoke College.

A single injection of thyroxin (0.005%) into the left ovarian capsule in a series of 14 albino mice invariably caused inhibition of proliferation in the germinal epithelium within a period of 24 or 28 hours after injection. Controls consisted of the right ovaries of the experimental mice and 6 animals which were either untreated or received systemic injections. Colchicine was given to all animals to facilitate counting mitoses. The thyroxin was injected in sesame oil or in Ringer's solution; control right ovaries were treated with the corresponding vehicle minus the hormone. Sesame oil stimulates proliferation but to a much lesser degree when accompanied by thyroxin. Ringer's treated right ovaries showed approximately the same number of mitotic figures as were found in normal untreated animals. Immature females had the largest average number, estrous females the next, and diestrous the least. The decrease in proliferation in the germinal epithelium of the left ovaries due to the thyroxin injections was as high as 70% in individual mice, averaged around 50%, and was never less than 30% (2 animals). The effect of thyroxin is interpreted as due to the direct action of the hormone on the cells because of the short time interval following injection and the localization of the effect to the thyroxin treated ovary in each mouse.

192a. *Embryonic bone transplants in drill hole defects of adult long bones.* Charles STEINMAN, Department of Anatomy, Vanderbilt University. (Introduced by Sam L. Clark.)

To study the process of bone repair, and the influence of embryonic bone upon this process, 22 tibiae of adult rabbits were exposed, stripped of periosteum and a groove made in each. Into each of 14 defects were placed small bone fragments of 18-day rabbit embryos. Nothing was put in the remaining 8 defects.

The development of firm exostotic nodules was noted over each defect containing embryonic bone transplants but not in the controls.

Microscopic sections of the controls revealed that the defects were first filled by blood and fibrinous clot, which was replaced by fibrous and granulation tissue by the seventh day. New bone from periosteum and endosteum bridged the gap by the twenty-eighth day and then remodeling of the cortex took place.

Sections through the region of the defects with transplants revealed the absorption of the embryonic bone by the seventh day, and the presence of many centers of proliferating cartilage in the fibrous callus. By the twenty-first day masses of cartilage filled the defect, extended into marrow cavity, and protruded onto the surface. The cartilage was slowly replaced by new bone and the gap was bridged by seventieth day. Some cartilage was present after 150 days in the exostosis.

While bone defects are commonly repaired by intramembranous bone formation, those filled with embryonic bone, showed cartilage proliferation, and endochondral bone formation.

193. Primitive carotid intersegmental arteries and the formation of the basilar in rabbit embryos injected while alive. Leon H. STRONG, Department of Anatomy, University of Michigan.

When the primitive internal carotid of the first aortic arch stage reaches the brain at the diencephalon to anastomose with the primary superficial plexus of the brain, it forks into a rostral and dorsal branch. The dorsal may be considered the first intersegmental. Just caudal to this is a second intersegmental supplying the trigeminal ganglion. This forks into a caudal and rostral branch. Intersegmentals behind this do the same. A ventral line is formed by anastomosis on either side, a longitudinal trunk from diencephalon to the last caudal segment, the primitive arterial tract of Adamkiewicz. With the rapid enlargement of the brain the primitive carotids move lateral and dorsal. The bases of their intersegmentals are carried lateral. With the degeneration of the first 2 aortic arches, the interarch segments which persist back to the third arch, now the internal carotid, give off intersegmentals for some time before these atrophy and disappear. But those intersegmentals from the aorta, serially continuous caudally, do not. The cranial ends of the 2 arterial tracts, by anastomosis and enlargement form a basilar trunk. This basilar segment of the arterial tract is developed from anastomoses between the carotid intersegmentals only. The basilar is not an extension forward from the vertebral arteries as previously supposed.

194. Establishment of differences in susceptibility to audiogenic seizures of 5 endocrinic types of mice. E. M. VICARI, R. B. Jackson Memorial Laboratory.

Five inbred strains of mice previously reported as different in the histochemistry of their adrenal glands were observed with respect to their susceptibility to audiogenic seizures followed by death. These strains are: dilute brown, albino-A, extreme dilution, C3H, c57 black. Both sexes in equal numbers were used involving 571 individuals. Ages tested were from 15 to 60 days and 3 to 9 months. Audiogenic convulsions were brought on by ringing of an electric bell.

Results show that the dba and black strains were most susceptible at 30 days, 78% and 83% individuals had seizures of which 67% and 58% died. But the blacks at 35 days dropped to 8% with 90% deaths while the dba maintained its high susceptibility with 44.4% deaths. Neither strain showed any susceptibility at ages 3 to 9 months. The extreme dilution strain showed a high susceptibility at age 55 days with 83% seizures, 100% recovery; at 7 to 9 months 100% seizures and recovery. The albino A had no convulsions at any age under 3 months. At 5 months 28.5% had seizures of which all recovered. The C3H strain, all under 3 months tested, showed no convulsions whatsoever.

The dba and black strains are less susceptible with age while the extreme dilution and albino-A strains are more so with age. The C3H are unaffected.

195. *The rate of renewal in man of the water and sodium of the amniotic fluid as determined by tracer techniques.* G. J. VOSBURGH, D. B. COWIE, L. M. HELLMAN, W. S. WILDE, and L. B. FLEXNER, Department of Obstetrics, Johns Hopkins University and Hospital and the Departments of Terrestrial Magnetism and Embryology, Carnegie Institution of Washington.

The amniotic fluid was considered a relatively stagnant body fluid until the availability of isotopes permitted measurement of the rate of renewal of its constituent substances. Such measurements in the guinea pig showed that the water of the fluid is replaced about once an hour whereas the rate of replacement of sodium is about 50 times slower.[1]

We have measured water and sodium transfer to the human amniotic fluid using deuterium oxide and radioactive sodium as tracer substances. The observations with sodium were made on 20 women, and with water on 5 women on whom hysterotomies were performed. We have found that on the average 0.345 ml of water is transferred to a ml of amniotic fluid per hour; i.e. the water of human amniotic fluid is turned over once every 2.9 hours. The amniotic fluid of man contains 2.8 mg sodium per ml[2] and sodium is transferred at an average hourly rate of 0.137 mg per ml; hence the sodium of the fluid is turned over once every 20.5 hours. It is evident that the membranes which are traversed by substances reaching the amniotic fluid from the maternal circulation are far more permeable to water than to sodium and that the water of the amniotic fluid is replaced with astonishing rapidity.

[1] Flexner, L. B., and A. Gellhorn. Am. J. Physiol., *136*, 757, 1942.
[2] Needham, J., Chemical Embryology. Cambridge, England. 1931.

196. *On the staining of fat and fatty substances with azo dyes of the Sudan series in aqueous solution.* Harold L. WEATHERFORD, Department of Anatomy, Harvard Medical School.

It is well known that certain substances insoluble, or soluble with difficulty, in water can be dissolved in aqueous solutions of other substances. Neuberg has called this phenomenon hydrotropism — and supposes that a weak addition-compound is formed between the substance dissolved and the dissolving or hydrotropic substance. The ideas of Neuberg have been applied to several members of the Sudan series of azo dyes in an attempt to produce aqueous solutions that will stain fat and fatty substances in tissues. More than 80 potential hydrotropic agents have been tested and a large number of them, when compared with 70% alcohol are found superior as solvents for Sudan dyes. Four criteria are required of hydrotropic solutions of these dyes: (1) they should stain fat or fatty substances more clearly than a solution of the same dye in 70% alcohol; (2) there should be no "bleeding" of the dye into the tissue; (3) they should produce no deleterious effects on the tissue; (4) the methods finally recommended should be practical for daily use. The report now in preparation contains a full discussion of hydrotropism, the principles involved and their applicability.

197. Progress of studies designed to determine whether the fetal hypophysis produces hormones that influence development. L. J. WELLS, Department of Anatomy, University of Minnesota.

Fetal rats with placental circulation intact were deprived of their hypophyses by decapitation. Of the 17 males that survived for periods of 28 to 102 hours before severance of the umbilical cord near term, 5 were given a series of 5 subcutaneous injections of equine gonadotropin (Gonadogen). Injections were made twice daily. In autopsying littermate controls, they were decapitated and their headless bodies weighed. Stained serial sections of adrenals and of sex glands were studied microscopically and then used in determining the volume of these organs (paper-weight method).

While the data are too few to show whether the fetal hypophysis produces adrenotrophin, they suggest that it actually does. Thus in the 2 uninjected fetuses deprived of their hypophyses for the longest periods (102 hours, 77 hours), the volume of the adrenal, in cubic millimeters per gram of headless body weight, was less than half that of littermate controls (39%, 48%).

As to whether the fetal hypophysis produces gonadotrophin, the preliminary observations do not warrant a definite conclusion. The volume of the reproductive glands, when expressed as percentage of body weight, was similar to that of controls (testis, seminal vesicle, bulbo-urethral gland). However in all injected fetuses the interstitial cells of Leydig were much larger and far more numerous than those of ''hypophysectomized'' controls.

Preserved organs of 6 ''hypophysectomized'' female fetuses require additional study.

198. Reaction of damaged hepatic parenchyma of mice to vital staining. W. Lane WILLIAMS, Department of Anatomy, University of Minnesota.

Ten male albino mice received daily subcutaneous injections of 0.2 cm^3 of trypan blue (0.5% aqueous solution) and either 0.03 cm^3 of chloroform or 0.04 cm^3 of carbon tetrachloride for 4 to 7 days. The mixtures of the latter contained 1 part of carbon tetrachloride or chloroform in 4 parts of mineral oil.

The type of hepatic parenchymal injury characteristically produced by chloroform or carbon tetrachloride resulted. Many foci of liver cells showed cytoplasmic vacuolation and eosinophilia, and either nuclear pyknosis or absence of the nucleus. The cytoplasm of the hepatic epithelium in such areas of injury was diffusely stained or colored by trypan blue. The dye was present in these cells as an amorphous suffusion and not in the form of granules and particles common to macrophages. Identical methods of histological technique demonstrated no deposition of dye within morphologically normal hepatic epithelium either in these mice or in animals not treated with carbon tetrachloride or chloroform. The dye in the cytoplasm of damaged liver cells was easily removed by subjecting the sections to water or aqueous solutions of alcohol whereas the dye within macrophages was relatively resistant to the same procedures.

The findings demonstrate that necrotic and/or damaged liver cells react to parenterally administered dye in a manner similar to certain other epithelial and nonepithelial cells.

DEMONSTRATIONS

(The numbers indicate the order of the demonstrations on the program of the meeting.)

D 32. Follicular atresia in the ovaries of newts injected with diethylstilbestrol. A. Elizabeth ADAMS, Department of Zoology, Mount Holyoke College.

The ovaries of adult Triturus viridescens given total doses of 5.0–7.0 mg of diethylstilbestrol[1] in sesame oil intraperitoneally over a period of 78–108 days were compared with those of controls injected with sesame oil only. At autopsy, the ovaries of the stilbestrol-injected newts often looked cystic and oil-filled. Only non-yolked oocytes were present together with masses of tissue resulting from the resorption of heavily yolked oocytes. The ovaries of the oil-injected controls were noticeably smaller, but also often looked as if they contained oil. They consisted of small non-yolked oocytes and typical pigmented corpora atretica. The average ovarian weight in alcohol for the 10 stilbestrol-injected newts was 224.9 mg and for the 10 oil-injected controls, 42.1 mg.

Histological examination revealed oogonia and non-yolked oocytes in varying sizes and numbers in the ovaries of both groups. The most striking difference between the groups was the appearance of the corpora atretica. Those in stilbestrol-injected newts were in general very much larger than those in controls and consisted largely of epithelioid cells, containing varying amounts of pigment and were slightly vascularized. Yolk granules were sometimes present, depending on the age of the corpus atreticum. In the oil-injected controls, the corpora atretica showed the typical vacuolated, pigmented and well-vascularized condition and the size characteristic of the urodele yolked oocyte undergoing resorption after the spawning season, or hypophysectomy or starvation.

[1] Diethylstilbestrol supplied through the courtesy of Dr. C. W. Sondern, George A. Breon and Co., Inc., Kansas City, Missouri.

D 38. Staining reactions with the Feulgen and dinitrophenylhydrazine reagents. S. ALBERT, Department of Anatomy, McGill University. (Introduced by C. P. Leblond.)

The distribution of the Feulgen and dinitrophenylhydrazine reactions was examined in formalin-treated normal and cancerous tissues of mice and rats of various ages. These techniques enabled a cytological study of these reactions which was especially clearcut in the case of the dinitropheylhydrazine reaction. The effects of strain and age differences will be especially emphasized.

D 27. The transparent chamber technique in the mouse in the study of vascular physiology and tissue growth. Glenn H. ALGIRE, National Cancer Institute, National Institute of Health, United States Public Health Service, Bethesda.

The transparent chamber technique as adapted to the mouse is presented by means of microscopic demonstration of the living tissues, photographs and models illustrating the technique, graphs, and photomicrographs to show applications in quantitative studies on vascular physiology and tissue growth, with special reference to cancer.

D 30. Dermal characteristics in the presence of basal cell epitheliomas of human beings and after regression following treatment with tissue extracts. Joseph C. AMERSBACH, Elsie M. WALTER, and George S. SPERTI, Skin and Cancer Unit of the New York Post-graduate School and Hospital and the Institutum Divi Thomae. (Introduced by Rucker Cleveland.)

Biopsy specimens obtained from 47 patients with basal cell epithelioma prior to and following treatment with human and beef spleen and liver extracts were fixed in Bouin's fluid and stained with hematoxylin and eosin. The lesions ranged as large as 4 cm in longest dimension. Some were as recent as 1 month in development, others had existed for several years. The occurred chiefly on the cheek, forehead, nose, and base of the scalp. The age range of the patients was from 33 to 82 years. Intradermal infiltrations using 0.5 to 1.0 cm^3 of tissue extracts (1.0 $cm^3 \cong$ to 1–15 gm tissue) were made weekly about the lesions, as few as 3 and as many as 20 treatments being required for complete regression. The detailed clinical report appeared in the "Archives of Dermatology and Syphilology," 54, 119, August, 1946, and a second is in press in the same Journal.

Study of the sections revealed a variety of histologic characteristics including some basophilia of the connective tissue components. There was an increased number of various kinds of connective tissue cells in specimens obtained before as well as after treatment, when these were compared with sections of normal skin fixed and stained in the same manner. The nature of the process of regression is undetermined.

D 42. Leukocyte migration into and through the epithelium of the trachea in the mouse.[1] Warren ANDREW and Marjorie R. BURNS, Department of Anatomy, Southwestern Medical College.

Slides and photomicrographs are presented to illustrate some of the phenomena of migration of leukocytes into and through the epithelium of the trachea. It has long been known that such a phenomenon occurs in the normal animal but little attention has been paid to the details. A variety of cell types are seen to enter the epithelium in the present material. They are primarily lymphocytes, but many polymorphonuclear neutrophils are present, in contrast to such migration in the intestine, where the cells are practically entirely lymphocytes. The presence of large vacuoles about cells of whatever type which have entered the epithelial layer is a conspicuous feature. While most of the leukocytes here are intercellular in position, they frequently cause a very great deformation of the epithelial cells and figures are seen suggestive of an intracellular penetration in some cases. The leukocytes frequently are greatly elongated and may be mistaken for modified epithelial cells in sections. Changes in lymphocytes such as pycnosis and fragmentation, common in the intestinal epithelium, are rare in the trachea but do occur in some instances.

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[1] This study was supported by grants from the Winthrop Chemical Company and the John and Mary R. Markle Foundation.

D 18. A simple method for obtaining photographic and cinematographic records of embryos and other small objects. Leonard-F. BÉLANGER, Department of Histology and Embryology, School of Medicine, University of Ottawa, Canada.

The necessity of securing records of the general aspect of human embryos which are to be sectioned serially, has prompted the development of the following apparatus. A Cine-Kodak Titler is fixed to a wooden support which permits its use in a mobile vertical position. For photography, a 35-mm camera is secured to the Titler with a mounting block. Records can be obtained this way of objects up to 60 mm in length and 20 mm in width and thickness. The object is easily framed and photographed along with 2 rulers at right angles to one another and a strip of paper bearing the necessary data. The film strip can be studied with a microfilm reader; it can be projected or an enlarged print, the size of a filing card, can be made of it, permitting accurate longitudinal and cross measurements. For cinematographic studies of the living chick, frog, fish and invertebrate embryos, a standard movie camera is substituted for the still camera and its support in the apparatus. Experiments can be performed while the embryo is easily maintained within the field of the instrument.

D 17. The developmental stages and anatomy of Xenobus laevis, African horned toad. T. W. M. CAMERON and Mrs. CAMERON, Institute of Parasitology, McGill University. (Introduced by S. M. Friedman.)

This toad was introduced to science by Professor Hogben and his colleagues in South Africa. It is bred extensively for use in pregnancy diagnosis and hormone research.

This demonstration shows the various stages of development and the anatomy of the living animal from the egg to the 2-year-old sexually mature toad.

D 40. Microscopic demonstration of neurosomes in skeletal muscle following atrophy of disuse. Eben J. CAREY, Leo C. MASSOPUST, Walter ZEIT, and Eugene HAUSHALTER, Department of Anatomy, Marquette University School of Medicine.

(See abstract of paper from platform.)

D 28. A comparison of growth and differentiation of avian femur rudiments on clots formed from embryo extract and plasma with those grown on clots of partially or completely known constituents. Esther CARPENTER and Ursula ROTHFELS, Department of Zoology, Smith College. (Introduced by Esther Carpenter.)

As controls, femur rudiments of 6-day chick embryos were grown by the watch glass method (Fell, '29) on the surface of clots composed of equal amounts of adult fowl plasma and 10% embryo extract from 10-day chick embryos.

For the experimental cultures the following combinations were tried: (1) White's ('46) synthetic medium for animal tissues to which washed agar had been added to produce a concentration of 0.75%; (2) a clot composed of bovine fibrinogen (Armour), bovine thrombin (Upjohn) and White's synthetic medium; and (3) a clot composed of fibrinogen, thrombin and embryo extract.

In cultures in which the synthetic nutrient replaced the embryo extract during the first week there was no growth in length, pycnotic nuclei were numerous and the femur rudiments were slightly opaque. After 2 weeks fewer pycnotic nuclei were present, the rudiment had increased in length and improved somewhat in shape. No bone was observed.

Femur rudiments on clots containing embryo extract but with the plasma replaced by bovine thrombin and fibrinogen, grew and underwent morphogenesis as well as those with plasma. The bone deposited was thinner, however, and the cartilage cells less vacuolated.

D 51. *A cell type possibly responsible for facultative salt excretion in the gills of Fundulus heteroclitus (a euryhaline fish).* D. Eugene COPELAND, Biology Department, Brown University. (Introduced by A. B. Dawson.)

A cell type is demonstrated that shows consistent cytological pictures, one specific to fresh water and the other to sea water (or concentrated NaCl). The cells are located in the gill filament and elongated in most cases to reach from the circulation to the surface of the filament. Regaud preparations show a rich profusion of apparent mitochondria. Prolonged osmication (Mann-Kopsch procedure for 1 month in 2% osmic tetroxide at room temperature) causes a specific blacking of these cells that is strongly retained after all other elements have been decolorized. Behavior is gauged by the presence of a vesical at the distal end of the cell in sea water and absence of the vesicle in fresh water. Mucicarmine and thionine tests are negative for mucus in this cell type.

D 36. *Selective impregnation of cells of the basophilic series of the anterior pituitary gland of the rat and chicken: their early recognition in the fetal rat and embryonic chicken.* Alden B. DAWSON, The Biological Laboratories, Harvard University.

In fully differentiated anterior pituitary glands of the chicken prepared by Bodian's protargol method without gold toning, typical basophiles and smaller cells, interpreted as partially differentiated basophiles, are deep black. Acidophiles of both types are reddish-brown and inactive chromophobes are bright yellow. In the rat only regular basophiles and less differentiated basophiles react strongly with silver. In all these cells of the basophilic series the elements reacting with silver appear as discrete spherical bodies and do not seem to be identical with the diffuse type of granulation demonstrated in basophiles with anilin blue. The history of the cells, blackened specifically with protargol, has been followed in progressively younger animals to determine the time of their first appearance. They were first found in the embryonic chick on the eighth day of incubation and in the fetal rat on the sixteenth day of gestation. It has not been possible to identify these argentophile cells at such early ages by the selective staining methods usually applied to the pituitary gland. Thyroidectomy cells of the chick, induced by thiouracil treatment, have little affinity for silver but a ring of blackened granules frequently surrounds the negative image of the Golgi apparatus. In the rat modified basophiles — thyroidectomy cells (thiouracil treatment) and castration cells — gradually lose their ability to react with protargol.

D 33. A histological study of eye abnormalities in the C57 black strain of mice. Patricia DOUGLASS and W. L. RUSSELL, Jackson Memorial Laboratory. (Introduced by Katharine Hummel.)

Non-genetic eye abnormalities occurring spontaneously in the C57 black strain range from corneal opacity to eyelessness. The abnormalities may be present in both eyes or only one, much more commonly the right, and are more frequent in females than in males.

A considerable variation in kind and degree of defects is shown by histological examination. The lower grades of abnormality are characterized by opacities of varying size, waviness and invagination of retinal layers sometimes accompanied by proliferation of the cells in one or more layers with or without invasion into adjacent layers. Fusion of the cornea with the choroid or lens, and of the retina with the choroid often occurs, also proliferation and disorganization of the choroid. In more extreme abnormalities the eyeballs are very small and consist of partially differentiated but disorganized connective, pigmented and retinal tissues imbedded in muscle. The most extreme cases are characterized by the presence of only muscle and lacrimal gland in the orbit.

The fact that the whole range of abnormalities fits into a single pattern of defects suggests that the C57 black strain is prone to irregularity in one particular phase of eye development.

D 47. The anatomical distribution of the ductules and islets of the rabbit's pancreas in alloxan diabetes. G. Lyman DUFF, Department of Pathology, McGill University. (Introduced by B. Kropp.)

The demonstration of the ductular ramifications in the normal rabbit pancreas is both difficult and unsatisfactory. However, it is possible to obtain a selective hydropic change of the epithelial cells of the ductules and islets of Langerhans by experimental means which renders their anatomical distribution and relationships apparent.

Moderate to extreme hydropic degeneration of the pancreatic ductules and islets was observed in rabbits rendered diabetic by alloxan when the diabetic state had persisted for several months. Such changes have not hitherto been described in alloxan diabetes in the rabbit. The earliest appearance of these alterations was at the end of 45 days after the injection of alloxan and hydropic changes were never absent after 90 days providing that the average fasting blood sugar level during the experiment had been 303 mg % or higher. The hydropic state of the ductules and islets persisted without histological evidence of any further change in association with more or less severe diabetes lasting for periods up to 1 year. The alpha cells of the islets remained histologically normal but appeared to be increased in number. Preliminary observations demonstrated that the hydropic degeneration of both ductules and islets is reversible by adequate treatment of the diabetic state with insulin.

D 21. Two unusual anomalies in the same cadaver (Negro male, age 85 years). John C. FINERTY and Mildred TROTTER, Department of Anatomy, Washington University.

I. Persistent right ischiatic artery. The anomalous artery is a direct continuation of the anterior division of the hypogastric artery and passes through the greater sciatic foramen closely associated with the tibial division of the sciatic nerve. After sending branches to posterior thigh muscles the vessel extends inferiorly as the popliteal artery with normal course and relations. A typical, but small, femoral artery is present and anastomoses with the popliteal artery.

II. Hourglass bladder. The urinary bladder is divided by a horizontal septum into 2 chambers of approximately equal capacity. The upper chamber is somewhat to the right of the midline and posterior to the lower chamber, which is in the usual position in the pelvis; these are connected by a centrally located, circular orifice 1 cm in diameter. The ureters terminate in the lower chamber on the extremities of a well-marked transverse ridge. The right ureter, larger than the left, traverses part of the septum. The remaining angle of the vesical trigone is represented by a typical urethral orifice.

D 34. The effect of oral administration of vitamin A on the expression of the recessive gene "rhino" in the mouse. F. Clarke FRASER, Department of Genetics, McGill University. (Introduced by J. C. B. Grant.)

The first visible abnormality in the skins of homozygous rhino mice is a hyperkeratosis of the epidermis and follicle neck walls at the time when active growth of the first hair generation ceases (14 days), which is associated with a widening of the hair canals and a subsequent falling out of the hair coat. This is followed by the development of cysts in the hair-canals, the sebaceous glands and the follicle-ends, a thickening and wrinkling of the epidermis and complete suppression of further hair growth.

The skins of mice fed massive oral doses of vitamin A (of the order of 500–1000 International units every other day) for several months are much thinner and less wrinkled than those of litter-mate controls. Preliminary histological examination shows a reduction in the number and size of the sebaceous gland and follicle end cysts, and signs of abortive attempts at new hair growth in the treated mice.

D 29. Methods for the experimental investigation of cardiovascular-renal dynamics in the rat.[1] S. M. FRIEDMAN and C. L. FRIEDMAN, Department of Anatomy, McGill University.

a. Previously described methods for the study of renal function in the rat have been simplified. The present procedure makes use of sodium-p-amino hippurate as a test substance to measure tubular excretion and renal blood flow, while inulin is used to estimate glomerular filtration. The demonstration consists of graphs and charts illustrating the development of the method and its final form.

b. A method for the direct reading of heart rate and heart sounds on the Cathode Ray tube is demonstrated.

[1] This work was supported by a grant from the Life Insurance Medical Research Fund.

D 4. Boutons terminaux. W. C. GIBSON, Montreal Neurological Institute. (Introduced by C. P. Martin.)

Synaptic endings in the spinal cord, sympathetic nervous system, cerebellum, basal ganglia and cerebrum will be demonstrated by slides stained by block impregnation or frozen silver methods. At present it appears that the *boutons terminaux* stain with much greater facility in the spinal cord, but the development of modifications for cortical material may show the same richness of synaptic endings in that area.

The method of preference for staining synaptic endings would seem to be Rio-Hortega's double impregnation technique, employing a preliminary staining with dilute silver nitrate followed by silver carbonate. The chief practical use of the method lies in determining the precise point at which a tract ends, as shown by its degenerating end-feet situated on the surface of the next neuron in the circuit. Criteria for determining the number of days following tract section, at which degeneration is best seen, have still to be worked out for cortex, cerebellum and basal nuclei.

D 9. An hypophysial portal system in certain amphibia, and nerve endings associated with neurohypophysial vessels. J. D. GREEN, Department of Anatomy, Wayne University. (Introduced by Gordon H. Scott.)

The hypophysis of Rana catesbiana is supplied by 2 large arteries from the main divisions of the internal carotid. These pass backward on the under surface of the hypothalamus in the pia-arachnoid. Just before they reach the gland each subdivides into a leash of arterioles and penetrates a small thickening of the neural stalk, which projects between the 2 bulging lobes of the pars intermedia, to form a rich capillary plexus within it. Blood issuing from this plexus is collected into from 1 to 4 large vessels passing backward on the under surface of the adenohypophysis, which is supplied by capillaries arising from them. Venous blood is collected into a large skein of veins issuing from the lateral aspect of the gland and discharging into the plexus of the saccus endolymphaticus. In Amblystoma tigrinum a somewhat similar arrangment occurs although the effluent channels from the neural stalk are diffuse. Though a hypophysio-portal system is not anatomically very apparent in this form, as stated by Roofe and by Craigie, a similar functional arrangement seems to exist. Similar observations have been made on Triturus torosus and Necturus. India ink preparations, dissections, and illustrations are demonstrated, and nerve endings associated with the neurohypophysial vessels are also shown.

D 31. Cytochemical evidence for the cessation of hormone production in the zona glomerulosa of the adrenal cortex during treatment with desoxycorticosterone. Roy O. GREEP, Harvard School of Dental Medicine, and Helen Wendler DEANE, Department of Anatomy, Harvard Medical School.

Ketosteroids (sudanophilic, Schiff-positive, acetone-soluble, birefringent, and fluorescent in formalin-fixed frozen sections) occur in both the zona glomerulosa and the zona fasciculata of the rat's adrenal cortex. Previous experiments showed that the ketosteroids in the fasciculata might increase or decrease independently of any change of those in the glomerulosa. The former experiments suggested that the fasciculata contains corticosterone-like hormones, whereas the glomerulosa possesses the desoxycorticosterone-like steroids.

Two mg desoxycorticosterone-acetate (Ciba) was injected daily into both intact and hypophysectomized male rats. In both groups this treatment induced a slight atrophy of the adrenal but no alteration in the size of the thymus. Cytochemically the fasciculata was unchanged. In the glomerulosa, however, the ketosteroid reactions diminished gradually, with a detectable loss of fluorescence by 7 days, a reduction in sudanophilia and birefringence by 12 days, and a total loss of all the reactions employed by 28 days.

These results support the view that hormones concerned with salt balance, such as desoxycorticosterone, are produced in the zona glomerulosa of the adrenal cortex. When this type of hormone is provided in adequate quantity, the zone undergoes a disuse atrophy and ceases forming its normal secretion. A morphological duality of the cortex thus parallels the physiological and biochemical duality already recognized.

D 43. Comparative pulmonic alveolar size in man, cat, rabbit, monkey, guinea pig, goat, dog, baboon, rat and mouse.[1] W. Stanley HARTROFT, Banting and Best Department of Medical Research, University of Toronto. (Introduced by Charles C. Macklin.)

Microsections and photomicrographs illustrating differences in pulmonic alveolar size of man and 9 species of laboratory animals as seen in 25 μ paraffin microsections will be shown. The lungs were preserved in expansion either by vascular perfusion with fixative while in the unopened thorax, or by bronchial filling with the same, the degree of expansion being controlled by comparing certain pulmonic measurements with the corresponding pleural cavity dimensions. A wide variation in alveolar size was found, the largest being in man, and the others diminishing in order as in the title. Methods of measuring the alveoli so preserved will be demonstrated. Graphic comparisons of the arithmetic means of several thousand measurements will be presented. A summary of 50,400 measurements on 25,200 alveolar outlines will be given.

[1] This work was done in the Department of Histology and Embryology of the University of Western Ontario Faculty of Medicine, London, Canada.

D 37. A method of studying dry smears of blood with phase microscopy followed by staining techniques. Oliver P. JONES, Department of Anatomy, University of Buffalo and Oscar W. RICHARDS, Research Division, American Optical Co., Scientific Instrument Division.

Unstained dry smears of embryonic blood on cover slips were inverted on a drop or two of either Baker's formol-calcium or 10% formalin and ringed with petrolatum. After studying cells with phase microscopy, it is possible to mark them for restudy with a bright field microscope following Wright's stain or Sudan black. Although bright and dark contrast phase microscopy may be used with Sudan black preparations, the latter combination brings out more detail than either technique alone.

D 22. Examples of anatomical specimens mounted in plastics. Otto F. KAMPMEIER and Thomas N. HAVILAND, Department of Anatomy, College of Medicine, University of Illinois.

Different kinds of anatomical preparations, such as dry and wet specimens, cleared specimens, celluloid, vinylite and Wood's metal corrosions, injections, dissections, gross sections, stained sections, whole mounts of fetuses, etc. are shown embedded in Castolite and Plexiglass.

D 35. Specialized argyrophilic (argentaffine?) cells in the anal canal of the rhesus monkey. Hadley KIRKMAN and Irene OW, Department of Anatomy, Stanford University.

As a normal epithelial component of the alimentary tract argentaffine cells are well known to be very widely distributed through all classes of vertebrates. They have been described repeatedly in simple columnar epithelia of all parts of the stomach and intestine, in the gall bladder, the pancreatic duct and the glands of Brunner. Reports of their presence in stratified epithelia, however, are rare.

In the stratified squamous epithelium of the ano-rectal junction there occurs, in 4 rhesus monkeys examined, a circular band of slender, bipolar cells. The cytoplasm of these cells contains argyrophilic granules similar, except for lack of polarized distribution, to those found in typical argentaffine cells. In many of these cells one or more fine, intracytoplasmic, argyrophilic fibrillae are recognizable, as is true also for many typical argentaffine cells of monkeys. While the cells are intimately related to fibers of the mucosal nerve plexus the relationship may be one of either continuity or contiguity. Many, and possibly all, of the cells appear to extend throughout the entire thickness of the epithelium. In the simple epithelium of the rectum a few transitions between them and typical argentaffine cells have been observed.

They are not the same as the tall, slender cells frequently observed in stratified epithelia elsewhere. They are absent in rats and hamsters and at cardiac-esophageal and cervico-vaginal junctions.

D 11. Reconstructions showing topography of fibers in medullary center of the human cerebellum. Wendell J. S. KRIEG, Institute of Neurology, Northwestern University Medical School.

Eight graphic reconstructions are demonstrated, representing the topography of fibers in the medullary center of the human cerebellum.

The appearance of the fibrous core of the cerebellum after removal of the cortex and the secondary folia is first presented in medial and oblique lateral views.

Incidentally, this is a useful method for visualizing the lobules and relation of vermis to lateral lobe folia. Then, representing the crest of the principal folia as segments of hoops, the course and relations of sample fibers are depicted stereographically, at first in their full extent, then cut at various levels to show other features.

Arcuate fibers are in 4 groups: A, a sagittal group connecting folia in the medial part of the vermis and extending a short way laterally in the superior semilunar lobule; B, a posterior group which forms an extensive connecting system between folia of the inferior semilunar, slender, and biventral lobules and tonsil; C, a system connecting lobules of pyramis and uvula in sagittal plane; D, connections between tonsil and posterior vermis.

Fibers from the vermis run forward in the anterior medullary velum. The medial ends of the semilunar lobules send fibers forward medial to the brachium conjunctivum and blending with it.

(See abstract of paper read by title for brachium pontis and restiform body.)

D 24. An appendix epiploica resembling a vermiform appendix. Homer B. LATIMER and Alfred H. HINSHAW, Department of Anatomy, University of Kansas.

This unusual appendix epiploica was first mistaken for the vermiform process by the medical students who were dissecting the cadaver. Later, a true vermiform process was found in the retrocaecal and retroperitoneal position.

In all of the texts on anatomy and on surgical anatomy consulted, no clinical significance was attached to these diverticulae. A search of the clinical journals showed that they are rather frequently the cause of sufficient abdominal pain to warrant operation. In none of the cases reported was the cause of the pain correctly diagnosed before operation.

This epiploic appendage was found in a 64-year-old, colored male who had died of heart disease. It was normal, not having undergone torsion which usually causes the necrosis of the contained fat. It was 3.5 cm in length and 7 mm in diameter. It was attached to the distal end of the caecum and lay parallel to, and slightly attached to the terminal part of the ilium. Because of its size, shape and location, it seems that during a laparotomy, if a similar condition should be found, the appendix epiploica might be mistaken for a true vermiform appendix.

Many cases of torsion of the epiploic appendages are reported in the literature and it does seem that a little more attention should be given to these epiploic appendages in the text books.

D 8. A new method for staining nerve fibers and endings using the Schiff reaction. H. M. LIANG, Department of Anatomy, Washington University. (Introduced by E. V. Cowdry.)

This is a simple and reliable method for staining nerve fibers and their endings of sensory, motor, cerebro-spinal or autonomic origin in whole mounts or sections. The staining principle is based on the assumption that certain chemical constituents of nerves (probably aldehyde in nature) will give a positive reaction with

Schiff's reagent. This premise was later sustained by actual observations. The procedure is a follows:

A. *Whole mount*

1. Fix tissue in 1% acetic acid or formic acid for 1 hr.
2. Wash in SO_2 water, 2 changes.
3. Transfer tissue into Schiff's leucobase for 1–3 hrs. or until the nerves are deep purple in color.
4. Wash thoroughly in several changes of SO_2 water.
5. Rinse with dist. water.
6. Counterstain in 1% methylgreen aq. sol. for 15 minutes.
7. Dehydrate, clear and mount.

B. *Section*

1–5. As in the above.

6. Embed, cut and remove paraffin as usual.
7. Counterstain in 1% methylgreen aq. sol. for 5 minutes.
8. Dehydrate rapidly through graded alc.
9. Clear and mount.

All glasswares and reagents should be Schiff negative.

D 44. *Detached respiratory squames from the lung of the albino mouse.* Charles C. MACKLIN, The University of Western Ontario Faculty of Medicine, London, Canada.

(See abstract of paper presented from platform.)

D 7. *Certain aspects of peripheral ganglion cells in Fundulus.* Samuel R. MAGRUDER, Department of Anatomy, Tufts College Medical School.

Histological preparations of peripheral ganglion cells of Fundulus prepared by the Nonidez method are presented. The ganglia studied are in the area of the anterior part of the stomach and liver, and are usually composed of large and small cells arranged in an irregular series along a nerve trunk. Many of the large cells contain complex polymorphic nuclei. Some nuclei are ring-shaped, some irregular elongate with folded nuclear membrane while some are five-lobed figures containing 8 or more nucleoli in addition to round droplet-like ''bodies.'' Others contain 2 or 3 nuclei connected by thin strands of nuclear material. In addition there are a few cells which contain 3 to 5 nuclei between which no connection can be seen. These are always comparatively small and have never been seen to contain droplet-like bodies. General cell appearance, nuclear configuration and the presence of droplet-like bodies in the nucleus cause these cells to resemble those of the central nervous system described by Speidel ('22), Palay ('43) and others. Palay interpreted the characteristics as being evidence of nuclear secretion. However, only rarely has evidence been presented which indicated possible neurosecretory activity in peripheral ganglion cells (Lennette and Scharrer, '46). The preparations demonstrated here may possibly indicate neurosecretory processes although a complete secretory cycle has not been established.

D 5. Nerves of human dura mater. C. P. MARTIN, F. L. McNAUGHTON and C. L. LI, Montreal Neurological Institute.

The nerve supply of the dura mater was studied by anatomists in the nineteenth century, but little attention has been given to the dural nerves in recent years, though these nerves are undoubtedly of great importance in the understanding of head pain.

Full-thickness preparations of dura mater from middle and posterior fossae, tentorium, and falx, will be demonstrated, stained by silver nitrate and cleared in oil of wintergreen. Full details of the nerves in dura of the posterior fossa remain to be worked out. The remainder of the cranial dura receives its chief supply from the trigeminal nerve. The main dural branches can be followed grossly in these preparations, and compared with the diagrams of dural innervation.

D 12. Brain-modelling as a method for teaching advanced neuroanatomy. F. L. McNAUGHTON and John LORD, Montreal Neurological Institute. (Introduced by N. L. Hoerr.)

The brain modelling method of Adolf Meyer and Louis Hausman has been used at the Neurological Institute as the basis for the teaching of advanced neuroanatomy for Graduate Fellows, and as an elective course for upper year medical students interested in Neurology and Neurosurgery.

Reconstruction of the model must be based on the studies and observations of the student himself and this idea is stressed throughout. As source material, serial sections stained by Weigert-Pal, Nissl and Bodian methods are provided to supplement the student's own brain dissection, which includes fiber dissections of brain stem and cerebellar peduncles (Rasmussen). Tract degenerations demonstrated in human material, and experimental animal material are available for detailed study, and a bibliography of important articles introduces him to some of the certainties and uncertainties of neuroanatomy.

Lecture-demonstrations are arranged to fit in with the schedule of the modelling class, and in future will be combined with lecture-demonstrations in Neurophysiology by the Institute Physiologist.

By this pleasant method, the student provides himself with proven facts and a dynamic approach to the structure of the nervous system and, at the same time, prepares himself to proceed with original work.

The course occupies 2 evenings weekly for a 4-month period.

D 19. The embalming of fetuses and infants through the superior sagittal sinus. James A. MILLER, Department of Anatomy, Emory University School of Dentistry.

This technique for embalming fetuses, used for the past 3 years, is presented at this time because of the current nation-wide shortage of anatomical material and the possibility of using fetuses for certain types of dissections.

Studies by Tobler ('15) indicated that a needle inserted in the posterior angle of the frontal fontanelle to a depth of 3 to 6 mm will enter but not pass through the superior sagittal sinus. Batson's experiments ('40) showed that the vertebral veins provide a set of valveless venous channels connecting all parts of the body including the dural sinus system. Using the superior sagittal sinus as a portal of entrance of embalming fluid and the cut umbilical vessels for exit of blood, full term fetuses have been completely filled in less than 2 minutes at 5 lbs. pressure. Histological preparations from these fetuses indicate excellent penetration of fluid into all organs.

The only essential piece of equipment is an hypodermic needle of large caliber with an adjustable holder which prevents penetration to a depth greater than 5 mm, controls the angle of penetration and has a slot for an adjustable strap passing under the fetal chin. The demonstration will consist of the needle holder, photographs of fetuses from 4 months on embalmed with the technique, and histological preparations made from fetuses embalmed in this manner.

D 23. Electromyographic recordings from the m. biceps brachii during movements of the radioulnar, elbow, and shoulder joints. O. A. MORTENSEN, W. E. SULLIVAN and Meryl MILES, Department of Anatomy, University of Wisconsin.

The myograms were made with a 6-channel ink writing electromyograph having a paper speed of 3 cm a second. A pair of small surface electrodes was fixed to the skin (electrode past and celloidin) over each head of the muscle. Normal voluntary movements of the radioulnar, elbow, and shoulder joints were made by a subject in the sitting position, at average speed, with and without added load.

The electrical potentials were recorded when the biceps was acting to accelerate or decelerate a movement and when it was influencing motion as a topographical antagonist.

Supination of the radioulnar joint, flexion of the elbow joint, and flexion, abduction, adduction, and rotation at the shoulder joint represent acceleration.

Extension of the elbow, with gravity acting as the accelerating force, was used to illustrate deceleration.

Pronation of the forearm and active extension of the elbow were the movements used to illustrate the activity of the biceps when the muscle was acting as a topographical antagonist.

The records show variations in the amplitude of the potentials during the movements described. There are also variations which are the result of changes in the position of ''accessory joints,'' load, and greater subjective effort.

D 20. Genetic growth differences as factors in the determination of bilateral symmetry in the rabbit. M. A. NACE and P. B. SAWIN, Department of Biology, Brown University.

The origins of the paired ilio-lumbar arteries in most races of rabbits, including races III and X, tend to be asymmetrical. The artery on the left side arises anterior to that on the right in approximately 60–70% of cases with 2–10% of the reversed type. Race V, in contrast, tends to have more symmetrical individuals and approximately 19% of the reversed asymmetrical type. The hybrid, F_2 and backcross generations between these races reveal the fact that the relative amount and direction of asymmetry appearing in these generations is correlated with the relative mean antero-posterior level of the origins of these arteries. Thus the asymmetry appears to be determined by the relationship of the same growth forces, as have been shown in previous studies to influence the positions of the thoraco-lumbar and lumbo-sacral borders, and of the ventral spinous processes.

D 10. Sectional atlas of the brain of the adult albino rat. Wendell NICKELL and J. H. GRAVES, Institute of Neurology, Northwestern University Medical School. (Introduced by Wendell J. S. Krieg.)

A detailed cross-sectional atlas of the adult rat brain, consisting of 30 pairs of drawings, is presented. Alternate drawings from sections stained by the Weil method and with thionin were made. Additional illustrative material indicating the plane of section is included.

D 25. Developmental topographic anatomy of the infant thorax. Gustave J. NOBACK, Department of Anatomy, Cornell University Medical College.

Reconstruction by orthoscopic drawing of the pleural reflections, diaphragm, lungs, pericardium, heart and its valves, thymus, great vessels, etc., as projected against the thoracic cage and the vertebral column. The ages of the specimens utilized are, newborn, 10 days, 2 months, 3 months, 8 months, 11 months, 1 year, 1 year and 1 week, 15 months, and 2 years.

D 2. Changes in fiber tracts associated with inclusion bodies in nerve cells. James W. PAPEZ, Zoology Department Cornell University.

Inclusion bodies in nerve cells provoke changes in axon and sheath cells (oligodenendroglia). Sheath cells produce mucoid droplets found crowded between fibers of fiber tracts and at terminals of tracts.

Serum exudates from small blood vessels appear to be allergic reactions toward the mucoid bodies or other material. Patches of exudate may be widely disseminated in the c.n.s., but when in fiber tracts they lead to sclerosis. These exudates of serum (globulin), globular or racemose in shape, move and surround neighboring mucoid bodies, and form large laminated bodies. When a patch of such exudate spreads through tract of nerve fibers it blocks lymph spaces. Axons, myelin and sheath cells are engulfed and lymph supply is blocked.

Patches of exudate produce sclerosis: (1) myelin is removed by scavenger cells, (2) axons break up and disappear, and (3) sheath cells (oligodendroglia) give rise to dense fibers — the final sclerotic patch. Occlusion of small blood vessels is often seen in or near the patch, and sometimes an organized thrombus. Primary demyelination is another feature of affected tracts but depends to a larger measure on other factors or patches in some other region of the tract.

For inclusion bodies in nerve cells see abstract in Anatomical Record, suppl., *88:* 34, April, 1944.

D 6. Crossed fibers in the facial nerve in human embryos and fetuses. Anthony A. PEARSON, Department of Anatomy, University of Oregon Medical School.

(See abstract of paper from platform.)

D 39. Autographic studies of the bony tissues with radioactive phosphorus and of the thyroid with radioactive iodine. W. PERCIVAL, D. FINDLAY, J. GROSS and C. P. LEBLOND, Department of Anatomy, McGill University.

The distribution of radioactive phosphorus in the bony tissues of newborn animals was examined by several autographic techniques and was found to extend to all bones, ossification centers and tooth germs. This distribution was found to be completed as early as 1 hour after injection and to coincide with the distribution of alkaline phosphatase and of von Kossa's reaction.

The distribution of radioactive iodine was examined in the thyroids of animals in different experimental conditions (variable dietary iodine, stimulation with thyrotropic hormone, etc.:). In the active glands, the passage of iodine through the cells into an organic combination in the colloid is so rapid that, at no time after injection, is there enough iodine in the cells to make it visible by the autographic method. In resting glands, iodine occasionally may be found in the cells, especially soon after injection.

D 13. A brain stem model for teaching purposes. G. L. RASMUSSEN and S. PIAZZA, Department of Anatomy, University of Buffalo.

This model is intended to help the student visualize the continuity and the topographical relationships of the principal nervous pathways — not only in respect to neighboring structures of certain levels but also to the exterior of the brain stem itself. The various levels are of plexiglass on which are outlined the fasciculi, tracts, nuclei, etc. and they fit within a large half shell model of the brain stem. Individual neurons are fashioned to simulate their essential structure and to permit easy synaptic hook-up and rearrangement in order to demonstrate any particular functional system under study.

D 26. Spiral arteries in the ovary: injection-corrosion preparations. S. R. M. REYNOLDS, Department of Embryology, Carnegie Institution of Washington.

The arterial blood supply of the ovary in the rabbit is derived from 1 or 2 spiral arteries lying along the hilus of the ovary. The spiral arteries stem from the ovarian artery at the caudal pole of the ovary. The result of this arrangement is speculative, but hemodynamic principles suggest that it may serve to reduce arterial blood pressure to approximately osmotic levels and to equalize it throughout the ovary. This effect would be accomplished by decrease in the diameter of the vessel, increased length of vessel to be traversed by blood, and by change in the direction of blood flow (cf. Am. J. Obst. Gynecol., in press, 1947).

Injection-corrosion preparations in which vinylite was used as the injection mass will be demonstrated. A number of specimens will be shown in which there are casts of blood vessels to, and within corpora lutea. The effect of rapid follicular growth, ovulation and early luteinization will be demonstrated. In the early stages, the coils are "paid out" either by becoming less tightly coiled, or by becoming undulating in character. By the sixth to the ninth day after injection of gonadotrophins, the original configuration is laregly restored provided there are no large follicular cysts or corpora hemorrhagica. Photographs, stereoscopic transparent pictures in color, stereoscopic photographs, and original specimens will be demonstrated.

D 15. ***On a human ovum of 17 days development.*** **G. Gordon ROBERTSON, Department of Anatomy, Baylor University College of Medicine.**

An implanted ovum was recovered from an amputated uterus on the thirty-third day after the beginning of the preceding menstrual period. A block of tissue which included the chorionic vesicle was fixed in 10% formalin, embedded in paraffin, and sectioned serially at 6 μ. The plane of sectioning was found to be 10° from the transverse plane of the embryonic disc. A model of the embryo was constructed.

Comparison of measurements and morphological details with those of other early human ova indicates that the age is about 17 days. The external dimensions of the chorionic vesicle are $3.81 \times 3.63 \times 2.68$ mm. The villi are branched, and the longest villus is 0.51 mm. Peripherally, the intervillous spaces have broad connections with venous sinusoids but no demonstrable connections with the spiral arterioles.

The embryonic disc is trilaminar. The ectodermal plate measures $0.46 \times 0.48 \times 0.04$ mm. A primitive node and streak, a head process, and a cloacal membrane are present in the disc. The amniotic cavity has a volume of 0.0086 mm^3, the yolk-sac cavity a volume of 0.0209 mm^3. A small allantoic diverticulum extends from the yolk sac into the body stalk. Angiogenesis is evident in the chorionic and body-stalk mesoblast. Both angiogenesis and hemopoiesis are evident in prominent blood islands of the yolk-sac wall.

D 48. ***Isolation of the A- and B-cells in the pancreas of newborn puppies by the method of early duct ligation.*** **Maria A. SERGEYEVA, Institut d'Anatomie Pathologique, Université de Montréal, and Department of Anatomy, McGill University. (Introduced by L. F. Bélanger.)**

Partial ligature of the pancreas was carried out in a number of newborn puppies. The technique consisted of ligating the duodenal and splenic regions of the pancreas without affecting the blood supply of these ligated areas. The animals were sacrificed from 2 to 16 weeks later.

In the ligated duodenal end, the normal acinar structure was at first transformed into coiling strands of oval or rounded cells with prominent, voluminous nuclei. Later on, an abundant proliferation of B-cells arose from these strands, while they themselves gradually became thinner. Maximum B-cell proliferation occurred between the eighth and twelfth week and was then followed by atrophy of the ligated segment.

In the ligated splenic end, the acini underwent a similar transformation. These, however, showed an A-cell proliferation which was maximal at 8 to 12 weeks after ligation.

This difference may be correlated with differences in innervation and embryological origin of these 2 different regions.

This operation reveals the potential ability of acinar cells to become transformed into endocrine cells.

D 49. *The origin of fat cells in the involuting thymus.* Christianna SMITH, Department of Zoology, Mount Holyoke College.

In age involution, the thymus of the mouse and the rat becomes replaced in varying degrees by fat cells. For the most part, the intrathymic, subcapsular fat cells develop in situ from lymphoid cells by the gradual accumulation of fat droplets which fuse finally to form 1 large globule in them.

D 3. *The "acid" phosphatase content of normal autonomic neurons and its increase following section of the cell processes.*[1] Wilbur K. SMITH and Charles LUTTRELL, Department of Anatomy, The University of Rochester, School of Medicine and Dentistry.

The phosphatase content of autonomic neurons was investigated by a modification of the histochemical method of Gomori ('41), the most important alterations being the fixation of the tissues by perfusion with 20% formalin in Ringer's solution and incubation of the sections in the glycerophosphate substrate at pH 4.7–5.0 for a longer time.

The neurons in normal stellate ganglia prepared in this manner show comparatively little phosphatase reaction, the cytoplasm being only lightly stained. A narrow zone around the nucleus shows a greater response. The nucleolus stands out darkly stained among smaller, scattered and more lightly appearing nuclear granules. Nerve cell processes react to approximately the same degree as the cytoplasm. The majority of the nuclei of the peri-neuronal cells appear to have a phosphatase content greater than that of any part of the neuron itself, while the nuclei of the cells of connective tissue enclosing the ganglion show less enzyme reaction.

Chromatolysis in neurons following section of a branch of the stellate ganglion is accompanied by a great increase in phosphatase content of the cytoplasm. In the central nervous system, the cells of the dorsal motor nucleus of the vagus show increased phosphatase following section of the vagus nerve.

[1] Aided by a grant from The National Foundation for Infantile Paralysis.

D 1. *Osmophilic granules in chromatolytic nerve cells.* W. A. STOTLER, Department of Anatomy, University of Oregon Medical School. (Introduced by O. Larsell.)

Distinct granules blackened by osmic acid are seen in chromatolytic cells whose axons have been severed in sections of the rat brain stained by the Swank-Davenport method and counterstained with neutral red-toluidene blue. The granules appear early in the chromatolytic process and form a critical method of identifying affected cells. Granules are demonstrated in the red nucleus following lesion of the rubro-spinal tract, in the ventral tegmental nucleus following lesion of the mammillary peduncle, and in the nuclei of cranial nerves following interruption of their axons. These granules constitute a possible source of error in identifying degenerated myelinated pathways.

D 14. Vascular patterns of development in the primitive neural tube of the rabbit. Leon H. STRONG, Department of Anatomy, University of Michigan.

(See author's abstracts under paper from platform and paper read by title.)

D 41. A comparison of smooth muscle myofibrillae in golden hamster, albino rat and rhesus monkey. Donald C. TANNER, Department of Anatomy, Stanford University. (Introduced by Hadley Kirkman.)

Slides of Bouin fixed muscle, stained by Bodian's protargol method, illustrate the great relative superiority of hamster material for the preparation of exceptionally beautiful demonstrations of mammalian smooth muscle myofibrillae.

D 46. A histological transformation of intestinal mucosa following administration of alloxan. Harry C. TRIMBLE and Harold L. WEATHERFORD, Departments of Biological Chemistry, and of Anatomy, Harvard Medical School.

Shaw Dunn and his associates observed that the primary effect after the administration of alloxan to animals was a selective necrosis of the islets of the pancreas, and the animals became diabetic. No significant changes were seen in other organs.

We have given alloxan to dogs over varying periods and find that when included in the food, there was a temporary elevation of blood sugar and no sugar in the urine. But when alloxan in solution is injected intravenously a severe diabetes ensues. Besides degenerative changes in the islets of Langerhans, we find a remarkable transformation of the epithelium, from columnar to flat, covering the villi of the small intestine. The epithelium of the intestinal glands seems less affected, there being no reduction in cell heights, but from the crowded appearances near the base of the glands there are observed evidences of cellular increase. The villi as a whole look swollen, and the bundles of smooth muscle fibers in them hypertrophic.

D 50. Accessory thyroid tissue and thymus IV in baboon. John H. VAN DYKE, Department of Anatomy, Washington University School of Medicine.

An *atrophic* thymus IV, infiltrated with eosinophiles and associated with developing thyroid tissue within the same connective tissue capsule, has been identified in an adult male baboon (Papio anubis; Subgenus Choerapithecus).

A large medullary cyst, possessing simple columnar epithelium (basophilic and serous secretions) and, in part, *hyperplastic* stratified squamous epithelium with pearl formations, intervened between thymus and accessory thyroid tissue. Cyst extensions, continuous with small (young), densely staining thyroid-like vesicles, were contiguous with large vesicles (older) undergoing transformation into a typical accessory thyroid lobe, the caudal end of which was attached by a strand of follicles to the main thyroid mass at the site of entry of parathyroid IV and ultimobranchial tissue. Although rich in colloid and colloid cells, both thyroid bodies appeared *active;* typical chief cells were rare. Clear cells, however, with large brown cytoplasmic granules, following potassium dichromate fixation, were conspicuous. Globular secretions (serous; frequently basophilic), derived only from such cells, mingled with acidophilic colloid.

The evidence has been interpreted as indicating that thymic cytoreticulum or ultimobranchial tissue left outside the thyroid gland during embryonic development (occasionally as cysts) may be induced to form accessory thyroid tissue ***after birth,*** as in sheep (Van Dyke, '45), and that resorption of colloid, apparently secreted by *colloid cells,* is facilitated by a proteolytic enzyme that may be derived from these granular cells.

D 16. *A study of closure of the pleuropericardial and pleuroperitoneal canals in the human embryo.* L. J. WELLS, Department of Embryology, Carnegie Institution of Washington and Department of Anatomy, University of Minnesota.[1]

In examining serial sections of 221 embryos of the Carnegie Collection (25 somite to 27 mm stages, Streeter's Horizons XII to XXII), it was found that both pleuropericardial canals are closed in 32 embryos of Horizon XVIII (12 to 17.2 mm). Both pleuroperitoneal canals are closed in 23 embryos of Horizon XX (16 to 25 mm).

To assist in studying the manner of closure, 3-dimension reconstructions of 4 embryos were prepared from sheets of cellulose acetate. These models will be demonstrated (4.0, 8.3, 9.2 and 14.5 mm embryos).

Pleuropericardial canals disappear when common cardinal veins fuse with primitive mediastinum. This process is completed earlier on left side than on right. Thereafter those paired membranes that separate pleural and pericardial parts of coelom and that contain phrenic nerves, resemble "mesenteries" of right and left portions of sinus venosus. These membranes contain mesenchyme of origin of part of that endothoracic fascia which loosely binds pleural sacs to pericardial sac of adult.

It would seem that pleuroperitoneal membranes, initially ridges of mesothelium plus mesenchyme they contain, owe their origin to fact that lung buds are too large to pass through very top of paired archways between posterior cardinal veins and sinus venosus. Narrowing of pleuroperitoneal canals involves at first the rapid growth of liver and Wolffian bodies (on left side, stomach as well). Closure occurs soon after suprarenals contact liver.

[1] Fellow of the John Simon Guggenheim Memorial Foundation.

D 45. Effect of alloxan on the pancreatic islets of the guinea pig demonstrated with janus green. Charles A. WOERNER, Department of Anatomy, Long Island College of Medicine.

Twenty guinea pigs were given alloxan in doses of 50 to 200 mg per kilo, intravenously. Twenty-four hours after the alloxan was given the animals were perfused with janus green using the method described by Bensley. Pieces of pancreatic tissue taken from the head, body, and tail of the pancreas of each animal were then washed in water, dehydrated in alcohol, cleared in benzol, and mounted in balsam. For comparison 8 apparently normal guinea pigs were perfused with janus green and the pancreatic tissue prepared in the same manner.

The larger islets appear to show more degeneration than the smaller islets in that many of the cells of some of the larger islets fail to stain with the janus green. While there appears to be some decrease in the number of large islets, the number of small islets does not appear to be greatly reduced.

MOTION PICTURE DEMONSTRATIONS

Dm 59. A new method of semen analysis.[1] Edmond J. FARRIS, The Wistar Institute of Anatomy and Biology.

A method of semen analysis based on counting the number of moving cells is shown. The new procedure correlates the factors of volume, count per cm^3, and actual percentage of motility, to give the absolute motility, which is the actual number of moving spermatozoa in the ejaculate.

The findings obtained from 200 semen analyses on over 150 individuals has resulted in an index of fertility.

Since fertility is dependent upon moving sperm, it is felt that a correlation of the many criteria rather than individual factors is the more reasonable explanation of fertility.

Males are classified as high fertile, relatively fertile, or infertile, as determined by the absolute motility.

Photomicrographs were made by phase microscopy.

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[1] This investigation was aided by grants from the Committee on Therapeutic Research, Council on Pharmacy and Chemistry of the American Medical Association, and the Samuel S. Fels Fund.

Dm 58. Serial levels through an embedded specimen recorded by direct photography of the block surface. Erling S. HEGRE and Alton D. BRASHEAR, Department of Anatomy, Medical College of Virginia.

Embryos prepared for block-surface staining (see Anat. Rec., *97:* 21–28) were mounted vertically in a sliding-knife microtome. A 16-mm movie camera supported above the specimen was used in recording the details of the embryo as seen on the cut surface of the block at successive levels. The completed film is suitable for continuous projection or it may be used in making reconstructions of the specimen.

Dm 53. A colored moving picture used in teaching gross anatomy. W. Henry HOLLINSHEAD and J. E. MARKEE, Department of Anatomy, Duke University School of Medicine.

The reel to be shown is an excerpt from approximately 15,000 feet of colored motion pictures prepared in our department as an aid in teaching gross anatomy. These pictures show demonstration dissections of most of the regions of the body. During the dissection, nerves and blood vessels were painted, so that they might show up more clearly. Then, under the movie camera, the important structures in a given region were pointed out, moved about to show their relations, and, when necessary, removed to display deeper structures.

This film has been used by us in teaching gross anatomy to nurses, physiotherapists and first year medical students, and also in presenting reviews of anatomy to both undergraduate and graduate students. The level of instruction, and the structural details stressed, can be varied within wide limits by supplementation with appropriate lantern slides.

Dm 57. Hypokinesia and the retention of superimposed posture in the primate as a result of neural lesion. Fred A. METTLER, Department of Neurology, College of Physicians and Surgeons, Columbia University.

Overactivity as a result of cortical removal and essential overactivity with fatuity, as a result of striatal ablation have been described by Hines and Richter, Mettler and most recently by Heath and Freedman. The motion picture to be presented illustrates the effect of bilaterally removing all cortex rostral to the middle of area 6, the heads of the caudate and most of the pallida.

Addition of pallidal removal converts overactivity into underactivity. There is a chronic loss of adaptive activity to disturbing situations, stereotypy of movement and apparent drowsiness. Flexor plantar responses may be obtained, the patellar reflexes are brisk, easily elicited and show good amplitude. In addition, superimposed postures are maintained. Such a condition does not depend upon hypothalamic damage.

The current interpretation of this condition suggests that cortical and striatal removal release thalamopallidal circuits dealing with stereotyped varieties of movement (associated movements) which depend to a large extent upon proprioceptive stimulation and that this circuit is interrupted by pallidal removal. Such an interpretation agrees with the observed loss of associated movements which is encountered in various clinical conditions such as lenticular degeneration.

Dm 55. Cinephotomicrograph of pure thymus epithelium.[1] Raymond G. MURRAY, Department of Anatomy, Tufts College Medical School. (Introduced by Benjamin Spector.)

This film combines representative portions of many cinephotomicrographs of epithelium from cultures of rabbit thymus ranging in age from 1 day to 1 year. Typical epithelial growths in the form of "tongues" projecting from thymus cultures are shown, as well as the subsequent activity of the pure epithelium resulting from cutting away the rest of the culture. After several days in vitro this pure epithelium is shown exhibiting pinocytosis and possibly phagocytosis, and at times growing in a manner resembling connective tissue. A more typical mode of growth, however, is that of anastomosing cords of flat cells. Two instances of what appear to be differentiating mitoses are shown. In each case 1 daughter cell returns to the epithelial cord and the other moves about in a manner somewhat suggestive of a lymphocyte, although much slower. Such mitoses occurred only twice, and both times while the culture was under low oxygen tension. Gall bladder epithelium, cultured and photographed under similar conditions is included for comparison.

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[1] Aided by a grant from the Abbott Memorial Fund of the University of Chicago.

Dm 54. *Dynamics of normal and pathological temporomandibular articulations.* William M. ROGERS and S. L. KATZ, Departments of Anatomy and of Dental and Oral Surgery, College of Physicians and Surgeons, Columbia University.

These pictures show the types and limitations of movement in a variety of disorders of the temporomandibular joint. Results following surgical interference in selected cases, are presented.

Laminagrams of normal and malfunctioning articulations are used as an aid in studying their anatomy and their function.

Dm 52. *Motion of the ankle and foot produced by individual muscles in the leg.* Charles E. TOBIN and Francis F. BAKER, Department of Anatomy and Division of Orthopedics (Department of Surgery), The University of Rochester, School of Medicine and Dentistry.

A dissected, unembalmed leg and foot were photographed on Kodachrome film to show the action of the muscles in the leg on the movements of the tarsus, metatarsus and digits.

The muscles were isolated, each identified by a label, and its contraction simulated by mechanical tension.

This film is offered as a technique to aid in teaching the action of these muscles in their functions as extrinsic muscles of the foot.

Dm 56. *Circulatory flow pattern in the bat's wing.* Richard L. WEBB and Paul A. NICOLL, Departments of Anatomy and Physiology, Indiana University Medical School.

Observation of peripheral blood circulation in the subcutaneous tissue of the wing membrane of the living bat is possible by use of the compound microscope. Records of vascular pattern and behavior are recorded by cinematography for this demonstration. The arrangement of the vascular system to the capillary bed is in the form of a series of arcades. The capillary system proper is a complex net of endothelial tubes, supplied by terminal arterioles and drained by coalescing venules connecting with the arcuate vessels. No preferential flow paths or arterio-venous anastomosis have been observed.

Active vasomotion on the arterial side of the system is more pronounced as the capillary bed is approached, being manifested mostly in the terminal arterioles and precapillary sphincters, where it is rhythmic in nature. On the venous side of the circulation, active vaso-motion is resumed at the location of the first venous valve. From here on the rhythmic movement is definite, apparently concerned with the propulsion of blood in its return to the heart.

By use of methylene blue vital stain, some of the muscular elements are shown. Of especial interest is the coiled or spiral muscle cells of the terminal arteriole. Arrangement of muscle fibers in the region of venous valves and junctions is shown.